Project Ascent: Measuring What a Small Language Model Actually Learns

Three Findings: a Calibrated Model, a Retargeted Model, and a Contaminated Ladder

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Abstract

Project Ascent builds decoder-only Transformer language models from scratch and measures what each design decision actually changes. The models are deliberately tiny — 1.5M, 2.6M and 6.3M non-embedding parameters at a 128-token context — and this paper reports the negative results with the same rigour as the positive ones. Three findings carry the paper, and Table 38 is the taxonomy of everything else: 89 rows in four populations, 10 *supported*, 26 *unsupported*, 15 *measurement failures* and 38 *declined claims*, every row traceable to a tracked file. The **first finding is the calibration result**. Over 4,064 held-out next-token predictions the model is accurate 52.95% of the time at a mean confidence of 52.55%, with an expected calibration error of 0.0233 and bins that track (confidence 0.979 $\rightarrow$ accuracy 0.962; confidence 0.054 $\rightarrow$ accuracy 0.074). The model is a competent, well-calibrated next-token predictor and an incompetent instruction follower, grounder and tool user, and every downstream failure in this paper is a failure to *use* the competence rather than a failure to have it. The **second finding is that instruction tuning retargets that competence rather than adding to it**. Tuning on 512 examples raises held-out instruction following from 0.0000 to 0.6016 while adding no next-token competence on the distribution it was tuned on: there the accuracy difference is -0.0009 (95% $[-0.0160, +0.0142]$, not significant), the mean confidence nevertheless rises from 0.3969 to 0.6261 with the accuracy flat, and the calibration error on that distribution more than doubles (0.1118 $\rightarrow$ 0.2772). Off target the trade is paid in accuracy: the tuned model loses 10.7 points of next-token accuracy (-0.1068, 95% $[-0.1284, -0.0852]$) and its calibration error rises from 0.0233 to 0.1306. The **third finding is the contamination**. The scaling ladder mixed two architecture families, was recorded on a parameter axis the published exponents do not use, and was trained by a pipeline that was never in the repository — so a mid-campaign change of model class was invisible to every downstream artefact, including the ledger — and its headline exponent is withdrawn. The study of what replaced it says that this ladder is *informative about β and uninformative about α*. E is the conditional entropy of an infinite model, so the corpus's unigram entropy of 4.9946 nats is a loose ceiling and never a target; the correct anchors are the token-space compression ceiling of 3.5918 nats per token and the model's own achieved held-out loss of 2.5152 nats, and E must lie below the achieved loss. Both fitted E values do — 1.964 on the total-parameter axis and 1.911 on the non-embedding axis — so the bound admits both conventions; the axis the paper uses is the published one, and the conclusion is unchanged. The β profile's compatible range $[0.38,\ 0.54]$ excludes Kaplan's 0.30 and contains Chinchilla's 0.50; the α profile's range $[0.34,\ 0.82]$ contains both published values and discriminates between none of them. The α width is not noise: it is a ridge, with $d\alpha/dE = +0.5542$. The refit is not a minor correction. The withdrawn fit's own parameters score $R^2 = +0.9957$ on the other family's points and -1.312 on the canonical models' own points, and that other family owned the entire top of the ladder, so the original "consistent with Chinchilla" claim was resting on it. A KV cache gives byte-identical output and 3.71$\times$ to 18.36$\times$; sharing KV heads buys an 8$\times$ cache reduction and 12.1% fewer FLOPs at a small

but statistically significant quality cost; speculative decoding is a bounded negative whose cause is arithmetic rather than implementation. Decoding is a clean quality/diversity trade-off whose quality axis is self-preferring, and self-consistency cannot manufacture competence that is absent. Supervised fine-tuning scales monotonically with no saturation in range, and in-context learning copies a demonstration's *format* without its operation. On grounding, the model identifies which number a passage asserts *at chance*, and the logit lens shows why: context numbers are represented in the residual stream and assertion-specific content is not, so the representation is not carried to the output. That act's control set was run after the claim was written, it went *against* the stronger version of the claim, and the stronger version is withdrawn here rather than quietly dropped. The lesson we want remembered: a test that compares two implementations cannot detect a bug that both of them share, so prefer an absolute sanity invariant — and check the provenance of your data set before you fit a law to it. A good-looking fit is not evidence of a good fit. All fifteen measurement failures this project recorded were caught by such an invariant or by a positive control, and not one of them by comparing two implementations.

1 Introduction and research questions

This is a course project in building a language model from scratch and *measuring* the effect of every design choice, rather than asserting it. The models are intentionally small: the largest has 6.3M non-embedding parameters and a 128-token context. That scale cannot demonstrate useful capability, and this paper does not claim that it does. What the scale can do is isolate effects, and several of the effects it isolates are null results with a mechanism attached.

The project is organised as ten *acts*. Each act removes one limitation that a previous act exposed, and each act ends in the same shape: a table with its uncertainty stated, one or two figures, and a conclusion that says both what the measurement supports and what it does not. A negative result gets a table, figures and a conclusion, exactly as a positive one does. There is no section in this paper that reports a number and moves on.

The thesis, stated by the measurement that carries it. The models built here are competent, well-calibrated next-token predictors and incompetent instruction followers, grounders and tool users. Over 4,064 held-out next-token predictions on C4, the 6.3M model is accurate 52.95% of the time at a mean confidence of 52.55%, with an expected calibration error of 0.0233 and confidence bins that track accuracy closely (confidence 0.979 $\rightarrow$ accuracy 0.962; confidence 0.054 $\rightarrow$ accuracy 0.074). That is the only positive result in the act that measures it, and every other act in this paper is a measurement of a failure to *use* that competence: a zero-shot instruction floor at exactly 0.0000, a grounding rate at chance, a tool-call template whose argument is always `s`, an in-context learner that copies a demonstration's format and not its operation. A failure to use a competence is a different kind of fact from a failure to have one, and this project's contribution is to separate the two. The consequence for the reader is a specific and slightly uncomfortable one: the floor results in Acts 7, 8 and 9 are not evidence that the models cannot do those things in principle, and they are not evidence that they can either. They are evidence that at this scale, with this training budget, on this corpus, the capability that is present does not travel.

The second-largest result is a negative one, and it is about the act the project was built around. The scaling ladder's exponent turned out to be unidentifiable at this scale, and the reasons are now known in detail rather than guessed at. The ladder mixed two architecture families because the training pipeline that produced one of them was never committed to the repository, so a mid-campaign change of model class was invisible to every downstream artefact including the ledger. What replaced the withdrawn fit is an **asymmetry**: on the total-parameter axis the ladder is *informative about β and uninformative about α*. The compatible β range is $[0.38,\ 0.54]$ — it excludes Kaplan's 0.30 and contains Chinchilla's 0.50, so it discriminates — while the compatible α range is $[0.34,\ 0.82]$, wide enough to contain every published exponent, so **on α the ladder is compatible across their entire range and**

discriminates between none of them. The mechanism is the design of the sweep: each run contributes interior checkpoints at fixed N and rising D, so within-run D variation pins β, while α depends on the number and spread of distinct N values, of which the ladder now has eight. The α width is not noise but a ridge: fixing E on a grid and re-optimising α traces a monotone path with $d\alpha/dE = +0.5542$. Both results are reported in Section 7.

The research questions, in order, are:

1. **Do the classical baselines behave as advertised?** (Acts 0–1.) How much does n-gram order matter relative to smoothing, and are recurrent architectures distinguishable at all on a corpus this small?

2. **Which Transformer components actually matter?** (Acts 2–3.) What do normalisation, positional encoding, head count and FFN activation do when measured one seed at a time, and how much of a tokenizer comparison survives a context control?

3. **Do the published scaling laws hold below their fitted range?** (Act 4.) We fit our own α and β, on the parameter axis the published exponents use, and report the profile likelihood rather than a point estimate: on α the ladder is compatible with published exponents across their entire range and discriminates between none of them, while β, which is two to three times better constrained, excludes one published value and contains the other. We also ask whether the fitted E is a measured entropy; it is not, it is an extrapolated asymptote, and the width of the α profile turns out to be an (E, α) ridge rather than noise.

4. **What do inference-time techniques buy, and what do they cost?** (Acts 5–6.) KV caching, shared KV heads, speculative decoding, and the decoding sweep, with quality measured against a fixed judge.

5. **Can a model this small be taught to follow instructions, and does it ground what it reads?** (Acts 7–8.) Supervised fine-tuning, in-context learning, parameter-efficient adaptation, retrieval-augmented answering and tool use.

6. **Where does the information go, and can the project audit itself?** (Acts 9–10.) Causal head ablation, attention attribution and a logit lens, each with the control that decides whether its claim survives; then the measurement-audit ledger and the taxonomy of the ways a measurement in this project failed to measure what it claimed to.

The answer, in one sentence per act, is in Sections 3–13. The taxonomy of every hypothesis the project put at risk — with the four populations, their counts and the source of every row — is Table 38, at the front of this paper and generated from one tracked file (`results/v2/findings_taxonomy.json`). The headline claims are tabulated with their n, their interval and their status in Section 13.1, the full per-claim ledger is the tracked artefact `report/claims.csv`, and Section 14.4 publishes the claims we deliberately do *not* make.

2 Method

2.1 Model and training setup

Two model families appear in this paper, and they are not interchangeable. The *canonical* set used for Acts 5–9 is a decoder-only `MiniGPT` with learned absolute positions, tied input/output embeddings, pre-norm residual blocks and a PyTorch scaled-dot-product attention path with a manual fallback. Its three rungs are pinned and exported as weights so that every figure in Acts 5–9 can be regenerated without retraining:

Rung	Non-embedding params	Total params	Steps	Tokens	Validation loss
1.5M	1,567,200	2,605,120	7,500	30,720,000	3.0571
2.6M	2,694,144	3,939,648	12,960	53,084,160	2.8111
6.3M	6,351,360	8,012,032	30,720	125,829,120	2.5152

Table 1: The canonical pinned model set used in Acts 5–9. Source: `checkpoints/canonical/INDEX.json`, whose counts are computed from the instantiated model and carry a SHA-256 per file. *These are measurements, not search targets*: the ladder aimed at 1,536,000 / 2,654,208 / 6,291,456 non-embedding parameters, and the models that were actually built have 1,567,200 / 2,694,144 / 6,351,360. The gap at the 1.5M rung is exactly 20,480, one 128×160 learned position table, which is the same difference that separates the two architecture families in Section 7.2. The 2.6M rung reproduced the MHA sweep loss exactly, from a different script on a different day, which is an unplanned determinism check. $n = 1$ seed per rung for these three checkpoints; the 2.6M rung additionally carries the six paired seeds of Act 5b.

The Act 4 scaling ladder is a different sweep with its own architecture record and its own corpus, and it turned out to mix *two* architecture families: its runs, their families, their parameter counts and their seeds are given in Table 6 and discussed in Section 7.

2.2 Data and tokenization

Three corpora are used, and results are never pooled across them. (i) TinyShakespeare (1,115,394 characters, 66-symbol vocabulary) is the corpus for Acts 0–3, with a 90/10 character-level train/validation split of 1,003,854 / 111,540 characters. (ii) A C4 English shard (356,317 documents, 769,299,382 characters, SHA-256 recorded in the corpus manifest) is the corpus for Acts 4–9; it is tokenised with a 6,487-token BPE fitted on a 50,025,065-character sample of the training split only, at 2.2726 characters per token. (iii) Two held-out evaluation sets that are never trained on: MNLI (300 items, used in Section 9) and a synthetic instruction family with 512 training and 128 held-out items (used in Section 10).

Two tokenizer corrections belong to the method, not to an act, because they changed what a measurement could mean. Tokenizers are now fitted on the *training* split only, and validation is computed by a seeded full sweep rather than by unseeded random batches. Both fixes were needed before any v2 number could be trusted.

2.3 Measurement protocol

Every number in this paper carries its sample size, its interval and its status. The rules are:

- **Seeds.** A claim that is meant to establish an effect is measured with $n \geq 3$ seeds and reported as mean $\pm$ sample standard deviation. Where a method is deterministic, the three "seeds" are bit-identical; that is stated, and the result is relabelled $n = 1$ in substance rather than being given a zero-width error bar.
- **Accuracy.** Proportions are reported with 95% Wilson score intervals. Perplexity, loss and log-probability contrasts are reported as mean $\pm$ standard deviation or as a paired difference with a bootstrap interval; a Wilson interval is never used for a continuous quantity.
- **Unit of analysis.** Where a claim is about an item, the item is the unit. Repeat generations of the same item are averaged per item before scoring, and contrast intervals resample *items*, because repeats of an item share its difficulty and are not independent draws. This

distinction changes a Wilson bound: 0/40 items gives an upper bound of 0.0876, not the 0.031 that a spurious $n = 120$ would give.

- **Controls before comparisons.** Every A-versus-B experiment ships with the control that makes the comparison mean something: an equal-character-span control for tokenizer work, a non-numeric passage control for grounding, a matched trainable-parameter budget for parameter-efficient adaptation, and a zero-gap configuration for speculative decoding.
- **Pre-registration.** A falsification condition is written down before the run that it judges. Act 4's condition was written on 2026-10-15 and is reproduced verbatim in Section 7. Anything run after a result was seen is labelled *exploratory*, and no exploratory result is presented as a headline. The convention for the whole project, and the acts that did not follow it, are stated in Section 2.4.
- **Status vocabulary.** *confirmed* (the measurement supports the claim at the stated n and interval), *n.s.* (measured, and the interval does not exclude the null), *exploratory* (single seed, or run after seeing results), *ceiling* (the metric is saturated, so the value cannot rank conditions), *invalid* (the measurement does not test what it claims to test), *superseded* (a later measurement replaces it).

2.4 The falsification convention, and where it was not followed

The convention this project holds itself to is that **every act should have a written falsifier before the run it judges**: a statement of what would have counted against the claim, written down while it is still cheap to be wrong. Where such a document exists, this paper names it. Where one does not, Table 38 says so, row by row, in its *preregistered* column, and the corresponding row of the tracked artefact (`results/v2/findings_taxonomy.json`) carries the note.

The honest accounting is this. **Exactly one act in this project has a written falsifier**: Act 4, whose condition is `results/v2/act4-scaling/FALSIFICATION_CONDITION.md`, dated 2026-10-15 and reproduced in Section 7.14. It was written before any run of the sweep, it is the document that makes Act 4's headline "exploratory" rather than "confirmed" in the places where the family separation and the axis convention were decided after the fact, and it is the row in Table 38 whose *preregistered* column reads *true*. Every other act's hypothesis row reads *false*.

And the most decisive control in the project was post-hoc. Act 7's zero-shot floor is assertable only because a positive control — an oracle emitting the correct answer, scored through the teacher-forced path, the greedy path and the exact in-context scoring path — scored 1.0 while the real model scored 0.0000. That control was run *after* the 0.0000 floor had been recorded, and it turned out to be the thing that made the floor assertable at all. The same-task control behind Act 7b was built after the first in-context measurement had been voided by truncation. Neither is marked as pre-registered, because neither was: they are marked *post-hoc* in Table 38 and in the ledger, which is more honest than pretending they were planned.

Two consequences follow, and both are limits rather than boasts. First, this project cannot claim pre-registration as a general practice: of the hypotheses in Table 38, one was falsifier-first. Second, the project's actual defence is not pre-registration at all but the taxonomy in Section 13.2: every measurement failure that was *recorded* was caught by an absolute invariant or a positive control, and the controls that were decisive are marked as post-hoc where they were post-hoc.

Table 38 uses a four-value status vocabulary of its own, which is deliberately coarser than the ledger's: *supported* (the hypothesis held), *unsupported* (it did not — the negative-result population), *measurement failure* (a defect that was caught and documented) and *declined claim* (a claim this paper does not make). The ledger in Section 13.1 uses a different and finer

vocabulary (*confirmed*, *n.s.*, *exploratory*, *ceiling*, *invalid*, *superseded*, *open-item*) because a ledger row describes the evidential status of a *number*, while a taxonomy row describes the fate of a *hypothesis*.

2.5 A note on what a 6.3M model can and cannot do

It is worth being explicit about the scale, because several results are only interpretable against it. A 6.3M-parameter model trained on 125.8M tokens reaches a validation loss of 2.5152 nats, or roughly 3.63 bits per token. It is a competent next-token predictor of web text and an incompetent instruction follower, because instruction following is not a next-token prediction problem at this scale. It can copy the surface form of a demonstration while failing the operation the demonstration performs, and it can emit a tool-call template while being unable to fill it. Several of the negative results in this paper (Sections 10, 11 and 12) are not failures of method; they are the correct measurement of a model that has learned a format and not a capability — and Act 9 measures where that information goes: it holds a generic number representation in the residual stream, and that representation does not reach the output. Reporting any of them as something other than what was measured would be the easy mistake, and this paper tries not to make it.

3 Act 0: n-gram models and smoothing

3.1 Setup

Thirty-six runs: three n-gram orders (1, 2, 3) times four smoothing schemes (Laplace, Good–Turing, Kneser–Ney, linear interpolation) times three seeds. The corpus is character-level TinyShakespeare with a 90/10 character split (1,003,854 training characters, 111,540 validation characters, vocabulary 65).

3.2 Results

Order	Laplace	Good–Turing	Kneser–Ney	Interpolated
Validation perplexity (mean ± s.d. over 3 seeds)				
1	28.427 ± 0.000	28.427 ± 0.000	28.427 ± 0.000	28.427 ± 0.000
2	11.964 ± 0.000	12.070 ± 0.000	11.956 ± 0.000	11.964 ± 0.000
3	7.913 ± 0.000	8.085 ± 0.000	**7.446** ± 0.000	8.778 ± 0.000
Validation bits per character (mean ± s.d. over 3 seeds)				
1	4.829 ± 0.000	4.829 ± 0.000	4.829 ± 0.000	4.829 ± 0.000
2	3.581 ± 0.000	3.593 ± 0.000	3.580 ± 0.000	3.581 ± 0.000
3	2.984 ± 0.000	3.015 ± 0.000	**2.897** ± 0.000	3.134 ± 0.000

Table 2: Act 0: validation perplexity and bits per character for 12 configurations, each run at 3 seeds. Source: `results/act0-ngram/summary.json`; table source data in `report/tables/act0_ngram_perplexity.csv`. Perplexity is a continuous quantity, so mean ± sample standard deviation is the interval reported here (a Wilson interval applies to proportions, not to perplexity). **The standard deviation is exactly 0.000 in every row**: count-based n-gram models have no stochasticity, so the three seeds are bit-identical and the effective n is 1 in substance. No error bar in this table is informative.

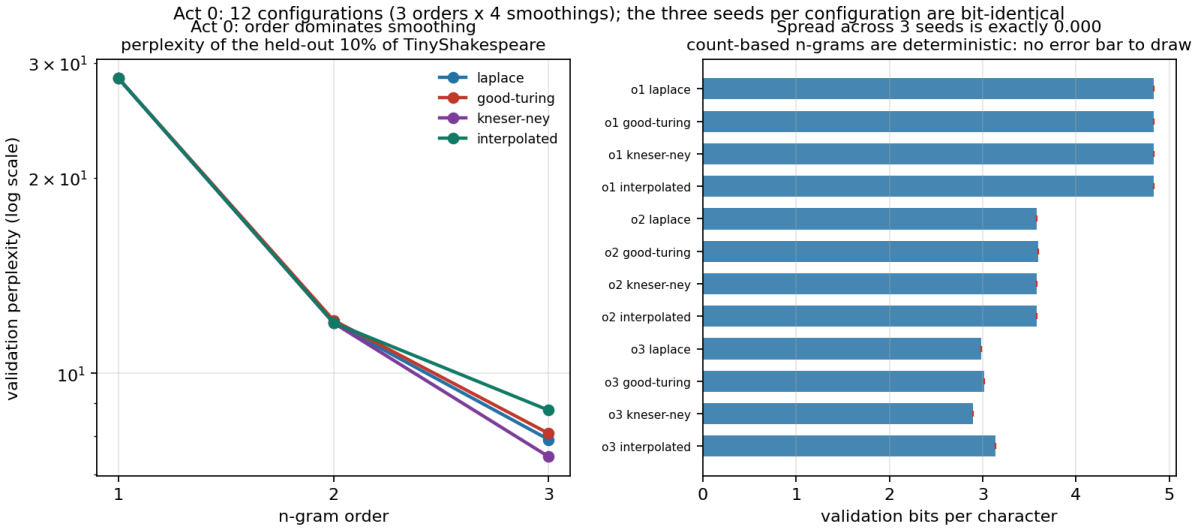


Figure 1: Act 0. *Left:* validation perplexity (log scale) against n-gram order, one series per smoothing scheme, over the held-out 10% of TinyShakespeare. *Right:* validation bits per character for all 12 configurations, with the seed spread drawn as an error bar. **What is plotted:** 12 configurations, each at $n = 3$ seeds. **Interval type:** the error bar is the sample standard deviation over the three seeds, which is exactly 0.000 because count-based n-grams are deterministic, so the effective n is 1 in substance. Source: `results/act0-ngram/summary.json`.

3.3 Embeddings: built, not measured

The syllabus for this act also lists dense representations: truncated SVD of a count matrix, word2vec in both CBOW and skip-gram form with negative sampling and frequent-word subsampling, GloVe, and fastText with character n-grams and boundary symbols. **None of this work is implemented in this repository and none of it is measured.** There is no `src/embed` package, no tracked result file and no figure. We therefore report it as absent rather than dressing up the absence as a result, and the coverage checklist in `AGENTS.md` records it as an open item.

3.4 Conclusion

What the measurement supports. Order dominates smoothing. Moving from order 1 to order 3 reduces validation perplexity from 28.43 to 7.45–8.78, a factor of 3.2–3.8, while the spread across the four smoothing schemes at a fixed order is at most 0.9 in perplexity at order 3 and is exactly zero at order 1. The best configuration in the table is trigram Kneser–Ney at 7.446 perplexity (2.897 bits per character).

What it does not support. It does not support a claim about smoothing methods in general, because the comparison is single-corpus, single-split and deterministic, and because the interpolation arm is handicapped by construction: its weights are fixed at (0.1, 0.3, 0.6) for every order, so at order 3 it is the only arm that is being asked to interpolate a distribution it was not tuned for, and it duly comes last (8.778). We do not claim that interpolation is a worse smoothing scheme; we claim that the arm as configured is the worst of the four here, which is a statement about the configuration. The three "seeds" give no uncertainty estimate at all, and the table is presented with the standard deviation of 0.000 visible for exactly that reason.

4 Act 1: recurrent baselines

4.1 Setup and results

A plain RNN, an LSTM and a GRU are trained on the same character-level TinyShakespeare split as Act 0 (500 steps, batch size 16, context 64), three seeds each, with parameter counts matched only loosely.

Model	Validation loss	Validation ppl	Parameters	n
RNN	2.438 ± 0.215	11.633	31,362	3
LSTM	2.590 ± 0.285	13.713	58,434	3
GRU	2.482 ± 0.254	12.231	45,954	3

Table 3: Act 1: recurrent language models, mean ± sample standard deviation over 3 seeds. Source: `results/act1-rnn/summary.json`; table source data in `report/tables/act1_recurrent_baselines.csv`. Validation loss is in nats per token; the interval is a standard deviation over 3 seeds, not a Wilson interval, because the quantity is continuous.

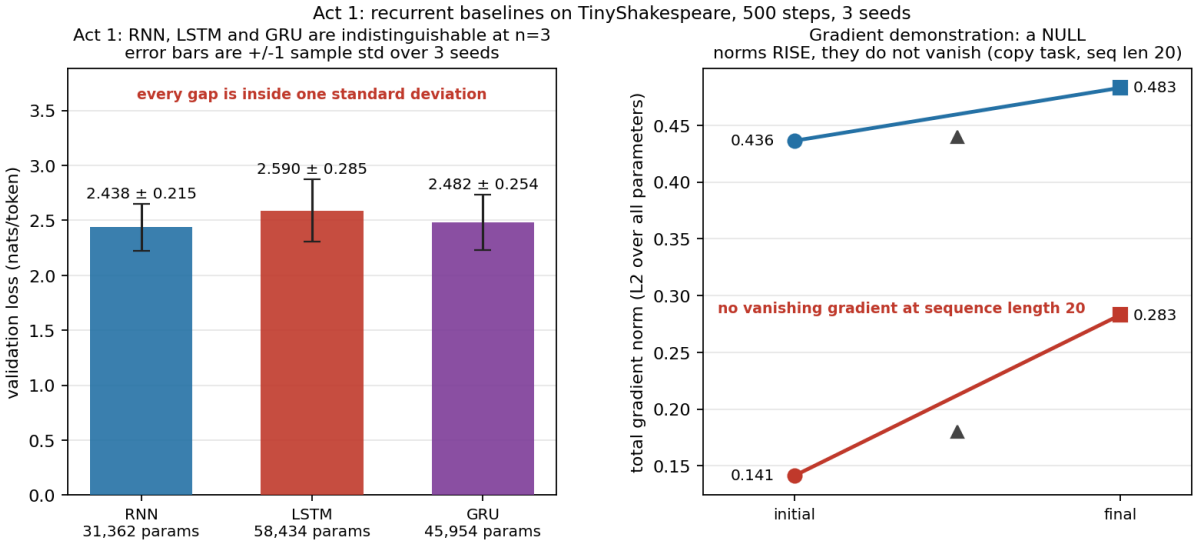


Figure 2: Act 1. *Left:* validation loss of the RNN, LSTM and GRU, with error bars of ±1 sample standard deviation over $n = 3$ **seeds**. *Right:* the gradient demonstration: the total L_2 gradient norm at the first and last of 100 steps of a copy task at sequence length 20, with the mean over the 100 steps marked by a triangle. **What is plotted:** 3 models at $n = 3$ seeds, plus a gradient measurement with $n = 1$ run of 100 steps per model. **Interval type:** standard deviation over seeds (left); no interval is available for the gradient norms, which are a single trace per model. The right panel is a **null**: the norms rise, they do not vanish. Source: `results/act1-rnn/summary.json`.

4.2 The gradient demonstration, which is a null

The experiment was built to demonstrate vanishing gradients, and it does not. The total gradient norm over all parameters rises over the 100 steps of the copy task: from 0.4363 to 0.4829 for the plain RNN, and from 0.1412 to 0.2828 for the LSTM (means 0.4400 and 0.1803 over the trace). The LSTM's norms are *lower* than the RNN's throughout, which is the opposite of the ordering the gating argument predicts. At sequence length 20 on a copy task, vanishing gradients were

not demonstrated at all. We report the measurement, not the narrative the experiment was designed to tell.

4.3 Conclusion

What the measurement supports. RNN, LSTM and GRU are *indistinguishable* at $n = 3$ on this corpus. The largest gap, LSTM minus RNN, is 0.152 nats, which is smaller than the smaller of the two standard deviations (0.215 and 0.285); the LSTM's mean is also worse than both alternatives, but not by an amount the seed spread does not already explain.

What it does not support. It does not support the ordering that recurrent architectures are usually reported in, and it does not support a general claim that gates do not help. The comparison is confounded by capacity: the LSTM carries 58,434 parameters against the RNN's 31,362, so the LSTM is being asked to do more with the same 500 steps and the same 1.1M characters. A parameter-matched comparison at longer training would be needed to say anything about gates, and we do not make that claim. Nor does this act demonstrate vanishing gradients: the demonstration measured the opposite, at a sequence length far too short to produce the effect.

5 Act 2: Transformer component ablations

5.1 Setup and a prominent caveat

Eighteen configurations, **seed 42 only**: seven normalisation variants (LayerNorm pre and post, RMSNorm pre and post, BatchNorm pre and post, and no normalisation), five positional encodings (learned absolute, sinusoidal, RoPE, ALiBi, none), three head counts (1, 2, 4) and three FFN activations (GELU, ReLU, SwiGLU). All are 2 layers, 64 dimensions, 4 heads, context 64, 500 steps, on the same character-level split as Act 0.

Family	Configuration	Validation loss	Parameters
Normalisation	layernorm-post	2.5615	108,416
	rmsnorm-post	2.5647	108,096
	rmsnorm-pre	2.6121	108,096
	layernorm-pre	2.6319	108,416
	none-pre	2.6380	107,776
	batchnorm-pre	2.6637	108,416
	batchnorm-post	2.6610	108,416
Positional encoding	pe-none †	2.4851	104,320
	pe-alibi †	2.4851	104,320
	pe-sinusoidal	2.4994	104,320
	pe-rope	2.5296	104,320
	pe-learned	2.6270	108,416
Attention heads	heads-4	2.6059	108,416
	heads-2	2.6161	108,416
	heads-1	2.6264	108,416
FFN activation	ffn-swiglu	2.5725	140,544
	ffn-relu	2.6174	108,416
	ffn-gelu	2.6272	108,416

Table 4: Act 2: validation loss (nats per token) for 18 ablations. **Single seed (seed 42, $n = 1$); exploratory; no error bars; no significance testing.** The values are validation *loss*, not bits per character: no bits-per-character figure is stored for this act. Source: `results/act2-ablations/summary.json`; table source data in `report/tables/act2_ablation_summary.csv`. † These two rows are numerically indistinguishable and the second of them did not test what it claims to test; see Section 5.3.

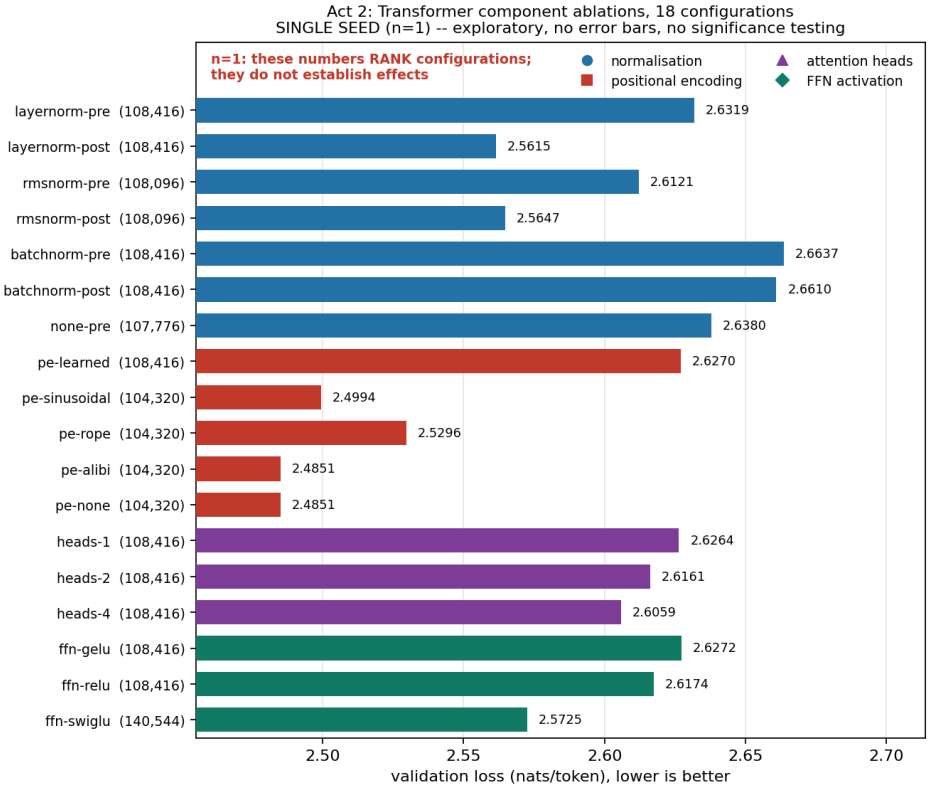


Figure 3: Act 2. Validation loss for all 18 ablation configurations, coloured by the component being ablated, with the parameter count beside each label. **What is plotted:** 18 configurations. **Interval type:** none — **single seed ($n = 1$), exploratory, no error bars and no significance testing**. The figure is a ranking of configurations, not an estimate of an effect. Source: `results/act2-ablations/summary.json`.

5.2 Why the numbers in this section are a ranking and not an effect

At $n = 1$ there is no variance estimate, so no comparison in Table 4 can be tested. The differences between configurations range over 0.179 nats across a set that is likely to have a seed spread of the same order as the smallest differences shown, and we have no measurement of that spread. Accordingly this section contains no $\pm$ columns, no p-values and no comparative adjectives except the word "suggests", which is used nowhere without it. The rows are reported so that a reader can see the shape of the ranking, and so that a future multi-seed rerun has a baseline to be compared against.

5.3 The ALiBi arm did not measure ALiBi

The two lowest rows of the positional-encoding block are not two results; they are one result counted twice, and we would rather say so than let the table imply a finding. The `pe-alibi` and `pe-none` configurations differ in exactly one field (`positional_encoding`), have identical parameter counts (104,320), and produce final validation losses of 2.4850885868 and 2.4850885391 — a difference of 4.8×10^{-8} nats, which is float32 round-off. Their loss trajectories agree to the same precision at every logged step.

The cause is in the source, not in the measurement. `src/model/ablations.py` defines `ALiBiPE.forward` to return its input unchanged, with the comment that the attention bias is applied in attention, and the ablation model's forward pass carries the comment *"ALiBi: apply attention bias (not implemented, skip)"*. The ALiBi arm therefore ran the no-positional-encoding model. The honest reading of Table 4 is that *no positional encoding scored best among*

the encodings that were actually implemented, and that ALiBi was not measured at all. The claim "ALiBi is competitive with learned absolute positions" is **invalid**; the claim "ALiBi was not measured" is confirmed. This is the same class of error as the Act 5 beam-cache claim in Section 8.4 and as Act 7's defect D11: an assertion about what the code does, with no test behind it.

5.4 Conclusion

What the measurement supports. A single-seed ranking of 18 configurations. Post-norm LayerNorm (2.5615) and post-norm RMSNorm (2.5647) occupy the top of the normalisation block, BatchNorm occupies the bottom (2.6610–2.6637), SwiGLU leads the FFN block (2.5725, at 32,128 extra parameters), and 4 heads beat 2 beat 1 in the head block. The absence of positional encoding leads the encodings that were implemented.

What it does not support. None of the above as an *effect*. It does not support any statement about ALiBi, RoPE, sinusoidal or learned positioning, because there is no variance estimate and one of the five arms did not run the thing it names. It does not support the claim that post-norm beats pre-norm, or that SwiGLU beats GELU, or that more heads are better; at $n = 1$ these are suggestions. And it does not support any comparison with the other acts in this paper, because the metric here is validation loss on a character-level corpus at a 64-token context, which is a different scale from every other act.

6 Act 3: tokenization

6.1 Setup and the control that makes the comparison mean something

Four tokenizer families — character, BPE, WordPiece and Unigram LM — were run under two context controls, at **seed 42 only**. The *equal character count* control gives every tokenizer the same 64-token window. The *equal character span* control gives every tokenizer the same number of *characters* of text, which is the control the v1 comparison needed and did not have: at a fixed 64-token window, the character model saw 64 characters and the BPE model saw about 104, so part of v1's BPE advantage may have been data exposure rather than tokenization quality.

Tokenizer	Control	Bits/char	Ppl/char	Chars/token	Vocab
Character	equal count	3.4680	11.0653	1.000	66
Character	equal span	3.4680	11.0653	1.000	66
BPE	equal count	3.0801	8.4569	1.606	166
BPE	equal span	3.1893	9.1214	1.606	166
WordPiece	equal count	3.4998	11.3122	1.001	166
WordPiece	equal span	3.5116	11.4049	1.001	166
Unigram	equal count	2.8021	6.9745	1.813	166
Unigram	equal span	2.8341	7.1309	1.813	166

Table 5: Act 3: validation bits per character for four tokenizer families under two context controls, **single seed (seed 42, $n = 1$); exploratory; no error bars**. Perplexity per character is shown for reference; per-token perplexity is not comparable across tokenizers and is not shown. Source: `results/act3-tokenizers/summary.json` and the per-run `tokenizer_stats.json` files; table source data in `report/tables/act3_tokenizer_comparison.csv`.

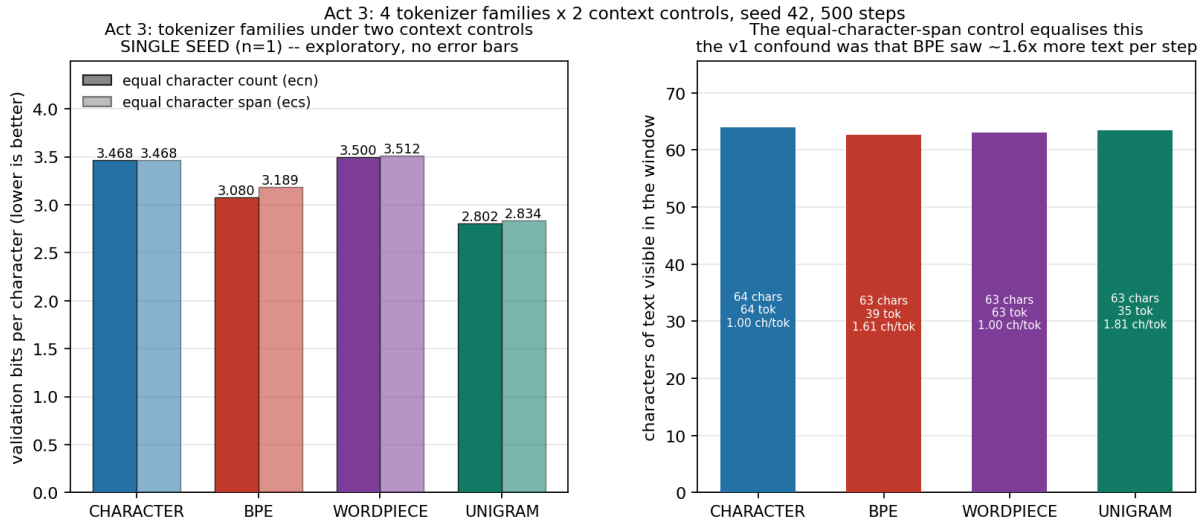


Figure 4: Act 3. *Left:* validation bits per character for the four tokenizer families under both context controls, with the control indicated by fill opacity. *Right:* the number of characters of text visible in the window under the equal-span control, which is the quantity the control equalises. **What is plotted:** 8 configurations (4 tokenizers × 2 controls). **Interval type:** none — **single seed ($n = 1$), exploratory, no error bars and no significance testing**. Source: `results/act3-tokenizers/summary.json`.

6.2 Conclusion, and what happened to the v1 headline

What the measurement supports. Under the equal-count control, Unigram LM is the strongest tokenizer at 2.8021 bits per character, followed by BPE at 3.0801, then character at 3.4680 and WordPiece at 3.4998. The equal-span control moves every tokenizer slightly and moves BPE most (3.0801 to 3.1893), which is the expected direction: shortening BPE's window from 64 tokens to 39 tokens costs it context, and it still beats the character model. WordPiece, whose tokens carry essentially one character, behaves like the character model and slightly worse, which is the expected direction as well.

What it does not support. It does not support the v1 headline, and the v1 headline is not reported here as a current result. The v1 comparison recorded 3.4525 bits per character for the character model against 3.0946 for BPE — about 10% lower for BPE — and it carried four self-declared confounds: a context confound (BPE saw more text per window), BPE merges fitted on the validation text, validation loss computed from 20 unseeded random batches, and a single seed with a single merge count. Act 3 supersedes that comparison structurally: it has the context control that removes confound (i) and a seeded full sweep that removes confound (iii). But Act 3 is *itself* run at a single seed, so it replaces a confounded single-seed result with a controlled single-seed result. The ordering of the two tokenizer families in Act 3 is the same as in v1 (BPE ahead of character), and the Unigram result is new. No claim about tokenizer families is established at $n = 1$.

7 Act 4: scaling laws

7.1 Method: one run, dozens of points

The sweep is a ladder over model size, trained on the C4 shard with the 6,487-token BPE described in Section 2.2. The method that made the act affordable is worth stating plainly, because it is the most transferable thing in the paper: **a training run is not one data point**. Every run logs its validation loss against tokens seen at each evaluation interval, so a single

run yields dozens of (N, D, loss) points along a fixed-N curve. The ladder is 23 runs over eight rungs; restricting to the interior region $D \geq 5$M leaves 5,176 (N, D, loss) points, of which 3,463 belong to the canonical family the fit uses — against the twenty-three endpoint values that a naive "one point per model" reading would give. The parameter axis those points are recorded on was itself a defect, and Section 7.5 settles it; the homogeneity of the model family those points belong to was a second defect, and Section 7.3 explains it.

7.2 The ladder was two architecture families, not one

Reading the run records rather than the ledger, the ladder is not one model family. Fourteen of the twenty-five run records are the canonical `MiniGPT` — 8 heads, learned absolute positional encoding — confirmed from the surviving checkpoints' state dicts and from the `architecture` block that the training code writes into every run's `metrics.json`. Five are a different family, `alibi_like`, with *no* learned position table. The two are trivially separable by the first logged validation loss, because the other family has no position table to learn: the canonical runs start at 8.72–8.91 nats, the other family at 129.57–223.77 nats. The remaining six records are excluded from every fit: three runs of a third architecture, which no $(\text{dim}, \text{layers}, \text{heads})$ combination reproduces, at the 0.8M rung; two "tiny" runs that are not rungs at all; and one 1.5M run superseded by the rescue of Section 7.12. The separation is not delicate, and it agrees with the architecture block wherever that block exists.

The two families do not sit on top of each other on the N axis either. The 1.5M rung contributes two different non-embedding parameter counts, 1,567,200 for the canonical run (which counts the $128 \times 160 = 20{,}480$-parameter position table) and 1,546,720 for the two other-family runs (which do not); the difference between them is exactly one position table.

Rung	Family	Non-emb. params	Total params	Runs	Final validation loss
0.3M	canonical	308,224	723,392	1	4.1695
0.4M	canonical	399,600	918,560	1	4.0563
0.8M	unidentifiable	694,144	1,524,480	3	3.4353–3.4408
0.8M	canonical	809,728	1,640,064	1	3.4157
1.5M	canonical	1,567,200	2,605,120	1	3.0468
1.5M	*other*	1,546,720	2,584,640	2	3.5558–3.5780
2.6M	canonical	2,694,144	3,939,648	6	2.8111–2.8382
4.1M	*other*	4,235,616	5,688,704	2	2.9957–3.0055
4.1M	canonical	4,264,288	5,717,376	1	2.6254
6.3M	canonical	6,351,360	8,012,032	1	2.5152
9.0M	*other*	8,992,224	10,860,480	1	2.8492
9.0M	canonical	9,029,088	10,897,344	2	2.4267–2.5436

Table 6: Act 4: the ladder, split by architecture family. Several rungs are *two* rows because they are two different models: the 1.5M rung holds a canonical and two other-family runs; the 0.8M rung holds the three unidentifiable runs and their canonical replacement; the 4.1M rung holds two other-family runs and their canonical replacement; and the 9.0M rung holds one other-family run and two canonical arms at $D/N = 10$ and 20. **What is measured:** the final validation loss of each run, from the run records, with both parameter axes shown. **Interval type:** none at the rung level; the spread shown is the range over that rung's runs, and the 2.6M rung's six runs carry six distinct values. The canonical family is MiniGPT with learned absolute positions; the other family has no learned position table. The **total-parameter** column is the axis the paper's fit uses (Section 7.5). **One provenance caveat:** the non-embedding counts in this table are the ones recorded in the ladder's own run records, and they are larger than the counts the Act 5 pin records for the same three models and the same final losses (1,567,200 against 1,536,000, 2,694,144 against 2,654,208, and 6,351,360 against 6,291,456). The ledger does not reconcile those two parameter counts; the fit's N axis is internally consistent because every one of its points comes from the same source. Source: `results/v2/act4-scaling/clean_refit.json` and `ladder_points.csv`; table source data in `report/tables/act4_ladder_rungs.csv`.

The consequence is that the fit that was previously reported was fitted to a mixture. The tracked fit consumed 2,776 interior points drawn from 12 runs as the ladder then stood — 7 canonical and 5 of the other family — so 27.6% of the ladder's present interior points, and **51.4% of the points in the fitted set itself**, came from a different model class. *Why* the two families coexist in one ladder is a separate question, and Section 7.3 answers it.

The decisive diagnostic is not an argument about which runs ought to count. It is what the tracked fit's *own parameters* do when they are evaluated on each candidate point set.

Candidate point set	n points	R^2	What the number means
As tracked ($D \geq$ 5M, 12 runs)	2,776	+0.2874	the tracked fit *claimed* $R^2 = 0.9950$ on this set
Other family only	1,428	+0.9957	the tracked fit's shape describes the *other* family
Canonical family only	3,463	−0.6661	anti-correlated with the canonical models' own points

Table 7: Act 4: the contamination diagnostic. The withdrawn fit's parameters ($E = 1.605$, $A = 394.4$, $\alpha = 0.445$, $B = 180.0$, $\beta = 0.290$), evaluated on each candidate point set. **What is measured:** R^2 against a constant-loss baseline on the three candidate sets, with the same parameter vector in every row. **Interval type:** none; each row is one deterministic evaluation of one fixed parameter vector on one point set. Source: `results/v2/act4-scaling/clean_refit.json` and `fit_results.json`; table source data in `report/tables/act4_contamination.csv`.

$\alpha = 0.445$ **is withdrawn.** It is reproduced almost exactly by the *other* family's points and is anti-correlated with the canonical models' own points, and the previously reported "consistent with Chinchilla" rested on that contaminated fit. The claim survives only in this form: the tracked fit describes a model family that is not the one the rest of this paper measures.

What the refit invalidated, in plain language. A reader who has not seen the run records will read the refit as a small correction, and it is not one. The original headline — that the leave-one-rung-out R^2 was at least 0.985 on *every* rung — was produced by the *other* family, and that family occupied the **entire top of the ladder**: the 4.1M and 9.0M rungs are entirely its own, and two of the three runs at the 1.5M rung are too. The fit that carried the headline was therefore a good fit of a different model class, sitting on a curve that looked right, at parameter values that looked right, with a residual plot that was flat in both $\log N$ and $\log D$ and therefore showed nothing at all. Three things were invalidated with it: the exponent $\alpha = 0.445$ itself, the "$R^2 \geq 0.985$ on every rung" headline, and the comparison with Chinchilla that rested on the fit. The comparison had to be re-established on the surviving fit, under the correct parameter convention, rather than assumed — which is what the rest of this section does.

A second, weaker, honest caveat belongs here: the tracked fit's exact 2,776-point set cannot be recovered, so the contamination is established by *parameter reproduction* rather than by re-running the original fit. The 2,776-point set is reconstructed from the 12 runs that existed when the fit was run, and on that reconstructed set the tracked parameters score $R^2 = 0.287$, not the $R^2 = 0.9950$ the tracked JSON records. The reproduction of the other family's points ($R^2 = 0.9957$) is the stronger evidence, and it does not depend on reconstructing the tracked set.

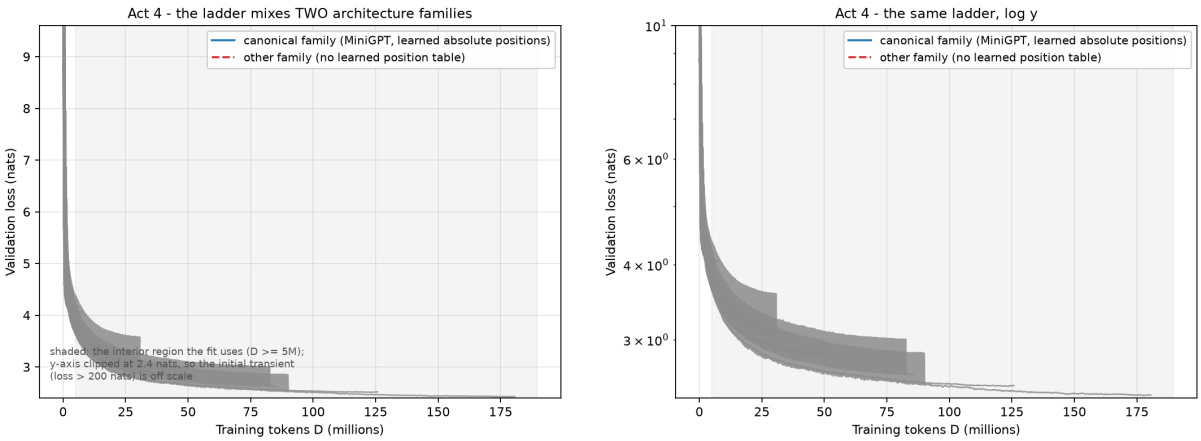


Figure 5: Act 4. The ladder, with the two architecture families drawn separately: the canonical family (solid blue, MiniGPT with learned absolute positions) and the other family (dashed red, no learned position table), one line per run. *Left:* validation loss against training tokens D on a linear y axis. *Right:* the same points on a logarithmic y axis. The shaded region is the interior region the fit uses ($D \geq 5$M). The y-axis is clipped at 2.4 nats, so the initial transient (loss above 200 nats) is off scale. **What is plotted:** 23 runs, 5,726 logged points in total, of which 5,176 are interior points ($D \geq 5$M): 3,463 canonical, 1,428 from the other family, and 285 from runs excluded from the fit. **Interval type:** none at the level of the curve; the seed spread within a rung is in Table 6. The title bar carries the paper's fit (the canonical family on the total-parameter axis) and the profile's compatible range. Source: `results/v2/act4-scaling/ladder_points.csv` and `clean_refit.json`.

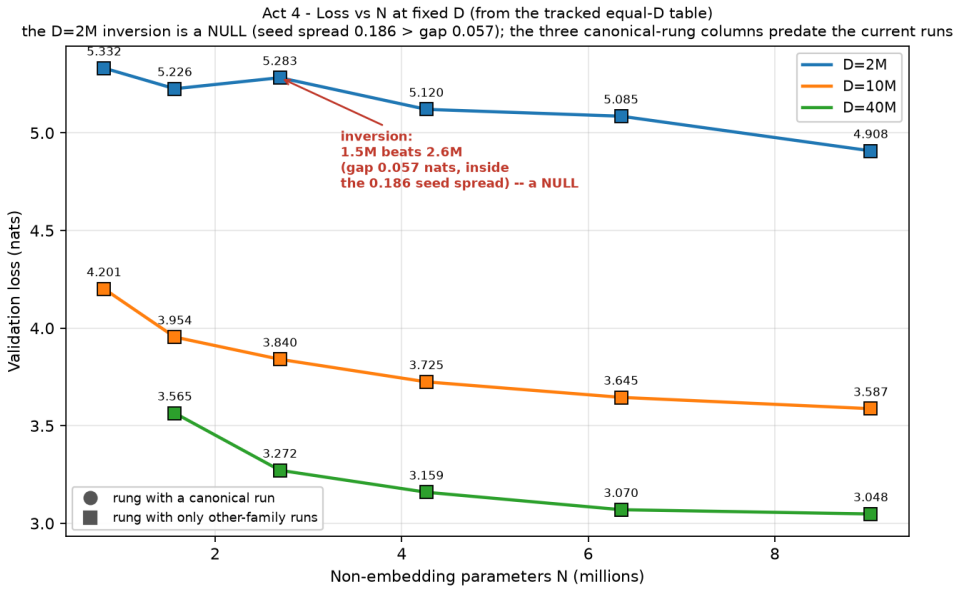


Figure 6: Act 4. Validation loss against non-embedding parameters N at three fixed training-token budgets D, read from the tracked equal-D table. Circles mark rungs that have a canonical run, squares rungs that have only other-family runs. **What is plotted:** one point per rung per budget, 6 rungs at $D = 2$M and $D = 10$M and 5 rungs at $D = 40$M. **Interval type:** none — one value per rung per budget. The visible non-monotonicity at $D = 2$M is a **null**: the 1.5M rung scores 0.057 nats better than the 2.6M rung, which is inside the 2.6M rung's 0.186-nat seed spread. **Caveat that travels with this figure:** the three rungs whose current runs are canonical carry values in the tracked table that do not match the current ladder's logged points at the same D; see Section 7.13. Source: `results/v2/act4-scaling/equal_D_table.csv`.

7.3 Why the architecture changed mid-campaign, and why it was invisible

The cause of the change is now known, and it is worth stating separately, because it is a *process* failure and not a modelling one. Git archaeology settles it.

- `src/scaling/` was **empty before commit `f1eaf77`**. `src/scaling/ladder.py` is *new* in that commit, and it builds `ALiBiTransformer` with 4 heads — the *second* family's architecture.
- The `ASPECT_LADDER` shape table that commit introduces is `dim` $= 32 \times$ `layers`, that is 128/4, 160/5, 192/6, 224/7, 256/8 and 288/9, which matches the shapes of the **canonical MiniGPT** runs exactly.
- So the committed ladder script encodes the *second* family's architecture with the *first* family's shapes, and the script that trained the canonical MiniGPT rungs **was never committed at all**.

The two families coexist in one ladder because **nothing recorded which script produced which points**. The runs that carry an `architecture` block are the ones re-run after `f1eaf77` (or after `0932d19`, which added the block); the runs without one are the older ones. The ladder's shape table, the fit, the ledger and the figures all treated the pooled points as one model family.

This is the taxonomy's largest entry (Section 13.2), and it is worth naming what it is *not*: it is not a hypothesis about position encodings that did not pan out, and it is not a bug fix. It is a training pipeline that was never in the repository, so a mid-campaign change of model class was invisible to every downstream artefact — including the ledger that was supposed to be the record of exactly this.

The generalisable lesson is the one this paper's abstract states: **a good-looking fit is not evidence of a good fit**. A fit is a statement about a *set of models*, and a set can be heterogeneous without anything in the fit complaining: the withdrawn fit's residuals are still flat in $\log N$ and $\log D$. You need the points to be one model family, and the only way to know that is to *tag every point with its configuration and check*.

7.4 The N audit: three N conventions in one table, none of them flagged

Every rung's N was audited against three independent sources: the column in the ladder's own CSV, the run's own recorded metric, and the architecture the run implies.

Rung	N in the ladder CSV	N in the run's recorded metric	N implied by the architecture	`LADDER` design target	D/N	All three agree?
0.8M	694,144	694,144	**809,728** (MiniGPT) / **793,344** (other family) — neither	786,432	22.7	**no**
1.5M	1,546,720; 1,567,200	1,546,720; 1,567,200	1,567,200 (canonical) / 1,546,720 (other)	1,536,000	19.6; 19.9	no (two families)
2.6M	2,694,144	2,694,144	2,694,144	2,654,208	19.7	yes
4.1M	4,235,616	4,235,616	4,235,616	4,128,768	19.5	yes
6.3M	6,351,360	6,351,360	6,351,360	6,291,456	19.8	yes
9.0M	8,992,224	8,992,224	8,992,224	8,957,952	10.0	yes

Table 8: Act 4: the per-rung N audit, against three independent sources. **What is measured:** the ladder CSV's own N column, the N each run recorded, the N the run's architecture implies, the `LADDER` design target, and the rung's D/N ratio. **Interval type:** none; the audit is a set of exact integer comparisons. **Source:** the first two columns are re-derived from `results/v2/act4-scaling/ladder_points.csv` and `clean_refit.json`; the architecture-implied and design-target columns are quoted from `docs/implementation.md`, which is where the audit was recorded. Table source data in `report/tables/act4_n_audit.csv`.

The audit was performed on the pre-rescue rungs, so the 4.1M and 9.0M rows show the other family's N; both rungs have since gained a canonical run, and `act4_n_audit.csv` lists both counts for every rung. Every row agrees across all three sources *except* the 0.8M rung, which does not. Its recorded N of 694,144 matches **no** MiniGPT or ALiBi shape at any (`dim`, `layers`, `heads`) in dims 16–256 and depths 1–10, while its embedding parameters ($830{,}336 = 6{,}487 \times 128$) show that the vocabulary and the width are as expected. The 0.8M rung is therefore a **third, unidentifiable architecture**: it has no `architecture` block, it ran on CPU, it sits at $D/N = 22.7$ rather than 20, and it carries the worst hold-out fold inside the fitted regime ($R^2 = 0.810$ on the non-embedding axis, 0.887 on the total-parameter axis, over the 0.8M–9.0M rungs; the 0.3M and 0.4M folds are worse still but lie outside the regime the claim is scoped to). It was also the cheapest rung to replace — about a minute of GPU — and it sat at the bottom of the N range, where a power law has the most leverage, so it *was* replaced: the canonical 0.8M run (Section 7.12) supplies the eight-rung fit, and the three unidentifiable runs are excluded from it.

One rung, four provenance stories: **694,144** (what the run recorded), **809,728** (the MiniGPT shape for dim 128, layers 4), **793,344** (the other family's shape) and **786,432** (the `LADDER` design target, which is a search target and not a measurement of anything). The taxonomy entry is therefore not one bad rung: it is *three N conventions in one table, none of them flagged*.

The D/N ratio, and what "the $D = 20N$ line" means. The audit's D/N column is not decoration: it is the design of the ladder, and it is where the next section's confounding argument comes from. The canonical rungs run at 19.6–20.0 against a nominal 20, so *"the $D = 20N$ line" is an approximation and not a constraint*, and the table gives each rung's own ratio rather than the design target's. The 0.8M rung's unidentifiable runs sit off the line at 22.7 on their recorded N, and the 9.0M rung's other-family run at 10.0 — the one place in the original ladder where N and D were not collinear across rungs, which is why the rescue added a second, novel ratio at the top rung, so that the 9.0M rung now appears at both $D/N = 10$ and $D/N = 20$ (Section 7.12).

7.5 The N-axis convention: a confound that was never stated

There is a third defect in the same act, and it is easier to miss than the family split because nothing about it is locally wrong. The ladder's recorded N is the run's *non-embedding* parameter count. The published exponents — Kaplan's and Chinchilla's — are fitted on *total* parameters. The two are therefore not on the same axis, so α was never directly comparable with a published value, and the confound was unstated: it was implicit in a column name in a CSV.

Quantity	Total-parameter N	Non-embedding N
Fitted irreducible loss E (nats)	+1.964	+1.911
Measured unigram entropy, held-out region (nats)	4.9946	
Fitted E as a fraction of it	0.39×	0.38×
Fitted α (point estimate)	+0.589	+0.407
Fold-wise α, mean of eight folds	+0.587	+0.407
Point estimate and fold mean agree?	yes	yes
R^2 on the fitted set	0.9960	0.9958
Profile: compatible α range	[0.34, 0.82]	[0.20, 0.60]
Contains Chinchilla's 0.34?	**yes**	**yes**
Profile: compatible β range	[0.38, 0.54]	[0.38, 0.54]
Contains Chinchilla's 0.28?	no	no

Table 9: Act 4: the two parameter conventions, on the same 3,463 interior points. **What is measured:** the fitted parameters of the law under each convention, the measured unigram entropy of the corpus, and the profile likelihood over α and over β under each convention. **Interval type:** the profile's compatible range is the set of values whose fit comes within 0.001 in R^2 of the best fit; it is a statement about what the data cannot distinguish, not an error bar. **Source:** `results/v2/act4-scaling/clean_refit.json` and `unigram_entropy.json`; table source data in `report/tables/act4_fit_results.csv`, `report/tables/act4_profile_likelihood.csv` and `report/tables/act4_unigram_entropy.csv`.

The paper uses total parameters, for the one ground that survives the widened ladder: **it is the published convention**, so it is the only axis on which a comparison with a published exponent is like-for-like. Two further grounds were available at four rungs and are not available at eight, and that movement is itself worth reporting. **(i) Stability**: under total N the point estimate (+0.589) and the fold-wise mean (+0.587) agree to 0.002, and under non-embedding N they agree to 0.001 (+0.407 against +0.407), so the four-rung instability of the non-embedding axis — its point estimate sat far below every one of its folds — disappeared when the two bottom rungs were added. **(ii) Plausibility of E**: both fitted values lie below the model's own achieved held-out loss of 2.5152 nats — 1.964 is 0.78× it and 1.911 is 0.76× it — so the physical

bound admits both axes instead of preferring one. The axis is therefore chosen by convention, and both conventions are reported side by side throughout, because the choice no longer changes the qualitative answer: *both* contain Chinchilla's $\alpha = 0.34$ and *both* exclude $\beta = 0.28$.

What changed in this revision. The physical ground was previously argued against the corpus's unigram entropy of 4.9946 nats, which is a ceiling on a conditional entropy rather than a competitor to it, and it was then re-made against the model's own achieved loss. On the widened ladder that check admits both conventions, so it is reported as a *bound that is respected* rather than as a ground for choosing. What the paper does not claim is that the two axes agree on the point estimate — they differ by 0.18 in α — only that widening the ladder removed the reasons for preferring one axis on fit quality.

Two checks confirm that the axis is not inflated by an artefact of the artefacts. The checkpoint file's parameter *sum* double-counts the tied output embedding — 3,643,040 against a recorded 2,605,120 at the 1.5M rung, a difference of exactly $6{,}487 \times 160$ — but the CSV's N column is the non-embedding count and is not that sum, so the axis is not inflated by the double count; and the vocabulary is shared (6,487 at every rung), so any correction for the embeddings would be a constant, not a per-rung error.

7.6 The irreducible loss is an extrapolated asymptote, not a measured entropy

E in the fitted law is meant to be the irreducible loss of the best constant predictor, so it cannot exceed the entropy of the text being predicted. That bound has now been measured rather than argued, and the measurement does not support the argument this paper used to make.

Region or quantity	Tokens	Entropy (nats)
Held-out region (the evaluation region)	1,000,128	4.9946
Training region (5M-token sample)	5,000,000	5.0068
Full corpus	338,513,094	5.0136
Fitted E, total-parameter axis	—	1.964 (0.39×)
Fitted E, non-embedding axis	—	1.911 (0.38×)

Table 10: Act 4: the measured unigram entropy of the corpus under the ladder's own 6,487-token vocabulary, against the two fitted irreducible losses. **What is measured:** the entropy of the unigram distribution over the 6,487-token vocabulary, computed on the held-out region, on a five-million token training-region sample and on the full corpus. **Interval type:** none; these are exact entropies of a counted distribution, not estimates. **Source:** `results/v2/act4-scaling/unigram_entropy.json`; table source data in `report/tables/act4_unigram_entropy.csv`.

The unigram entropy of the corpus under the ladder's own 6,487-token vocabulary is 4.9946 nats on the held-out region, 5.0068 on a five-million-token training-region sample and 5.0136 on the full 338,513,094-token corpus. **Both** fitted E values lie *below* the held-out figure: 1.964 is 0.39× it and 1.911 is 0.38× it, and both also lie below the 2.5152-nat loss a trained model of this ladder actually reaches. The vocabulary is small and highly compressible (2.27 characters per token), so its unigram entropy is about five nats, and Chinchilla's published $E = 1.69$ was never this corpus's entropy to begin with.

Three consequences, and they are the honest reading of the table. First, the "physically possible E" argument in its original form is withdrawn, and the parameter-axis decision rests on the published convention (Section 7.5). Second, E **here is an extrapolated asymptote from an eight-rung ladder, not a measured entropy**, and this paper does not imply that

it is the corpus's floor. Third, the two conventions no longer disagree about E — they agree to within 0.06 nats on the fitted value — and they now agree about α's compatibility too; what they still disagree about is α's *value*, which is the rest of this act.

7.7 The clean fit, on the total-parameter axis

We fit the Chinchilla-style form

$$L(N, D) = E + \frac{A}{N^{\alpha}} + \frac{B}{D^{\beta}} \tag{1}$$

by nonlinear least squares, using a Levenberg–Marquardt solver with an analytic Jacobian carried in log-parameterised coordinates and a multi-start grid, so that a shallow minimum cannot be mistaken for the optimum. On the canonical family alone — 3,463 interior points, 8 rungs, 8 distinct total-N values — the law gives $\alpha = +0.589$, $\beta = +0.461$ and $E = +1.964$ at $R^2 = 0.9960$. The fitted E is an extrapolated asymptote and not a measured entropy: the corpus's measured unigram entropy is 4.9946 nats.

Quantity	Value (total-parameter N)	Uncertainty or diagnostic
α (point estimate of the full fit)	+0.589	*not* the uncertainty statement; see the profile below
β (point estimate)	+0.461	—
E (irreducible loss)	+1.964 nats	an *extrapolated asymptote* from an eight-rung ladder, not a measured entropy: the corpus's unigram entropy is 4.9946 nats
A / B	+2,654.11 / +1,705.21	—
R^2 on the fitted set	0.9960	against a constant-loss baseline
Fitted points	3,463	8 rungs, 8 distinct total-N values
Residual correlation with $\log N$	−0.00081	negligible curvature
Residual correlation with $\log D$	−0.0062	negligible curvature
Jacobian condition number	4.00×10^3	—
Profile: best R^2 / best α	0.995976 / 0.58	75-point grid, α from 0.02 to 1.50
Profile: compatible α range	[0.34, 0.82]	within 0.001 in R^2 of the best fit
Profile: contains Chinchilla's 0.34?	**yes**	at the lower end of the range; the range also contains 0.50 and discriminates between none of the published exponents
Profile: best R^2 / best β	0.995978 / 0.46	75-point grid, β from 0.02 to 1.50
Profile: compatible β range	[0.38, 0.54]	within 0.001 in R^2 of the best fit
Profile: contains Chinchilla's 0.28?	**no**	the range EXCLUDES it
Leave-one-rung-out R^2, 0.3M–1.5M	−435 / −71 / 0.887 / 0.964	the first two lie outside the declared 0.8M–9M regime
Leave-one-rung-out R^2, 2.6M–9.0M	0.981 / 0.995 / 0.992 / 0.988	inside the regime
Fold-wise α, 0.3M–1.5M	+0.628 / +0.509 / +0.592 / +0.580	the same folds, quoted individually
Fold-wise α, 2.6M–9.0M	+0.691 / +0.590 / +0.604 / +0.501	the same folds, second half

Table 11: Act 4: the clean fit, on the canonical family's 3,463 interior points and the total-parameter axis. **What is measured:** the fitted parameters of the law above, the profile likelihood over α and over β, and the leave-one-rung-out diagnostics. **Interval type:** the profile's compatible range is *not* an error bar and *not* a fit covariance; it is the set of values the data cannot distinguish from one another. The fold-wise α values are quoted individually, and no standard deviation over the eight folds is given: over eight folds of a model whose α is not identified, a standard deviation is false precision, and the tracked JSON's covariance errors of ± 0.0032 (for the withdrawn fit) are not credible at a condition number of 1.9×10^6. The fit is *exploratory*: both the family separation and the axis convention were decided after the provenance was read. Source: `results/v2/act4-scaling/clean_refit.json`; table source data in `report/tables/act4_fit_results.csv`.

7.8 What the profile likelihood says, and what it does not

A point estimate of a parameter that the data cannot identify is not a confidence statement, so the paper does not quote one. Instead it profiles α: α is held fixed on a 0.02-wide grid from 0.02 to 1.50, the other four parameters are re-optimised at every grid point, and the R^2 is recorded. The *compatible range* is the set of α values whose fit comes within 0.001 in R^2 of the best fit. On the total-parameter axis it is [0.34, 0.82] and it contains Chinchilla's 0.34 at its lower end; on the non-embedding axis the same procedure gives [0.20, 0.60], which also contains it. Neither range excludes anything the paper compares against.

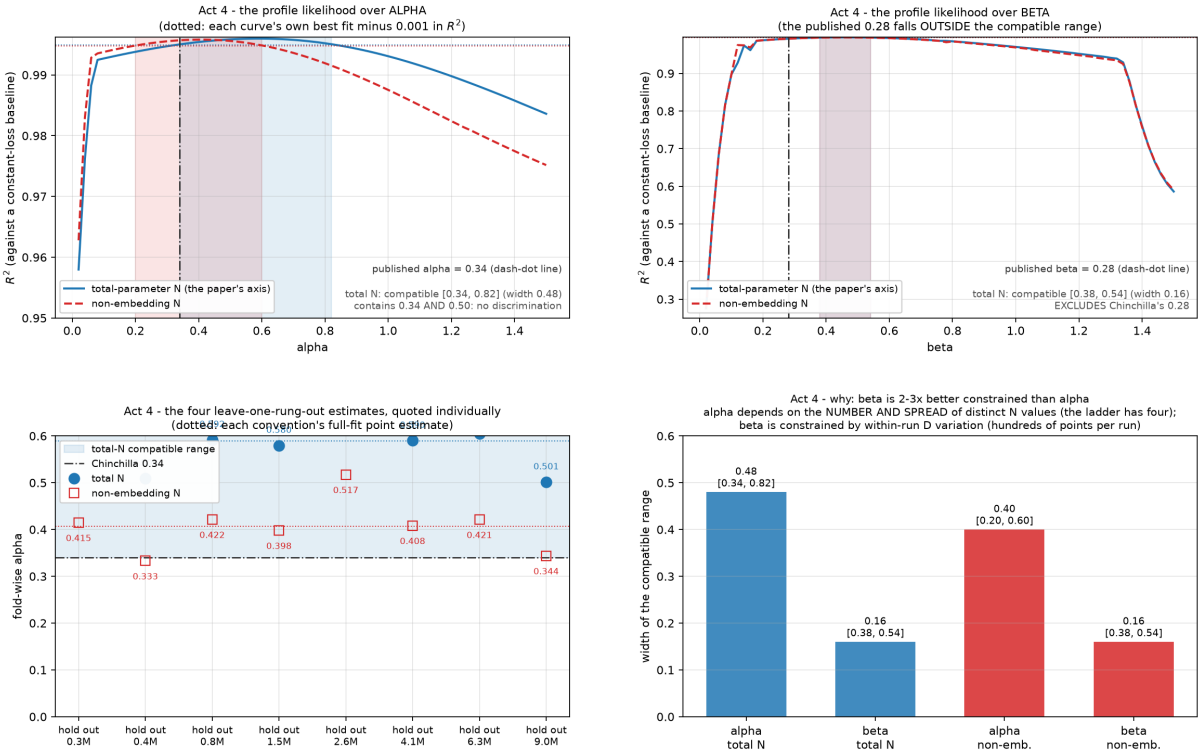


Figure 7: Act 4. The profile likelihood over α and over β, under both parameter conventions, and the diagnosis of the asymmetry between them. *Top left:* R^2 against α on the total-parameter axis (solid blue) and the non-embedding axis (dashed red); the dotted lines are each curve's own best fit minus 0.001 in R^2, and the shaded bands are the compatible ranges; the dash-dot line is Chinchilla's 0.34. *Top right:* the same panels for β, with the dash-dot line at Chinchilla's 0.28, which falls *outside* the compatible range. *Bottom left:* the eight leave-one-rung-out α estimates, quoted individually, with the total-N compatible range as a band and each convention's full-fit point estimate as a dotted line. *Bottom right:* the width of the compatible range for each parameter and each axis. **What is plotted:** 3,463 interior points ($D \geq 5$M) of the canonical family, on a 75-point grid over α and over β from 0.02 to 1.50, and eight leave-one-rung-out folds per convention. **Interval type:** the bands are the profile's compatible ranges — a statement about what the data cannot distinguish, *not* error bars; the fold-wise points are single deterministic re-fits with no interval attached. Source: `results/v2/act4-scaling/clean_refit.json` and `unigram_entropy.json`.

What this licenses, and what it does not. The honest sentence is still *this ladder is compatible with both published exponents and therefore does not discriminate between them.* The compatible α range on the total-parameter axis is [0.34, 0.82]: it contains Chinchilla's 0.34, at its lower end, and it also contains 0.50. A range that wide does not support the weaker wording "consistent with", because "consistent" suggests that something outside the range would have

been inconsistent, and here both published values sit inside it. What the range supports is the stronger and more defensible statement: the ladder excludes neither of the two published exponents and adjudicates between neither of them.

It is emphatically **not** a point claim. The paper does not say that our α is 0.34, and it does not say that our α agrees with 0.34. The point estimate is +0.589 and the fold-wise estimates run from +0.501 to +0.691; the compatible range spans a factor of 2.4 in α, which is exactly why it cannot be used to adjudicate between published values.

The eight fold-wise estimates are reported individually, because quoting their mean and standard deviation would manufacture a precision the model does not have. Under the total-parameter axis the folds give +0.628, +0.509, +0.592, +0.580, +0.691, +0.590, +0.604 and +0.501, and the full fit's +0.589 sits inside that range. Under the non-embedding axis the folds give +0.415, +0.333, +0.422, +0.398, +0.517, +0.408, +0.421 and +0.344, and the full fit's +0.407 sits inside them too. On the four-rung ladder the non-embedding axis's full fit fell far below every one of its folds; adding the two bottom rungs removed that signal, which is why the axis is now chosen on convention alone (Section 7.5).

Parameter	N convention	Best R^2	Best value	Compatible range	Contains published value
α	Total parameters	0.995976	0.58	[0.34, 0.82]	0.34: **yes**
α	Non-embedding	0.995811	0.40	[0.20, 0.60]	0.34: **yes**
β	Total parameters	0.995978	0.46	[0.38, 0.54]	0.28: **no**
β	Non-embedding	0.995811	0.46	[0.38, 0.54]	0.28: no

Table 12: Act 4: the profile likelihood over α and over β on the canonical family's 3,463 interior points, under both parameter conventions. **What is measured:** the compatible range, defined as the set of values whose fit comes within 0.001 in R^2 of the best fit, on a 0.02-wide grid from 0.02 to 1.50. **Interval type:** this range is *not* an error bar and *not* a confidence interval; it is a statement about what the data cannot distinguish, and the grid is 0.02 wide, so the endpoints are reported to that resolution. **The asymmetry is the point:** the α range is compatible with the published 0.34 on *both* axes, and the β range excludes the published 0.28 on *both* axes. Source: `results/v2/act4-scaling/clean_refit.json`; table source data in `report/tables/act4_profile_likelihood.csv`.

7.9 The beta profile, and the asymmetry that is the diagnosis

The same procedure, with β held fixed instead of α, gives a sharper answer, and the difference between the two profiles is the most useful thing this act has to say. On the total-parameter axis the compatible β range is [0.38, 0.54], a width of 0.16 against the α range's 0.48; on the non-embedding axis it is also [0.38, 0.54], a width of 0.16 against 0.40. **β is three times better constrained than α on the total-N axis and two and a half times on the non-embedding axis, on the same 3,463 points.**

The mechanism, which tells the reader what to do next. Each run contributes interior checkpoints at **fixed N and rising D**, so the within-run D variation constrains β tightly — a single run supplies hundreds of points at one N and a hundred different D values. α, by contrast, depends on the **number and spread of distinct N values** the ladder contains, and the ladder contains eight. So α has little to work with and β has hundreds of points per run. The consequence for the experimental design is direct: to identify α, widen the range of distinct N values; to identify β, add D. This is the reason the rescue in Section 7.12 added a 0.3M and a 0.4M rung rather than more tokens to the rungs it already had.

And the asymmetry fixes what the paper may claim about β. The total-N β range [0.38, 0.54] **excludes** Chinchilla's 0.28, and so does the non-embedding range [0.38, 0.54]. *No compatibility claim may be made for β at all*, on either axis. A beta profile that shows the published point outside the range is informative rather than embarrassing, and it is shown: Figure 7, top right.

7.10 The spread across fit variants is the result

The honest headline of this act is not a number. It is that a ladder of this size does not identify α at all, and the spread across defensible fit variants is the measurement that says so.

Fit	Points	Rungs	α (point)	Fold-wise α	β	R^2
Canonical only, total N *(the paper's fit)*	3,463	8	0.589	0.501–0.691	0.461	0.9960
Canonical only, non-embedding N	3,463	8	0.407	0.333–0.517	0.459	0.9958
Other family only, non-embedding N	1,428	3	0.511	0.346 / 0.350 / 0.540	0.421	0.9976
Mixed, as tracked	2,776	6	0.057	0.098–1.778	0.223	0.7087
Withdrawn tracked fit	2,776	6	0.445	—	0.290	0.9950 claimed

Table 13: Act 4: α across every fit variant we can defend, under both parameter conventions. **What is measured:** the fitted exponents on each point set. **Interval type:** the fold column gives the range over the leave-one-rung-out estimates, and no standard deviation is given over them; the individual values are in Table 11, and the mixed fit's own folds range from 0.098 to 1.778. The last row is the fit this paper previously reported, and its R^2 is the value it *claimed*; evaluated on the reconstructed point set it scores 0.2874. Source: `results/v2/act4-scaling/clean_refit.json` and `fit_results.json`; table source data in `report/tables/act4_fit_results.csv`.

Two things follow. First, **the mixture is visible directly as a collapse in fit quality**: one law on the mixed set reaches only $R^2 = 0.7087$, while either pure family fits at $R^2 \geq 0.995$. No statistical test is needed to see a difference that large, and no amount of reweighting would rescue it. Second, **the spread across variants is the measurement that says α is not identified here**: it runs from 0.057 to 0.589 across the defensible fits, and on the paper's own axis the profile's compatible range spans [0.34, 0.82]. A ladder of twenty-three runs over eight rungs cannot pin down an exponent four orders of magnitude below the published regime, and the statement in Section 7.8 is a statement about that width rather than a point result.

The methodological lesson generalises past this act, and it is the same lesson that Act 5c teaches from the other direction. A fit is a statement about a *set of models*, and a set can be heterogeneous without anything in the fit complaining. The ladder's homogeneity was an assumption, never a measurement, and it was wrong in a way a residual plot could not have shown: the residuals of the mixed fit are still flat in $\log N$ and $\log D$ (-0.0047 and -0.0047), because two different learning curves can be fitted by one curve without leaving a visible signature in the residual correlation. What exposed it was a provenance check — reading the run records — not a better statistic, and the cause of the change is now known too (Section 7.3). The generalisable form of the lesson is the one the taxonomy in Section 13.2 records: **a good-looking fit is not evidence of a good fit**. You need the points to be one model family, and the only way to know that is to tag every point with its configuration and check.

7.11 The other family's 129.6–223.8 nats are an initialisation property, not a metric bug

The other family's first logged validation losses range from 129.57 to 223.77 nats — the five values are 129.57, 134.70, 181.65, 183.96 and 223.77 — which is fifteen to twenty-five times the uniform-over-vocabulary loss of $\ln 6{,}487 = 8.78$ nats. A number that absurd on its face deserves the same treatment this project gave defect D18, which returned 11.98 nats for a model whose real loss is about 2.5. Three candidate causes, and the measurement.

Is the loss summed rather than averaged? If it were summed over positions and batch it would be about $8.78 \times 32 \times 128 = 36{,}000$ nats. The observed 129.6 is neither 8.78 nor 36,000, so the metric is an average.

Is the metric misaligned, a D18-style target shift? The same code path computes the canonical family's step-1 loss as 8.80, against a uniform-vocabulary cross-entropy of 8.78; a shift would corrupt both families equally. The metric is correct.

Therefore it is the model. Building that model class from the repository's own code, with the repository's own seed, on the same data, reproduces the effect to within 2%: 133.25 against 129.57 at the 1.5M rung, 184.49 against 183.96 at 4.1M and 224.99 against 223.77 at 9.0M, while the canonical family reproduces at 8.82 against 8.80.

This makes the contamination more serious, not less. Those five runs were not broken and not divergent: they were good models that trained normally and finished at 3.578 nats at the 1.5M rung against the canonical family's 3.057. They were invisible because their loss trajectory started fifteen to twenty-five times above uniform and ended half a nat worse — a plausible-looking curve, at a plausible-looking N, with a plausible-looking R^2, and nothing in the artefacts distinguished it from a canonical run until the provenance was read by hand. The defect was not a broken run that a reader could have spotted; it was a working run that a fit could happily describe.

7.12 The rescue: six arms, eight rungs, and a null that is now credible

The 4.1M and 9.0M rungs were entirely of the other family, so the clean fit had only four rungs and four distinct N values, which was the main reason α was weakly identified. The rescue is now **complete**: `src/scaling/rescue.py` re-ran six arms with the canonical architecture, with every logged point tagged by a config hash computed from the live model at the moment it is logged rather than retroactively from a rung table, because retroactive tagging is exactly what let the mid-campaign architecture change hide in plain sight.

The plan was set by the asymmetry in Section 7.9. α's identifiability rests on the **number and spread of distinct N values**, not on D: the interior checkpoints already give each run about 230 points at *fixed N and rising D*, and the $D = 20N$ line makes N and D collinear across rungs. Adding a point to a line that already holds four N values buys little; widening the N range buys a lot.

Arm	Cost	What it bought	Measured outcome
0.3M @ $D = 20N$	∼3 min	+1 distinct N at the very bottom: extends $\log N$ from $\ln(9.0/0.8) = 2.42$ to $\ln(9.0/0.3) = 3.40$, a **41%** increase in the N range for minutes of GPU	4.1695 nats
0.4M @ $D = 20N$	∼4 min	a further small N, the same leverage argument	4.0563 nats
0.8M @ $D = 20N$	∼1 min	replaces the unidentifiable third architecture at the *bottom* of the N range, where a power law has the most leverage	3.4157 nats; the fit's worst in-regime fold is now 0.887
4.1M @ $D = 20N$	92 min	+1 distinct N, the whole 4.1M rung replaced with a canonical one, and a comparison against the discarded 4.1M family that is **matched in N with D within 3.3%** (same N, same data, a different position table)	2.6254 nats against 2.9957–3.0055 for the other family: **0.37 nats lower at the same N**
9.0M @ $D = 10N$	∼120 min	half the cost of $20N$, a **novel D/N ratio** so the design is no longer a single line, and a **direct matched-in-N comparison with D within 0.4%** against the discarded 9.0M family, which sat at exactly 9.0M / $D = 10N$	2.5436 nats against 2.8492: **0.31 nats lower at the same N**
9.0M @ $D = 20N$	∼240 min	a second D ratio at the top rung, constraining the within-run shape as well as the level	2.4267 nats

Table 14: Act 4: the rescue, by arm. **What is measured:** the final validation loss of each rescued run, and the matched-in-N comparisons the 4.1M and 9.0M arms were designed to make. **Interval type:** none — every rescued rung is a single seed, so each outcome is a point value and not an error bar, which is the pre-registered acceptance check that a single-seed rung may not be presented with one. The costs are quoted from the project ledger, which is where the spending decision was recorded. Source: `results/v2/act4-scaling/ladder_runs.json` for the outcomes and `docs/implementation.md` for the costs.

The pre-registered gate was stated in advance, and it is answered. The question was no longer whether α varies widely across fit variants; it was **does the α profile narrow materially from** $[0.08,\ 0.54]$**?** It did not. After adding four distinct N values at the bottom and a second D/N ratio at the top, the total-parameter range is $[0.34,\ 0.82]$ — a width of 0.48 against the four-rung width of 0.46, and shifted upward rather than narrowed. The null is therefore **credible**: the right experiment was tried, the N range widened by 41%, the D/N collinearity was broken, and α is still unidentified. "We had four rungs" is an excuse; "we added rungs and α is still unidentified" is a result. The withdrawn $\alpha = 0.44$ does not come back either: the fit lands at 0.589, and the other family's own points still give 0.511.

The recipe was validated before any rescue spend. Re-training the 1.5M rung with the recovered recipe gave a final validation loss of 3.0468 against the then-canonical 3.0571 (a difference of 0.0103 nats), and the step-2,000 trajectory matched: 3.4821 derived against 3.4838 recorded. Without that check none of the rescued rungs would be comparable to the existing

ones, because the artefacts record no learning rate and the rescue had to use the recipe in the current training code. The validated run is the one now in the ladder; the older record is marked superseded.

What the rescue did and did not change. It changed the point set from four rungs and 2,292 points to eight rungs and 3,463 points, moved the fitted α from +0.209 to +0.589, moved E from +1.348 to +1.964 nats, and left the qualitative verdict untouched: the range is still too wide to discriminate between published α values, and β still excludes 0.28. It also turned the contamination into a controlled comparison: at matched N the canonical architecture trains 0.31 to 0.37 nats below the other family, which is the size of the effect the withdrawn fit had been absorbing.

7.13 The equal-D table, and an inversion that is a null

D (tokens)	0.8M	1.5M	2.6M	4.1M	6.3M	9.0M
2,000,000	5.3317	5.2261	5.2828	5.1205	5.0854	4.9080
5,000,000	4.4917	4.3701	4.2404	4.2158	4.1739	4.0527
10,000,000	4.2008	3.9541	3.8402	3.7247	3.6453	3.5873
20,000,000	4.1059	3.6495	3.5215	3.4004	3.3302	3.2975
40,000,000	—	3.5646	3.2717	3.1593	3.0699	3.0480
80,000,000	—	—	3.0023	2.8700	2.8560	—

Table 15: Act 4: validation loss (nats) at fixed training-token budgets, by rung, as recorded in the tracked equal-D table. Source: `results/v2/act4-scaling/equal_D_table.csv`; copy in `report/tables/act4_equal_D_table.csv`. A dash marks a rung that did not reach that budget. **What is measured:** one loss per rung per budget. **Interval type:** none. **Provenance caveat that travels with this table:** the values for the three rungs whose current runs are canonical (0.8M, 2.6M, 6.3M) do not match the current ladder's logged points at the same D, while the three rungs whose current runs belong to the other family match them to within 0.013 nats; see the paragraph below.

The provenance caveat, in numbers. At D = 10M the table records 4.2008, 3.8402 and 3.6453 for the 0.8M, 2.6M and 6.3M rungs, against 3.5173, 3.3052 and 3.2149 in the ladder's logged points at D = 10,035,200 tokens — a gap of 0.43–0.68 nats. For the other three rungs the same comparison gives 3.9541 against 3.9552 (1.5M), 3.7247 against 3.7121 (4.1M) and 3.5873 against 3.5873 (9.0M). The table therefore describes an earlier state of the ladder for the three canonical rungs, and it is reported here because it is a tracked artefact, not because it is a measurement on the canonical runs of this paper.

At D = 2M the 1.5M rung (5.2261) scores *better* than the 2.6M rung (5.2828): a 0.057-nat inversion, against a per-seed spread of 0.186 nats at the 2.6M rung. The inversion is therefore 3.3× smaller than the noise, and it is driven by a single high-loss seed. We report it because it is in the table and it would have been easy to omit; we do not report it as a finding, because it is not one, and with the provenance caveat above it is not even a measurement on the models this paper describes. This is the worked example of checking the seed spread before declaring a result, and it is why the equal-D table is in the paper at all.

7.14 The pre-registered falsification condition

The condition was written before any run of the sweep (2026-10-15) and is reproduced here in the form it was recorded. The claim under test was: *"Loss scales predictably with N (non-embedding parameters) and D (training tokens) in the ∼0.8M–9M regime ($L(N, D) = E +$*

$A/N^{\alpha} + B/D^{\beta}$)." It would be **falsified** if either (i) the fitted model failed to beat a constant-loss baseline on held-out points ($R^2 \leq 0$ against the rung-mean baseline), or (ii) the residuals showed systematic curvature in $\log N$ or $\log D$ indicating a missing lower- or higher-order term rather than uncorrelated noise. Three additional acceptance checks were recorded at the same time: every rung must produce at least 50 distinct D values, the embedding share must be reported per rung, and a single-seed rung may not be presented with an error bar. Anything decided after seeing the fit is labelled exploratory in this report.

The condition was NOT FALSIFIED for the clean fit, inside the regime it was scoped to. Every held-out rung between 0.8M and 9.0M is predicted with $R^2 > 0$ (0.887, 0.964, 0.981, 0.995, 0.992 and 0.988 on the total-parameter axis), and the residual correlations with $\log N$ and $\log D$ are -0.00081 and -0.0062 — negligible curvature. The two rungs the rescue added below 0.8M are a separate matter: holding out 0.3M or 0.4M gives $R^2 = -435$ and -71, because the fit is being extrapolated below the range it was fitted on. That is a statement about *where the law applies* rather than a falsification of the claim, and it is reported because the ledger's own condition makes it visible: the power law holds across the 0.8M–9M decade this act is about, and stops holding one bin below it.

The same protocol applied to the *mixed* set would have falsified it: the mixed fit's leave-one-rung-out R^2 is negative on five of its six rungs (-61.20, -0.34, -0.49, -0.03, -1.55 and $+0.42$). That is the condition doing its job in hindsight, and it is the reason the mixed fit's $R^2 = 0.7087$ is reported as a collapse rather than as a result. The ledger's earlier bracket $\alpha \in [0.445, 0.552]$, $\beta \in [0.209, 0.290]$ belonged to the withdrawn fit and is not carried forward; what this paper quotes instead is the profile likelihood's compatible range, with the fold-wise estimates reported individually beside it and the other parameter convention reported alongside rather than instead of it.

7.15 Caveats on the α comparison, and the 1.5M rung

The tokenizer caveat. The ladder's BPE has a 6,487-token vocabulary at 2.2726 characters per token, and its base alphabet contains 2,486 characters, most of them rare Unicode. A byte-fallback scheme would compress better *and* preserve a 0% unknown-token rate, which the current scheme does not: it is a legitimate improvement we did not make. The exponents are internally consistent with the points they were fitted to, but they are weakly identified at this scale (see below), and any comparison of our α with a published value is therefore **caveated**; we do not present it as a like-for-like comparison.

The entropy caveat. $E = +1.964$ is an *extrapolated asymptote* from an eight-rung ladder and not a measured irreducible loss. The corpus's measured unigram entropy under this vocabulary is 4.9946 nats on the held-out region, and the fitted E is 0.39× that, so the parameter must not be read as the corpus's floor (Section 7.6).

The schedule caveat. Interior ladder points are mid-schedule: the learning rate has not been annealed at the point where the run stopped, so those points carry an unquantified bias. This is the reason the annealed-endpoint sensitivity analysis exists, and it applies to the clean fit exactly as it applied to the withdrawn one.

The 1.5M rung is not one configuration, and now we know why. Its three recorded runs carry two different non-embedding parameter counts (1,567,200 and 1,546,720) and final validation losses spanning 3.0468–3.5780, a spread of 0.53 nats. The reason is not seed noise: two of the three runs are a different architecture, one with a learned position table and two without, and the 20,480-parameter difference between the two counts is exactly one position table. We therefore present the "seed spread" and "noise floor" statements of this act as **exploratory**: the fit is reported as measured, and the rung heterogeneity is a known heterogeneity in the input data, resolved for the fit by dropping the other family rather than by pooling it.

The 0.8M rung carried a third architecture, and that was the worst fold. Its recorded N of 694,144 matches no known shape, it has no architecture block, it ran on CPU,

and it sat at $D/N = 22.7$. It carried the act's worst hold-out fold on the four-rung ladder ($R^2 = 0.508$ non-embedding, 0.779 total), which is why it was the first rung the rescue replaced: the canonical 0.8M run now in the fit gives 0.810 and 0.887, and the three unidentifiable runs are excluded from every number this act reports (Section 7.4).

Identification. α is weakly identified at this scale, and the evidence for that statement is in the act itself: eight distinct N values, five rungs with a single seed, a profile's compatible range that spans [0.34, 0.82], and a spread across defensible fit variants that runs from 0.057 to 0.589. The uncertainty to quote is the profile's compatible range and the eight fold-wise estimates, not the point estimate and not the tracked JSON's covariance errors, which are not credible at a Jacobian condition number of 1.9×10^6. β is better identified (a compatible range of width 0.16 on the total-N axis) but its range *excludes* Chinchilla's 0.28, so no compatibility claim may be made for it. A third, independent check agrees and is worth one sentence: the ladder's local exponent — computed from adjacent rungs without the five-parameter fit at all — is made flat by $E = 0.20$, where it averages 0.157 with a spread of 0.053, so the check does not reproduce either published exponent either.

7.16 Conclusion

What the measurement supports. The ladder was two architecture families, and that is now a measurement rather than a story: 14 canonical runs and 5 of the other family, separable by a factor of fifteen in first logged loss and by the presence of a learned position table. The cause of the change is also known now, and it is a process failure — a training pipeline that was never in the repository — rather than a modelling one (Section 7.3). The ladder was also recorded on a parameter axis the published exponents do not use, and the paper now states which axis it uses and why; on the widened ladder the two conventions agree on everything except the value of α, so the choice is made on the published convention (Section 7.5). On the canonical family alone, on the total-parameter axis, a Chinchilla-form law fitted to 3,463 interior points over 8 rungs and 8 distinct N values gives $\alpha = +0.589$, $\beta = +0.461$ and $E = +1.964$ at $R^2 = 0.9960$, with a leave-one-rung-out R^2 of 0.887–0.995 across the 0.8M–9.0M regime. The pre-registered falsification condition was not falsified for that fit, and the residuals show no curvature. The statement the act supports about the published exponents is a **compatibility-and- indiscriminability statement**: the α profile's compatible range is [0.34, 0.82], it contains 0.34 and also 0.50, and the ladder is therefore compatible with both published exponents and discriminates between neither. The rescue is finished, and it made the null credible rather than overturning it: four more distinct N values and a second D/N ratio moved the range's width from 0.46 to 0.48 without narrowing it. The contamination is visible directly as a collapse in fit quality: one law on the mixed set reaches $R^2 = 0.7087$ while either pure family fits at $R^2 \geq 0.995$.

What it does not support. It does not support the withdrawn claim: $\alpha = 0.445$ describes the other family, not these models, and the "$R^2 \geq 0.985$ on every rung" headline was produced by that family too. It does not support a point claim about Chinchilla: the paper does not say that our α is 0.34, nor that our α agrees with 0.34, because the compatible range is too wide to say either thing. It does not support *any* compatibility claim for β, whose compatible range [0.38, 0.54] excludes the published 0.28 on both axes. It does not support a claim that this ladder identifies α at all: across the defensible fits α runs from 0.057 to 0.589, and the honest result of the act is that a ladder of twenty-three runs over eight rungs cannot pin down an exponent four orders of magnitude below the published regime. It does not support the claim that the two parameter conventions give the same point estimate, because they do not (0.589 against 0.407). It does not support reading E as the corpus's irreducible loss: the measured unigram entropy is 4.9946 nats and the fitted E lies below it under both conventions. And it does not support extrapolation: the ladder spans 0.3M–9.0M non-embedding parameters and 6.2M–180M tokens, and nothing here licenses a claim about models larger than the ones we trained.

8 Act 5: inference efficiency

8.1 5a: the KV cache

Context	Uncached (ms)	Cached (ms)	Speedup
128	51.7	14.0	3.71×
256	85.1	19.1	4.46×
512	198.2	22.7	8.72×
1024	630.7	34.3	18.36×

Table 16: Act 5a: generation latency with and without a KV cache, 32 new tokens, 4 layers, $d = 128$, 4 heads, best of three runs. **What is measured:** wall-clock latency, $n = 3$ timing repetitions per cell. **Interval type:** none; these are best-of-three wall-clock timings, not a mean with a confidence interval. Source: `results/v2/act5/kv_cache.json`; table source data in `report/tables/act5_kv_cache.csv`.

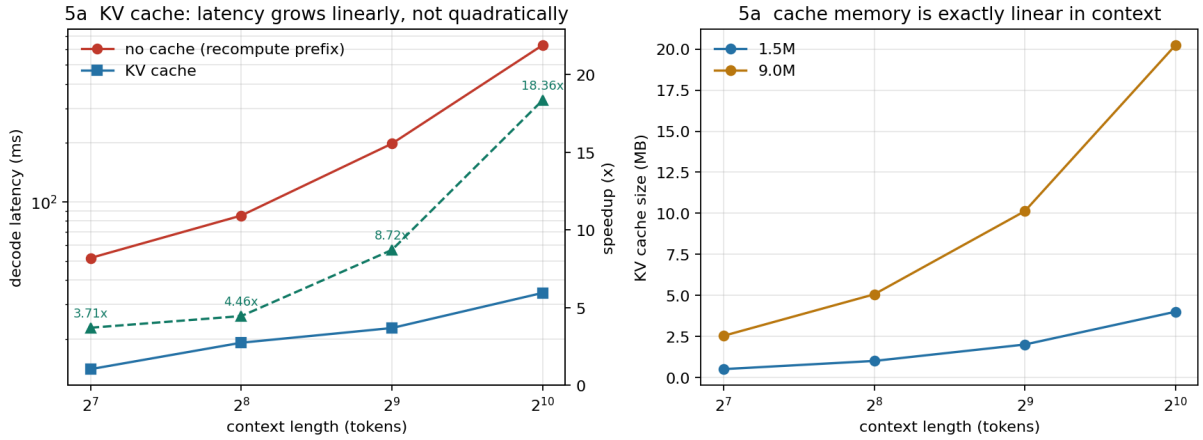


Figure 8: Act 5a. *Left:* generation latency with and without a KV cache against context length. *Right:* cache memory against context length for two model shapes. **What is plotted:** 4 context lengths for latency and 2 models × 4 context lengths for memory. **Interval type:** none (±1 sample standard deviation over 3 timing repetitions, shown as error bars). **Caveat that travels with this figure:** the 18.36× figure is a microbenchmark of the mechanism on a *purpose-built context-1024 model*; it is not a property of the canonical context-128 models used elsewhere in this paper, and it is not quoted as one. Source: `results/v2/act5/kv_cache.json`.

The cached and uncached paths produce byte-identical output: the maximum logit difference is 1.64×10^{-7} nats, which is 1.4 ULP in float32 and is attributable to BLAS accumulation order, not to a defect. Cache memory is exactly linear in context, doubling with every doubling of the window (0.50/1.00/2.00/4.00 MB for the 1.5M model and 2.53/5.06/10.13/20.25 MB for the 9.0M model at $T = 128/256/512/1024$). Caching only repays its memory once the sequence is longer than the embedding dimension, which is about 288 tokens for the 9.0M model.

8.2 5b: sharing KV heads

Seed	MQA (G=1)	MHA (G=8)	Difference	Provenance
42	2.8558	2.8111	+0.0447	resumed in 0.73 s †
43	2.8474	2.8331	+0.0143	full run
44	2.8661	2.8382	+0.0279	full run
45	2.8464	2.8205	+0.0259	full run
46	2.8429	2.8129	+0.0300	full run
47	2.8474	2.8190	+0.0284	full run
Primary analysis ($n = 5$, seed 42 excluded): mean +0.0253, two-sided $p = 0.0009$, 5/5 direction-consistent				
Sensitivity check ($n = 6$, all seeds): mean +0.0285, s.d. 0.0097, $t = 7.182$ on 5 d.f., two-sided $p = 0.0008$, 6/6 direction-consistent				

Table 17: Act 5b: paired validation loss at the 2.6M rung, MQA ($G = 1$) against MHA ($G = 8$), on the C4 corpus at $D = 53.08$M tokens. **What is measured:** a paired difference per seed. **Interval type:** the p-values are two-sided paired t-tests on the per-seed differences; the hypothesis was framed as three open outcomes before the data, so a one-sided test was never available. Source: `results/v2/act5/mqa_quality_2.6m.json`; table source data in `report/tables/act5_mqa.csv`.

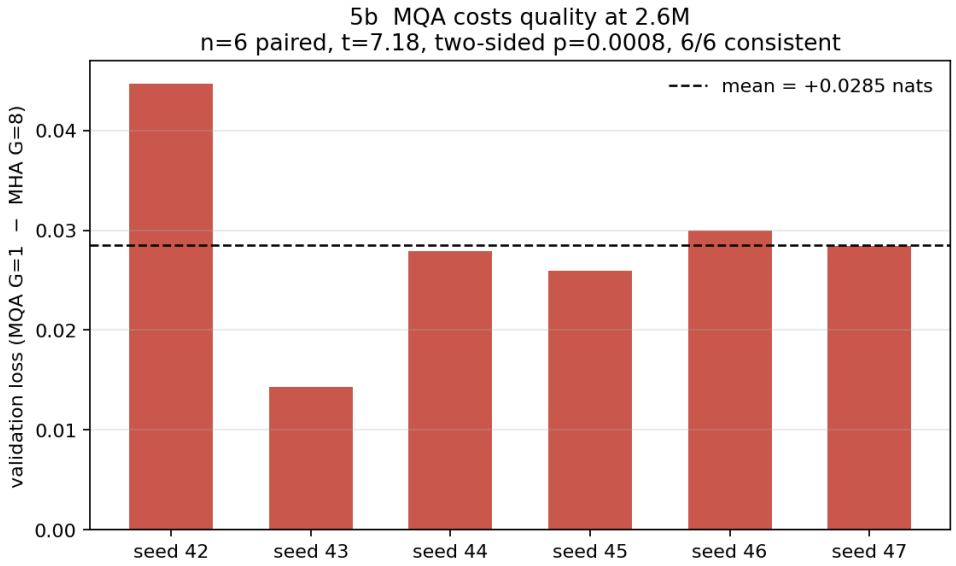


Figure 9: Act 5b. The six paired differences in validation loss between MQA ($G = 1$) and MHA ($G = 8$) at the 2.6M rung. **What is plotted:** 6 paired seeds, one point each, with the mean and the interval. **Interval type:** the error bar is the mean ± 1 sample standard deviation over the 6 paired seeds ($n = 6$); the $n = 5$ sensitivity check that excludes seed 42 is reported in the text and in Table 17. Source: `results/v2/act5/mqa_quality_2.6m.json`.

At $G = 1$ the KV cache at $T = 1024$ is 0.5 MB against 4.0 MB at $G = 8$: an 8× reduction. The FLOP accounting shows something that contradicts a claim made in the course material, and we state it plainly because it is a real finding: at 8 heads, dimension 128 and 4 layers, total FLOPs per token fall by **12.1%** going from $G = 8$ to $G = 1$, entirely from the shrinking KV projection (57,344 to 8,192 FLOPs per token), while the attention matmul cost is identical

(8,388,608 FLOPs per token) at every G. The statement that MQA and GQA "neither reduce compute" is false read literally, and true only of the attention matmul in isolation.

The quality cost of sharing KV heads is real. At the 2.6M rung the paired mean difference is +0.0253 nats with two-sided $p = 0.0009$ at $n = 5$, and +0.0285 nats with two-sided $p = 0.0008$ at $n = 6$, with the direction consistent in every pair.

We report the $n = 5$ analysis as primary. Seed 42's $G = 1$ arm resumed from an already-complete checkpoint in 0.73 s, so its full-run provenance cannot be re-verified: the artifact that would allow it no longer exists, because the canonical $G = 8$ model now occupies that directory. The provenance is nonetheless established *by inference*, not by artifact: `load_state_dict` defaults to `strict=True` and raises on shape mismatch, as the $n = 6$ chain crash demonstrated when it reported `size mismatch for blocks.5.attention.kv.bias: torch.Size([48]) vs torch.Size([384])` for the 2.6M model's head dimension of 24; a $G = 8$ checkpoint therefore *could not* have loaded into a $G = 1$ arm, so the 0.73 s resume necessarily loaded a shape-compatible $G = 1$ checkpoint already at step 12,960, and the resulting loss 2.8558 sits inside the $G = 1$ family (2.8429–2.8661, mean 2.8477). This is an inference from load semantics and value plausibility, *not* a re-verification, and it is not re-checkable without retraining seed 42 ($G = 1$) at roughly 39 minutes. The conclusion does not rest on it: excluding seed 42 entirely gives $n = 5$, $p = 0.0009$, 5/5 consistent.

The $G = 1/2/4/8$ quality curve (2.349 / 2.354 / 2.363 / 2.346) comes from the v1 path: TinyShakespeare, 500 steps, the character tokenizer, a single seed. It is **context only** and is never used as a result; it is reported here solely because the difference between it and the C4 result is the project's own finding that quality differences between attention variants only become visible after real training.

8.3 5c: speculative decoding, a bounded negative

K	Speedup, 1.7× gap	Acceptance	Speedup, 4.2× gap	Acceptance
1	0.502	0.857	0.610	0.857
2	0.441	0.661	0.481	0.576
3	0.399	0.569	0.454	0.556
5	0.346	0.473	0.339	0.391
7	0.300	0.395	0.228	0.233

Table 18: Act 5c: median speedup relative to plain cached decoding, and median acceptance, against the number of drafted tokens K, for a 1.5M draft against a 2.6M target (1.7× parameter gap) and against a 6.3M target (4.2× gap). **What is measured:** 50 cells in total (5 real C4 prompts × 5 values of K × 2 gaps), best-of-3 CPU wall clock. **Interval type:** none; the values are medians over 5 prompts of a best-of-3 timing, with durations of 0.03–0.11 s, and are a consistent comparison *within* one run, not a portable speedup figure. **0 of the 50 cells exceeded plain decoding**; the best cell was 0.622×. Source: `results/v2/act5/speculative_1.7x_multiprompt.json` and `speculative_4.2x_multiprompt.json`; table source data in `report/tables/act5_speculative.csv`.

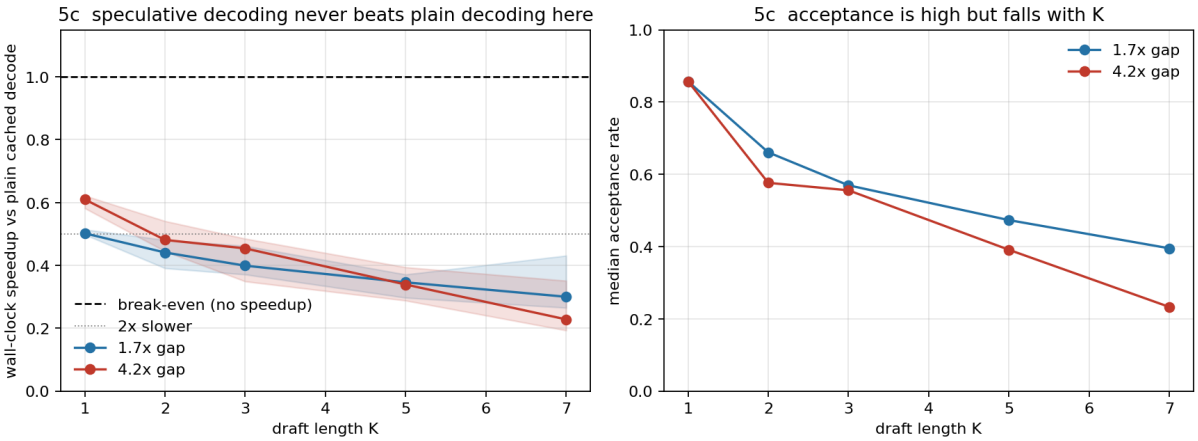


Figure 10: Act 5c. Wall-clock speedup relative to plain cached decoding (left) and acceptance rate (right) against the number of drafted tokens K, for the 1.7× and 4.2× model gaps. **What is plotted:** 2 gaps × 5 values of K, 5 real C4 prompts each, best-of-3 CPU timings. **Interval type:** the error bars are the min–max range over the 5 prompts, not a confidence interval; the timings are CPU best-of-3 with durations of 0.03–0.11 s and are not a portable speedup figure. The dashed line at 1.0 is parity with plain decoding; every point is below it. Source: `results/v2/act5/speculative_*_multiprompt.json`.

The negative result has a mechanism, and the mechanism is arithmetic rather than incidental. With per-token acceptance α, one round yields $(1-\alpha^{K+1})/(1-\alpha)$ tokens for $2K$ forwards. That ratio is below 1.0 for every $K > 1$ even at $\alpha = 1$, because $2K$ forwards buy at most $K+1$ tokens: $K > 1$ is a net loss on forward count before any hardware consideration enters. A real speedup additionally requires the target forward to be far more expensive than the draft forward. At 1.5M–6.3M non-embedding parameters neither forward is compute-bound; both are dispatch-latency-bound, so the target's batching advantage is negligible and the extra draft forward is pure cost.

Acceptance was *high*, which is what makes this interesting rather than worse: the median acceptance at $K = 1$ is 0.857 for both gaps, and the ledger records a first-token acceptance of 0.90 for the 6.3M target against 0.82 for the 2.6M target over 50 samples — the *larger* gap accepts more, because the 6.3M target is the better-trained model (2.5152 against 2.8111), not because the draft is closer to it in size. All three rungs share corpus, tokenizer and recipe, which keeps their next-token distributions close. The negative result is therefore not "speculative decoding fails"; it is that at this scale the forward-count budget dominates acceptance.

The general lesson, stated so that it outlives our models: speculative decoding pays when the target forward is compute-bound, so that verifying $K+1$ positions in one pass is materially cheaper than $K+1$ sequential passes. That is the regime of memory-bandwidth-bound decoding at large scale. Our models are too small to enter it, so the correct conclusion is about our scale, not about the method, which is why this is reported as a bounded negative rather than a refutation.

8.4 Methodology: the bug that a differential test could not find

The double-fed context. Speculative decoding re-fed the entire sequence at the top of every round, on top of an already-prefilled KV cache, in both the draft loop and the target verification loop. The context was therefore double-counted, and the target's distribution p differed from the draft's q even when draft and target were the *same weights*. The symptom was an acceptance rate of 15–40% at a zero draft-target gap, where it must be exactly 100%.

Why it failed silently. The cached and the uncached equivalence tests both passed, because both paths shared the bug and therefore agreed with each other while both were wrong. **A test that compares two implementations cannot detect a fault that both of them share.** What finally localised the fault in minutes was an absolute sanity invariant that no differential test would have suggested: acceptance must be exactly 100% at a zero draft–target gap, because $p = q$ makes $\min(1, p/q) = 1$ for every token. The regression test `test_zero_gap_acceptance_is_total` asserts 100% acceptance at $K \in \{1, 2, 3, 5, 7\}$ for both sampling and greedy.

What it cost. Before the fix, zero-gap acceptance at $K = 3$ was 15.6% and throughput was 28.9 tokens/s; after the fix, acceptance is 100% and throughput is 309.8 tokens/s, a 10.7× improvement. **All 5c numbers taken before that commit are void.** The zero-gap configuration is itself a negative result and is expected to be one: at zero gap no target forward pass is saved, while each round still costs $2K$ forwards against K for plain decoding, so it runs at 0.41–0.50× plain cached generation. Its purpose is to prove correctness end-to-end on a real model, not to measure performance.

The same class of error, in the same act. The Act 5a ledger originally asserted that "greedy/top-k/top-p/beam all use it [the cache]". That was false when written: `beam_search` had no cache parameter in any commit, and `tests/test_kv_cache.py` covered greedy, sampling and top-p but never tested beam at all. The assertion survived into a report because nothing behind it could fail. After the fix, beam does use a per-beam cache and the conclusion held, but the number moved: beam-4 is 4.26× greedy's time for 0.061 bits, where the uncached measurement had said 4.9×. Instrumenting forward passes and cache lengths, rather than rereading a ledger sentence, is what found it.

8.5 Conclusion

What the measurement supports. A KV cache produces byte-identical output, converts autoregressive decoding from quadratic to linear in context, and repays its memory once the sequence exceeds the embedding dimension. Sharing KV heads buys an 8× cache reduction and a 12.1% FLOP reduction — contradicting the course material's claim that these techniques do not reduce compute — at a small but statistically significant quality cost of about 0.026–0.029 nats at the 2.6M rung. Speculative decoding is implemented correctly and is nonetheless a 2–4× loss at every configuration tested, for a reason that is arithmetic.

What it does not support. It does not support quoting 18.36× as a property of the models in this paper: that is a mechanism microbenchmark on a purpose-built context-1024 model, and the canonical models have a 128-token context. It does not support a portable speedup figure for speculative decoding from CPU best-of-3 timings of 0.03–0.11 s. It does not support a claim about sliding-window attention at long context, because that sweep was dropped for cost (a measured 42.1 minutes per run against a 1.8–2.4 hour estimate) and the surviving measurement is at context 128 only, where $W \geq$ context is a no-op and only $W = 64$ is non-trivial. It does not support treating the zero-gap total-variation comparison as evidence that the operator is correct on a real model: at zero gap speculative decoding *is* plain sampling, so that comparison is near-tautological and proves only that the code runs.

9 Act 6: decoding

9.1 Setup and the sweep

The 6.3M canonical model, 40 real C4 prompts of 32 tokens, 32 new tokens, three seeds, giving 120 generations per configuration. Quality is bits-per-character of the *generated* text under a

fixed 6.3M reference model; diversity is the distinct-2 rate and the repeated-4-gram rate.

The quality axis is self-preferring, and this caveat travels with every number in this section. The reference model is the model under test. The axis is therefore valid for *ranking* decoding configurations against one another under a fixed judge, and it is *not* an absolute measure of text quality. A configuration could be scored badly by a judge that shares its own biases.

Configuration	BPC (ref.)	s.d.	rep-4	distinct-2	Seconds
beam4	1.3760	0.4523	0.3095	0.5944	104.0
beam8	1.3790	0.4354	0.4052	0.5177	232.9
beam2	1.3985	0.4733	0.3293	0.5750	61.4
greedy	1.4366	0.5245	0.4147	0.4952	24.4
top-p 0.3	1.5878	0.5349	0.1181	0.7785	31.3
top-k 5	1.6012	0.5335	0.0773	0.8075	30.9
temp 0.5	1.6083	0.5382	0.1293	0.7613	33.8
top-p 0.5	1.6624	0.5605	0.0687	0.8473	31.3
top-k 20	1.7153	0.5918	0.0233	0.8960	31.1
temp 0.7	1.7251	0.5225	0.0503	0.8661	33.7
top-k 50	1.8103	0.6113	0.0155	0.9290	31.1
top-p 0.8	1.8381	0.5898	0.0138	0.9368	32.6
top-p 0.95	1.9701	0.6014	0.0095	0.9618	35.3
temp 1.0	2.0074	0.5968	0.0014	0.9688	33.7
temp 1.3	2.3661	0.5810	0.0000	0.9895	34.6

Table 19: Act 6a: the decoding sweep. **What is measured:** 120 generations per configuration (40 prompts × 3 seeds). **Interval type:** the s.d. column is the standard deviation of the per-generation bits-per-character across the 120 generations, not a seed-level interval; configurations were not re-run with a fresh seed set, so the sweep has no seed-level error bar. The BPC axis is scored by the model under test and is therefore self-preferring: valid for ranking configurations against each other under a fixed judge, not an absolute measure of text quality. Source: `results/v2/act6/6.3M_decoding.json`; table source data in `report/tables/act6_decoding_sweep.csv`.

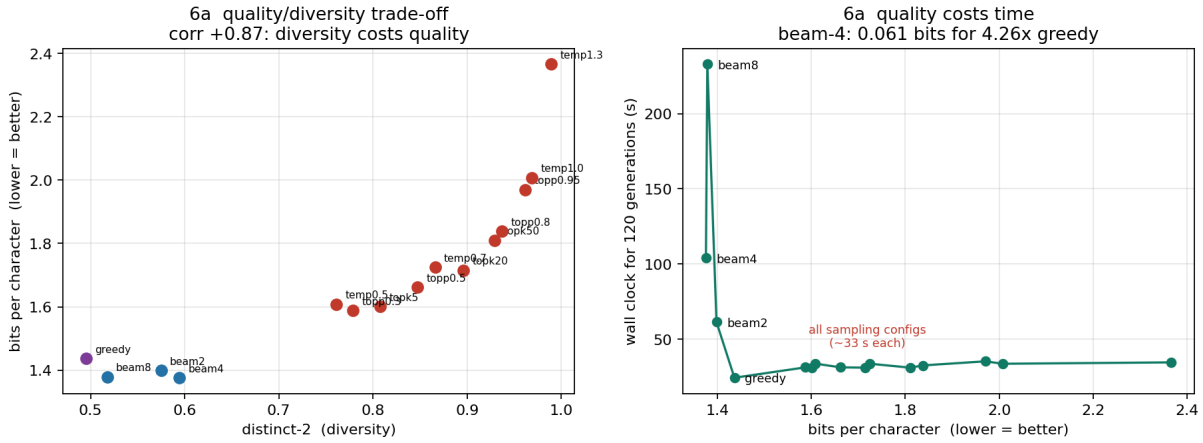


Figure 11: Act 6a. The quality/diversity and quality/cost trade-offs across the 15-configuration decoding sweep. **What is plotted:** 15 configurations, each from 120 generations (40 prompts × 3 seeds). **Interval type:** the error bars in the quality panel are the standard deviation of the per-generation bits-per-character across the 120 generations; the cost panel shows best-of-3 wall-clock timings after an explicit warm-up. The BPC axis is scored by the model under test and is self-preferring — it ranks configurations against each other and does not measure absolute text quality. Source: `results/v2/act6/6.3M_decoding.json`.

The trade-off is clean and strictly monotone: the correlation between distinct-2 and bits-per-character is +0.87, and the correlation between repeated-4-gram rate and bits-per-character is −0.79. Diversity costs quality essentially monotonically across the whole sweep.

Beam search beats greedy on quality — beam-4 is 0.061 bits per character better (1.3760 against 1.4366) — and beam search finds more *typical* text than greedy, which is the expected result rather than a surprise. But it costs 4.26× greedy's time for those 0.061 bits, and beam-8 is worse than beam-4 (1.3790 against 1.3760) at 2.24× beam-4's cost, the classic diminishing return. **Greedy is the quality/cost knee**: no other configuration is both cheaper and better. Every stochastic configuration is cheaper than beam-4 but at least 0.15 bits per character worse.

One determinism check is worth a sentence: all fifteen bits-per-character, repeated-4-gram and distinct-2 values were bit-identical across a repeat run of the whole sweep, which independently confirms that the cache fix changed cost only and never output.

9.2 Best-of-n, and why a self-referential reranker cannot win

n	BPC of reranked pick	BPC of mean sample	Gain	Generations
1	2.0074	2.0074	+0.0000	120
2	1.8684	1.9581	+0.0897	120
4	1.7888	1.9538	+0.1650	120
8	1.7459	1.9760	+0.2302	120
16	1.6959	1.9677	+0.2717	120

Table 20: Act 6b: best-of-n reranking of temperature-1.0 samples, reranked by mean log-probability under the model itself. **What is measured:** 120 generations per row. **Interval type:** none; the values are point estimates over the generation set, and the BPC axis is self-preferring as described in Section 9.1. Source: `results/v2/act6/6.3M_decoding.json`; table source data in `report/tables/act6_bestofn.csv`.

Reranking recovers +0.272 bits over the mean sample at $n = 16$ — a large, monotone, log-shaped gain, exactly as theory predicts for a maximum-of-n statistic. **And best-of-16 is still 0.259 bits worse than greedy** (1.6959 against 1.4366). The reason is structural, not a tuning failure: the reranker is the model itself, so it can only select text the model already finds likely. Best-of-n with a self-referential scorer converges toward the model's own mode asymptotically and never exceeds it, while greedy sits on that mode and sixteen random draws rarely reach it. Beating greedy here would need an *independent* scorer, which a 6.3M model does not have.

9.3 Self-consistency, a negative with an exact mechanism

Votes	Accuracy	95% Wilson CI	Correct	Total
1	0.3789	[0.3478, 0.4110]	341	900
3	0.3556	[0.3250, 0.3874]	320	900
5	0.3556	[0.3250, 0.3874]	320	900
10	0.3611	[0.3304, 0.3930]	325	900
20	0.3567	[0.3260, 0.3885]	321	900

Table 21: Act 6c: majority voting on 300 MNLI items, three seeds, 900 scored predictions per row. **What is measured:** accuracy of the majority label. **Interval type:** 95% Wilson score intervals on the proportion; the chance level for a balanced three-way set is 0.3333. The model was trained on C4 and never on MNLI. Source: `results/v2/act6/6.3M_decoding.json`; table source data in `report/tables/act6_selfconsistency.csv`.

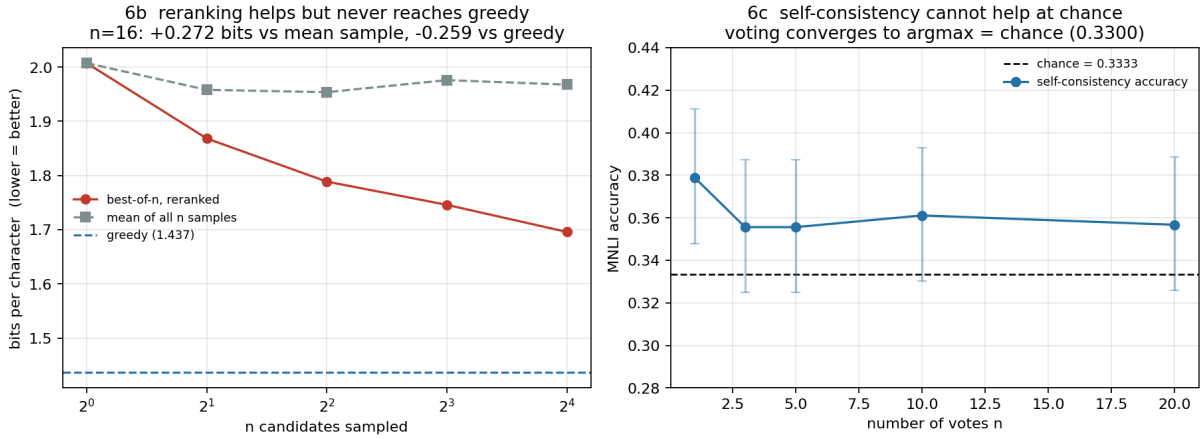


Figure 12: Act 6b and 6c. *Left:* best-of-n reranked bits-per-character and the mean sample, against n, with greedy drawn as a reference line. *Right:* self-consistency accuracy on MNLI against the number of votes, with 95% Wilson intervals and the chance line. **What is plotted:** 5 values of n for reranking (120 generations each) and 5 vote counts for self-consistency (300 items × 3 seeds each). **Interval type:** Wilson score intervals on the right; on the left the vertical bars are the range across the generation set, not a confidence interval. The BPC axis is scored by the model under test and is self-preferring. Source: `results/v2/act6/6.3M_decoding.json`.

Majority voting gives no gain at any n, and the mechanism is measured rather than guessed. The deterministic argmax accuracy is exactly chance: 0.3300 (99/300), with a 95% Wilson interval of $[0.2792, 0.3851]$, which contains 0.3333. Majority voting converges to the argmax as n grows, so there is nothing for it to improve on. The model *never* predicts the neutral class (recall 0.000): it predicts entailment 226 times against 111 gold and contradiction 74 against

105, giving per-class recalls of 0.712, 0.000 and 0.190. On the 216 items it can ever answer, its argmax accuracy is 0.4583 — *below* the 0.5139 coin-flip baseline for that binary subset. Its mean maximum probability is 0.4949 and its mean entropy is 0.7574 against a uniform 1.0986, so the distribution is biased and over-confident rather than uniform noise.

The $n = 1$ figure of 0.3789 must *not* be read as partial competence. It exceeds chance only because sampling is an unbiased draw from a biased distribution, while the argmax is dominated by the entailment over-prediction.

9.4 Conclusion

What the measurement supports. Across a 15-configuration sweep, diversity costs quality essentially monotonically (correlation +0.87 with distinct-2). Beam-4 produces more typical text than greedy for 0.061 bits at 4.26× the time; beam-8 is worse than beam-4 at 2.24× the cost; greedy is the quality/cost knee. Best-of-n reranking recovers +0.272 bits but remains 0.259 bits worse than greedy, and the reason is structural. Self-consistency gives no gain at any vote count, because the argmax it amplifies is exactly at chance and is systematically biased towards one class.

What it does not support. It does not support any absolute claim about text quality, because the judge is the model under test. It does not support a claim that self-consistency is useless in general: it supports the narrower claim that self-consistency cannot manufacture task competence that is not there, and it does so by measuring the competence and finding it absent. It does not support a claim that beam search is the right decoder choice, because the 0.061-bit gain costs 4.26× the time and the comparison rests on a self-preferring judge.

10 Act 7: instruction tuning and parameter-efficient adaptation

10.1 Setup

The 6.3M canonical model, a synthetic seven-task family, 512 training and 128 held-out items, three seeds, 24 epochs, teacher-forced exact match with 95% Wilson intervals. The held-out split is disjoint from the training split by construction, and the disjointness is asserted by the harness.

10.2 7a: the zero-shot floor, and the control that makes it assertable

The base model scores 0.0000 on all 128 held-out items, so every number above zero in this act is attributable to adaptation. That floor is only worth stating if the metric could have detected a nonzero score, and a positive control establishes it: an oracle emitting the correct answer scores 1.0 through the teacher-forced path, through the greedy path and through the exact ICL scoring path, while the real model scored zero on identical items. The harness is sound and the 7a floor is assertable.

10.3 7b: in-context learning copies the format, not the operation

Measurement	Value	n	Interval
First generated token is the demonstration's answer, with demo	18/18	18	exact count
First generated token is the demonstration's answer, zero-shot	0/18	18	exact count
Exact match on the query's answer, with demo	0/18	18	Wilson [0, 0.1759]
Exact match on the query's answer, zero-shot	0/18	18	Wilson [0, 0.1759]
Items whose answer changed because of the demonstration	0/18	18	exact count
Zero-shot generations that were only punctuation	18/18	18	exact count

Table 22: Act 7b: the same-task control on the 18 held-out addition items (unseen operand pairs), paired A/B with and without a demonstration in context. **What is measured:** a paired binary outcome per item. **Interval type:** Wilson 95% intervals on the proportions, plus exact counts; the deterministic model makes the three "seeds" bit-identical, so this is $n = 1$ in substance. Source: `results/v2/act7/6.3M_sft_peft.json`; table source data in `report/tables/act7_icl_control.csv`.

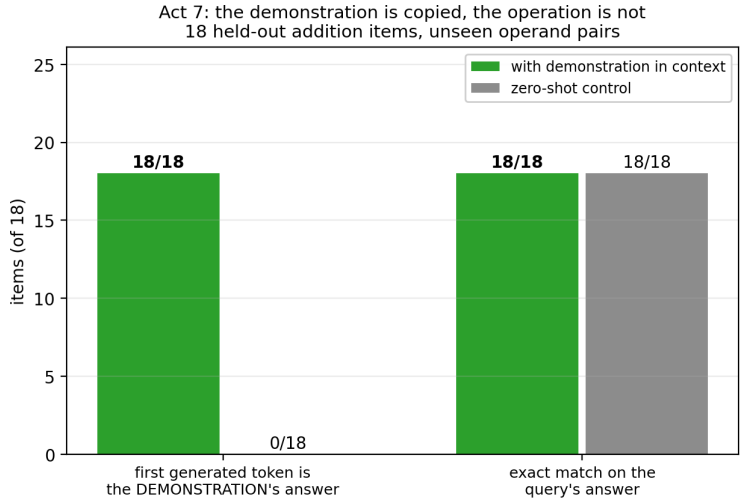


Figure 13: Act 7b. On the 18 same-task held-out addition items: the first generated token is the demonstration's answer in 18/18 items with a demonstration and 0/18 without, while the exact match on the query's own answer is 0/18 in both conditions. **What is plotted:** 18 items. **Interval type:** exact counts; the model is deterministic, so the three seeds are bit-identical and the effective n is 1. Source: `results/v2/act7/6.3M_sft_peft.json`.

The result is not that in-context learning is absent. On the 18 same-task addition items, the first generated token is the *demonstration's* answer on 18/18 items, against 0/18 for zero-shot, where the model emits only punctuation. The model is a competent pattern completer that cannot perform the operation: it continues the surface pattern using the most salient number in context, which is the demonstration's number, not the query's. The paired A/B is 0/18 with a demonstration against 0/18 without (Wilson $[0, 0.1759]$), with 0 of 18 items changed.

Three honesty notes travel with this number, and the first is the reason the number is a rewrite rather than a softening. (i) The three ICL "seeds" are bit-identical, because the model is frozen and the forward pass is deterministic: this is $n = 1$, not $n = 3$. (ii) Shot selection is cost-driven — only one or two demonstrations fit the 128-token window, so 110 of 128 items receive the single cheapest pool example, which is an addition demonstration and therefore *cross-task* relative to most queries. The clean same-task test is the 18 addition items, and only they are reported as a same-task result. (iii) ICL is scored teacher-forced, so it is not comparable to the free-running generation numbers elsewhere in this paper.

The failure that voided the first attempt of this measurement belongs in the paper, because it is the same class as Act 8's defect D14 and because the metric was sound while the prompt was not. The conventional 4-shot prompt was 328 tokens against a 128-token context, so truncation deleted the demonstrations; the positive control records the original prompt at 328 tokens and the visible-shot count as "2/4 (truncated away)", and the ICL caveat records that the demonstrations were removed from the model input. The demonstration was not ignored — it was not there. ICL is now fitted to the window, with every selected demonstration asserted present in the model input, and re-measured.

10.4 7c and 7d: a matched budget, and a rank that is pinned by construction

Method	Size	Trainable	Final train loss	Accuracy (Wilson 95%)
Full fine-tune	—	8,012,032	0.0043	**0.5755** [0.5256, 0.6240]
Adapters	4	18,464	1.1365	0.2136 [0.1755, 0.2573]
LoRA	1	14,336	1.3097	0.1823 [0.1469, 0.2240]
Prompt tuning	64	16,384	2.0697	0.0859 [0.0618, 0.1182]
Prefix tuning	5	20,480	3.2576	0.0000 [0.0000, 0.0099]

Table 23: Act 7c: the parameter-efficient family at a matched budget of 20,000 trainable parameters, except full fine-tuning, which is shown for reference and is *not* budget-matched. **What is measured:** teacher-forced exact match on 128 held-out items. **Interval type:** 95% Wilson score intervals; the interval is about ±0.04, which is why the ordering of the two middle rows is not established. Source: `results/v2/act7/6.3M_sft_peft.json`; table source data in `report/tables/act7_peft_matched_budget.csv`.

Rank r	Trainable	Accuracy	Per-seed runs
1	14,336	0.1823	0.1797, 0.1953, 0.1719
2	28,672	0.1927	0.1953, 0.1719, 0.2109
4	57,344	0.2838	0.3281, 0.2734, 0.2500
8	114,688	0.3516	0.3672, 0.3203, 0.3672
16	229,376	0.4245	0.4375, 0.3828, 0.4531
32	458,752	0.4713	0.4609, 0.5000, 0.4531
64	917,504	**0.5078**	0.5000, 0.5234, 0.5000

Table 24: Act 7d: the LoRA rank sweep. **What is measured:** teacher-forced exact match on 128 held-out items at three seeds. **Interval type:** the per-seed runs are shown rather than a single interval; the seed spread is at most 0.05 throughout, and the seeds vary optimisation *order* only, since the base weights are pinned, so between-initialisation variance is unmeasured. Source: `results/v2/act7/6.3M_sft_peft.json`; table source data in `report/tables/act7_lora_rank_sweep.csv`.

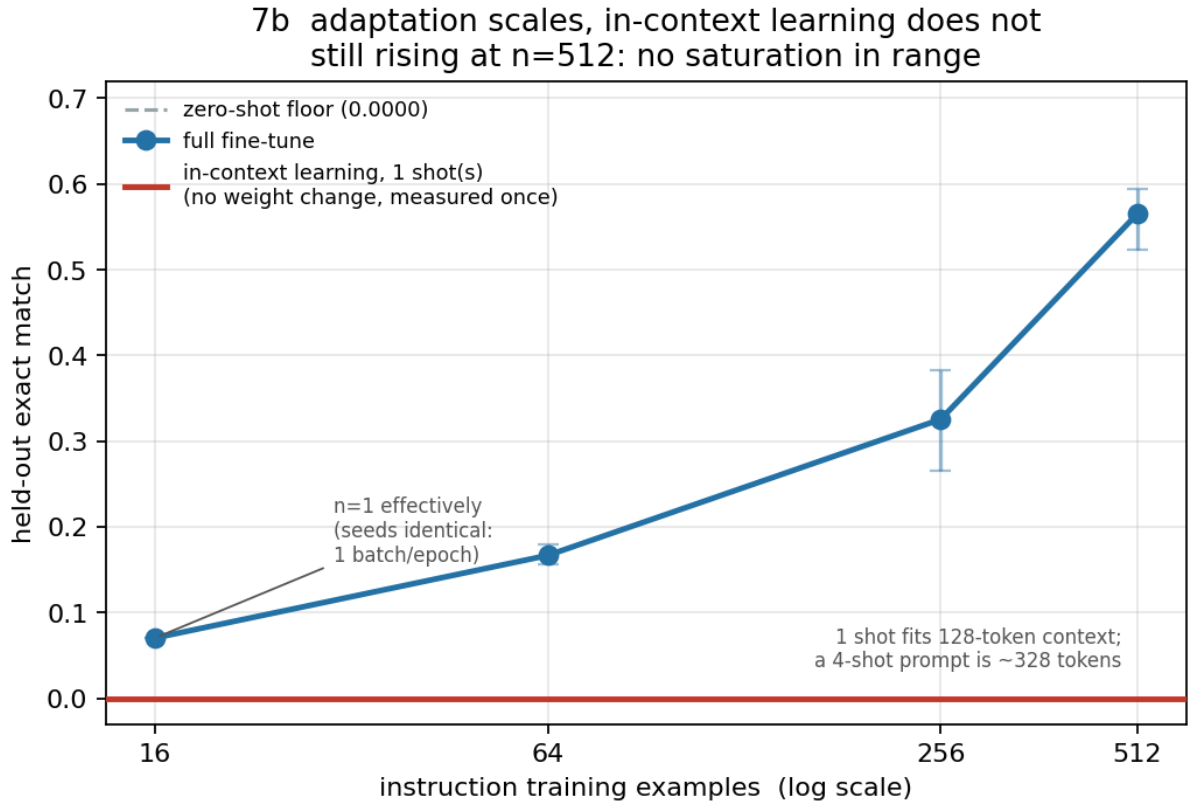


Figure 14: Act 7a and 7b. Held-out exact match against the number of instruction training examples for full fine-tuning, with the zero-shot floor and the in-context-learning arm drawn as horizontal lines, and the per-seed spread shown for each point. **What is plotted:** 4 training-set sizes at 3 seeds, plus a zero-shot floor and a single ICL measurement. **Interval type:** the vertical bars are the range across the three seeds; the $n = 16$ point is **effectively** $n = 1$ and is annotated as such, because with 16 examples and a batch size of 16 there is one batch per epoch and the seeded shuffle has nothing to shuffle. The ICL arm is measured once, because its prompt does not depend on the training-set size, and is therefore drawn flat. Source: `results/v2/act7/6.3M_sft_peft.json`.

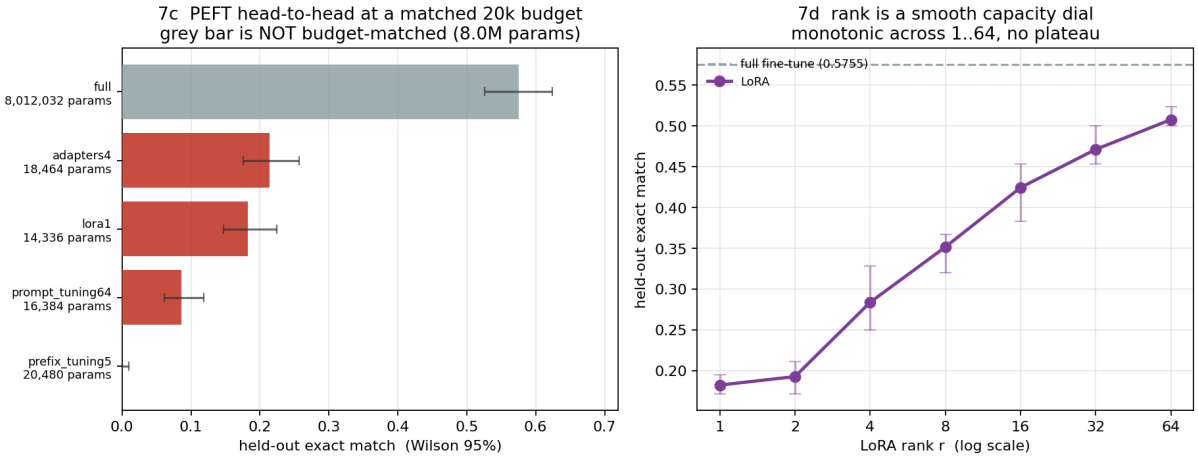


Figure 15: Act 7c and 7d. *Left:* the parameter-efficient family at a matched 20,000-trainable-parameter budget, with full fine-tuning shown in grey because it is *not* budget-matched. *Right:* the LoRA rank sweep against the full fine-tuning reference. **What is plotted:** 5 methods (7c) and 7 ranks (7d), each at 3 seeds on 128 held-out items. **Interval type:** 95% Wilson score intervals in the left panel; the right panel shows the per-seed range and the full fine-tuning reference, with no interval on the reference because it is a single configuration. Source: `results/v2/act7/6.3M_sft_peft.json`.

The two tables must be read together. At a matched budget, adapters score 0.2136 and LoRA 0.1823, and the Wilson intervals overlap by a wide margin (about ±0.04 on 128 items), so the ordering is *suggestive, not established*. But the LoRA arm at rank 1 is pinned low *by construction*: the budget forces a small rank, and Table 24 shows that this model's LoRA accuracy has not flattened at rank 1 — it rises monotonically to 0.5078 at rank 64. Comparing LoRA at rank 1 against adapters at size 4 is comparing two methods at two points on their own curves, and only one of those points is where that method is strongest. The honest reading is that the matched-budget comparison is a comparison of points on curves, and that the curves, not the points, are the result.

Prefix tuning's 0.0000 is a genuine failure to fit rather than a scoring artefact: its final training loss is 3.2576, and the training-loss column exists precisely so that "did not fit" can be told apart from "fit but scored zero". That distinction is only visible because the column was recorded.

Rank, by contrast, behaves like a smooth capacity dial rather than a threshold: accuracy rises monotonically from 0.1823 at $r = 1$ to 0.5078 at $r = 64$ with no plateau, and $r = 64$ recovers 0.5078 against full fine-tuning's 0.5755 using 11.5% of the trainable parameters.

10.5 7b revisited: the LIMA premise, not the LIMA claim

The full fine-tuning curve is 0.0703 → 0.1667 → 0.3255 → 0.5651 at $n = 16/64/256/512$, with no saturation: it is still rising by 0.24 from 256 to 512. The $n = 16$ row is identical across all three seeds (0.0703 three times), because one batch per epoch gives the seeded shuffle nothing to shuffle; that row is effectively $n = 1$ and is annotated as such in the table, in the JSON and in the figure, and carries no error bar.

This is *not* a contradiction of LIMA. LIMA argues that the capability already exists in the base model and that supervised fine-tuning merely elicits it. Here both zero-shot and in-context learning are 0.0000, so there is no latent instruction-following to elicit: the curve is *learning a skill*, not aligning one. The supported claim is that LIMA's "less is more" finding is not supported at 6.3M, together with the mechanism sentence above.

One negative inside the positive curve is retained rather than smoothed away. The `add` task

scores 0.9452 on *train* items (69 of 73) and **0.0000 on held-out items** (0 of 18). The held-out guesses are off by one — $8 + 2 \to 11$, $7 + 7 \to 15$, $6 + 8 \to 15$, $1 + 8 \to 10$ — which is the signature of memorising seen sums rather than learning addition. It is a result, not a failure of the harness.

10.6 Conclusion

What the measurement supports. The zero-shot floor is 0.0000 and is assertable, because a positive control confirms the metric registers 1.0 when the answer is present. Supervised fine-tuning scales monotonically with the instruction-set size, with no saturation in range, reaching 0.5651 at 512 examples. That figure carries two separate spreads and they should not be confused. Across the three seeds the runs are [0.5938, 0.5234, 0.5781] — s.d. 0.0370, range 0.0704 — and those seeds vary optimisation *order* only, because the base weights are pinned, so between-initialisation variance is unmeasured. Separately, re-running the identical recipe on a different device returned 0.6016, which is 0.0078 above the best of the three seeds (0.21 s.d.) and so just outside their range; because it is *the same seed* on different hardware it is not a fourth independent draw but a device-level reproducibility band of about 0.008 on a fixed seed. At a matched 20,000-parameter budget, adapters score 0.2136, LoRA 0.1823, prompt tuning 0.0859 and prefix tuning 0.0000 — and prefix tuning's zero is a failure to fit, not a scoring artefact. LoRA rank is a smooth capacity dial across $r \in \{1, \ldots, 64\}$, reaching 0.5078 at 11.5% of full fine-tuning's trainable parameters. In-context learning is not absent: the model copies the demonstration's answer as its first token on 18/18 same-task items.

What it does not support. It does not support a claim that in-context learning is absent at 6.3M. It does not support the ordering of adapters above LoRA. It does not support a claim about natural instructions: the task family is synthetic. It does not support a claim that the $n = 16$ row is an $n = 3$ result. And the error bars throughout are within-initialisation optimisation-order variance, because the base weights are pinned, so between-initialisation variance is unmeasured.

11 Act 8: grounding, hallucination, and tool use

11.1 8a: one prompt builder, and a context budget denominated in tokens

Two defects in the v1 retrieval path are fixed here, and the cost of one of them is worth a number. The v1 code passed `context_length // 2` as a limit in *characters*; on the canonical 128-token window that admitted 64 characters, about 32 tokens, of a 302-token chunk — so 9.2% of the characters, and 10.6% of the tokens, of the retrieved context survived. The token-based budget admits 77 tokens instead of 32, a factor of 2.4 more, and the assembled prompt measures 120 tokens, within the 128-token window. `fit_context()` binary-searches the longest context prefix whose *assembled* prompt fits in tokens while holding back an answer reserve, and then re-encodes to verify the result rather than trusting the arithmetic; it raises rather than returning a prompt that cannot be answered in. The second defect was that retrieval and supervised fine-tuning used different prompt strings, so a model tuned on the instruction contract was evaluated on a different one; one formatter now serves both.

11.2 8b: the choice rates, which are the result

The claim format was declared before any generation was sampled: *"the supplied passage states `<NUMBER>`"*, with condition-agnostic classes. Numbers are the claim currency because they are the one atomic fact that can be extracted and verified mechanically from C4 without an annotator.

Condition	Count	Rate	Wilson 95%	Chance-adj.	Bootstrap 95%
Oracle	9/40	0.2250	[0.1232, 0.3750]	+0.0412	[−0.078, 0.170]
Closed-book	11/40	0.2750	[0.1611, 0.4283]	+0.0913	[−0.037, 0.234]
Retrieved	8/40	0.2000	[0.1050, 0.3476]	+0.0162	[−0.101, 0.143]
Non-numeric passage	9/40	0.2250	[0.1232, 0.3750]	+0.0413	[−0.078, 0.166]
Counterfactual: chose the original number	9/40	0.2250	[0.1232, 0.3750]	+0.0413	[−0.084, 0.172]
Counterfactual: chose the *edited* number	0/40	0.0000	[0.0000, 0.0876]	n/a	[0.000, 0.000]

Table 25: Act 8b: the share of items on which the model's preferred answer is the item's asserted number, under five conditions, on one shared item set. **What is measured:** the argmax over the candidate set of the teacher-forced likelihood of each candidate answer. **Interval type:** 95% Wilson score intervals on the proportions, and a percentile bootstrap that resamples *items* for the chance-adjusted contrast. **The unit of analysis is the item,** $n = 40$, because the instrument is deterministic; a 0/40 therefore gives a Wilson upper bound of 0.0876, not 0.031. The per-item uniform chance is **0.1838**, not 1/3: the candidate set is deduplicated and holds 3, 4, 5 or 6 distinct numbers (1 item has 3 candidates, 3 have 4, 8 have 5 and 28 have 6), so the chance level is per item. 10 of 120 candidate passages were dropped for number truncation. Source: `results/v2/act8/hallucination.json`; table source data in `report/tables/act8_choice_rates.csv`.

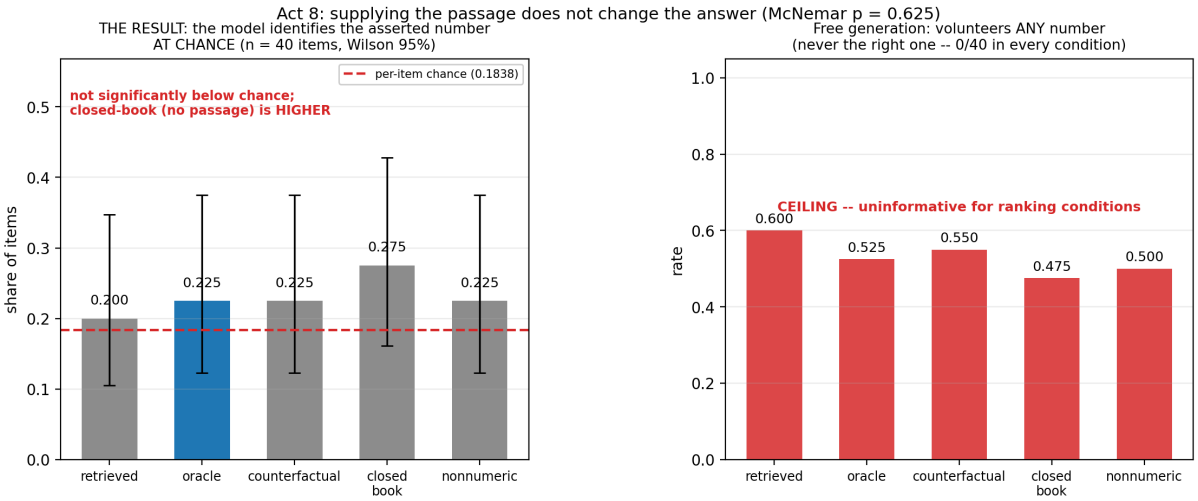


Figure 16: Act 8b. *Left:* the choice rate — the share of items on which the model prefers the asserted number — under each condition, with the per-item uniform chance level of 0.1838 drawn as a dashed line and 95% Wilson intervals on the bars. *Right:* the free-generation instrument, which shows how often the model volunteers any number at all; in every condition the model never volunteers the number it was shown (0/40), so the rate of *not* producing the right number is 1.000 everywhere. **What is plotted:** 5 conditions, $n = 40$ items each. **Interval type:** 95% Wilson score intervals (left); the right panel has no interval, because it is a single rate per condition over the same 40 items. The right panel is a **ceiling**: the absolute free-generation rate reads 1.000 in every condition (97 of 120 oracle generations contain no number at all), so it is reported and labelled and never used to rank conditions. Source: `results/v2/act8/hallucination.json`.

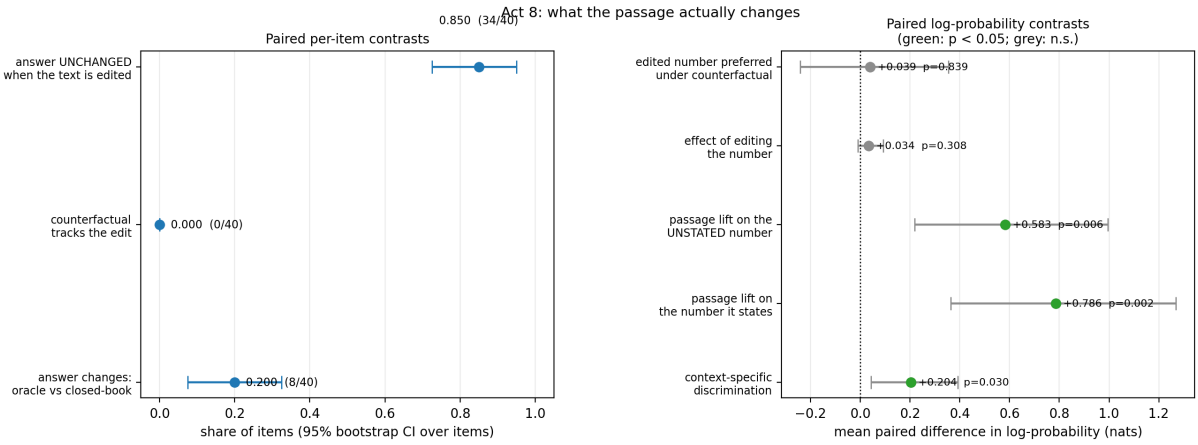


Figure 17: Act 8c. *Left:* the paired per-item contrasts in rate, with 95% bootstrap intervals that resample items. *Right:* the paired log-probability contrasts in nats, with bootstrap intervals; points are coloured green where $p < 0.05$ and grey where the contrast is not significant. **What is plotted:** 3 rate contrasts and 5 log-probability contrasts, each on $n = 40$ items. **Interval type:** percentile bootstrap over items, 10,000 resamples; the log-probability panel also reports a paired t-test p-value. Source: `results/v2/act8/hallucination.json`.

The result, in the order it should be read: the model identifies which number a passage asserts **at chance**. The oracle condition selects the asserted number on 9 of 40 items (0.2250, Wilson $[0.1232, 0.3750]$); closed-book, with no passage at all, selects it on 11 of 40 (0.2750,

Wilson $[0.1611, 0.4283]$); and the McNemar test between them gives discordant counts of 1 and 3 with $p = 0.625$, not significant. The retrieved condition selects it on 8 of 40 (0.2000) and the counterfactual condition follows the edit in 0 of 40 items (bootstrap $[0.000, 0.000]$), while under the counterfactual it still picks the *original* number on 9 of 40. Every chance-adjusted rate is not significant: +0.0162 retrieved, +0.0412 oracle, +0.0413 counterfactual (original number), +0.0913 closed-book, +0.0413 non-numeric.

Removing the passage changes the model's preferred answer on 8 of 40 items (0.20, bootstrap $[0.075, 0.325]$) — and not in a consistent direction, because the closed-book point estimate is *higher* than the oracle's. The symmetric test fails as well: editing the number in the text changes the preferred answer on 6 of 40 items, leaves it unchanged on 34 of 40 (0.85, bootstrap $[0.725, 0.950]$), and the edited number is not preferred under the counterfactual (+0.039 nats, $p = 0.839$).

Two different counts must not be conflated here. The "8 of 40" is the number of items on which the *preferred answer* differs at all between the oracle and the closed-book conditions, including items that switched between the counterfactual number and the "neither" class. The "4" is the McNemar *discordant* count: the items on which the two conditions disagree specifically about whether the asserted number was preferred, namely $1+3 = 4$ items. The first is a measure of how much the prompt moves the answer; the second is the measure the test is powered for. Neither is 8, and neither is the number that would be quoted if the two were swapped.

The wording matters and we have chosen it deliberately. *At chance* is correct. *Below chance* is not: 9 of 40 against a per-item chance of 0.1838 is not significantly below chance, and the closed-book point estimate is higher, so a directional claim would be reading noise.

11.3 8c: the mechanism, which cuts against grounding

Statistic	n	Mean (nats)	p	Bootstrap 95%
Passage lift on the number it states	40	+0.786	0.0017	[+0.364, +1.269]
Passage lift on the number it does *not* state	40	+0.583	0.0057	[+0.219, +0.996]
Context-specific discrimination (the difference)	40	+0.204	0.030	[+0.044, +0.391]
Oracle margin, asserted minus counterfactual	40	+0.352	0.011	[+0.098, +0.604]
Closed-book margin (control)	40	+0.148	0.199	[−0.072, +0.375]
Edit effect on the counterfactual number	5	+0.034	0.308	[−0.010, +0.091]
Edited number preferred under counterfactual	5	+0.039	0.839	[−0.242, +0.354]

Table 26: Act 8c: the paired log-probability contrasts that explain the choice rates. **What is measured:** the difference in the teacher's log-probability of two candidates, paired per item. **Interval type:** percentile bootstrap over items, 10,000 resamples, with a paired t-test p-value; the two contrasts with $n = 5$ are restricted to the items in which the counterfactual number was a candidate. Source: `results/v2/act8/hallucination.json`; table source data in `report/tables/act8_logprob_contrasts.csv`.

Having a passage raises the likelihood of numbers *appearing in it* by roughly 0.6–0.8 nats, and it raises it almost as much for a number the passage does not state. The tracked partner number is *absent* from the passage in 40 of 40 oracle items and 40 of 40 counterfactual items, because a

scored sentence contains exactly one number by construction of the claim format, so the +0.583 lift is a generic context effect on the candidate set rather than evidence about passage content. The lift rows also confound "has a passage" with "has a longer prompt": the closed-book prompt is 32 tokens and the oracle prompt about 85. The claim is therefore a **context-number bias**, not an assertion-reading ability.

The only content-specific trace is the residual +0.204 nats

(bootstrap $[+0.044, +0.391]$, $p = 0.030$), and it is marginal, unaccompanied by any behavioural change, and pointed the other way at the behavioural level. The two instruments disagree, and the behavioural one is the one that matters.

The strongest single piece of evidence for the bag-of-numbers reading is the new **non-numeric control**: a different sentence of the same document, containing no number, topic- and length-matched to the oracle passage. It scores **9/40 — exactly the oracle's 9/40**. Swapping the asserting passage for a non-asserting one of the same document changes nothing at all.

11.4 The retrieved condition, and the retriever that was broken

The first version of the retrieved condition queried the retriever with the instruction text, whose terms (`what`, `number`, `does`, `the`, `passage`, `state`) cannot come from a passage about a number. Term-overlap ranking degenerated to a constant and the tie-break returned the same chunk for every item, so the earlier 0/40 was a *query-design failure*, not a retriever failure and emphatically not a finding. The diagnosis was performed before the claim was revised, and the query is now the item's *lexical anchor*: content words from a different sentence of the same document, excluding the number, so the retriever has a topical handle but never sees the number or the answer-bearing sentence. The condition is therefore answer-agnostic rather than a disguised oracle.

Retrieval metric	Before the fix	After the fix
Distinct documents returned	1 (degenerate)	39
Document hit rate	—	0.975
Retrieved chunk contains the asserted number	0/40	0.475
Chose the asserted number under `retrieved`	—	8/40 = 0.2000
Follows the retrieved number when the chunk contained one	—	0.2857

Table 27: Act 8b: the retriever diagnosis and repair, and the re-measured retrieved condition. **What is measured:** retrieval hit rates and the choice rate on 40 items. **Interval type:** the choice rate's 95% Wilson interval is $[0.1050, 0.3476]$; the followed-context rate's is $[0.1172, 0.5465]$; the retrieval hit rates are proportions over the same 40 items. The anchor is derived from the target document, so the document hit rate is optimistic relative to a user's paraphrase; the retriever never sees the number or the answer-bearing sentence, so the condition is not a disguised oracle. Source: `results/v2/act8/hallucination.json`.

The retrieved condition now behaves like the others: it is at chance, with a chance-adjusted rate of +0.0162 and a bootstrap interval of $[-0.101, +0.143]$. **This is a measured result, not an open item**: the retriever is answer-agnostic (it never sees the number or the answer-bearing sentence), it returns 39 distinct documents instead of one, and the condition is scored with the same instrument and on the same 40 items as the other four. What it does not show is a benefit from retrieval; what it shows is that supplying a passage and supplying a passage that was *retrieved* both leave the model at chance.

11.5 8d: tool use, format acquired and operation absent

Rate	Greedy ($n = 40$) Count	Wilson 95%	Sampled ($n = 120$) Count	Wilson 95%
Well-formed `CALC(...)` call	0/40	[0, 0.0876]	0/120	[0, 0.031]
Echoes the call template	19/40	—	3/120	—
Malformed call	19/40	—	3/120	—
Direct answer without a tool, and correct	0/40	—	1/120	—

Table 28: Act 8d: model-initiated tool use. **What is measured:** whether the model emits a parseable `CALC(expression)` call, under greedy decoding (40 items) and under sampling at three seeds (120 generations). **Interval type:** 95% Wilson score intervals on the two well-formed-call rates; the remaining rows are counts, and no interval is quoted for them. Source: `results/v2/act8/tool_use.json`; table source data in `report/tables/act8_tool_use.csv`.

The bare rate is 0%: no well-formed `CALC(...)` call, under either decoding regime. The mechanism is not. **19 of 40 greedy generations emit the literal `CALC(`, and in all 19 the argument is exactly `"s"` — the call is `CALC(s)`.** The template evidence is at the token level: `CALC(` tokenises to $[49, 4083, 49, 22]$, that is `['C', 'AL', 'C', '(']`, and after that prefix the next-token distribution has the literal character `'s'` (id 97, $p = 0.0278$) at the argmax, with `'express'` (id 6369, $p = 0.0214$) as runner-up — a fragment of the word *expression*, which appears in the prompt's own template. The model is completing the instruction's template, not writing an arithmetic expression. The argmax probability is low, so the echo is a *modal* effect rather than a confident one, and the greedy/sampled gap (47.5% against 2.5%) is reported rather than smoothed over: the echo is a mode phenomenon that $T = 1.0$ sampling dilutes.

This is the same shape as Act 7's in-context-learning result, and the paper says so explicitly: **the format is acquired and the operation is absent**. The model's own "expression" template, echoed back to it, is enough to produce a tool call it cannot complete.

11.6 Conclusion

What the measurement supports. At 6.3M, supplying a passage does not measurably change what the model says. The model identifies which number a passage asserts at chance (9/40 under oracle against 11/40 closed-book, McNemar $p = 0.625$), it follows a counterfactual edit in 0 of 40 items, and a topic-matched passage containing no number scores exactly the same 9/40 as the asserting passage. The mechanism is a generic context-number bias: a passage raises the likelihood of the numbers in it by 0.6–0.8 nats whether or not it asserts them, and the only content-specific residual is +0.204 nats at $p = 0.030$, which comes with no behavioural change. Model-initiated tool use is 0/40 well-formed calls, with 19/40 greedy generations echoing the template and the argument `s` in all of them.

What it does not support. It does not support a claim that the model is *below* chance: the interval contains the chance level and the closed-book point estimate is higher. It does not support any comparison between conditions on the absolute hallucination rate, which is a ceiling of 1.000 everywhere. It does not support a claim about factual truth: numbers proxy for atomic claims and there is no fact base, so this measures *groundedness* against a supplied source, not truth. It does not support a generalisation about retrieval-augmented generation at any other scale.

12 Act 9: interpretability

This act asks where the model stores what it fails to use. It uses the pinned canonical 6.3M model — 8 layers, 8 heads, 64 heads in total, dimension 256, context 128 — and a held-out slice of 64 sequences of 128 tokens from the C4 shard at document offset 300,000, far beyond anything the training sweep touched.

The intact baseline is measured in this run, on that slice: cross-entropy 2.8551 nats, perplexity 17.376. Every delta in this section is therefore *internal* to the experiment and is **not** comparable to the published 2.5152 validation loss of the same checkpoint on the full validation set.

This act was measured twice, and the second measurement changed one of its claims. The first version of 9c reported that the model encodes which number a passage asserts and then destroys it before the output. A control set was then run — five conditions for the lens, a chance anchor for the attribution method, and three checks on the head claim — and the controls went *against* that version of 9c. The strong claim is withdrawn below, in the form the controls support, and the withdrawal is part of the result rather than a footnote to it.

12.1 9a: causal head ablation with a random-ablation null

Every head is ablated in turn — its attention output is zeroed before it is mixed back into the residual stream — and the held-out loss is re-measured. The essential control is a *random-ablation null*: the same number of heads removed at random, with the same budget, so that a large Δ can be distinguished from the fact that some heads matter more than others by position alone. Without that control the experiment would measure the head *index*, not the head. The null is 40 random single-head sets, with random 2-, 3- and 4-head sets for the combinations.

The finding of this subsection is a redundancy, and it comes first because it changes what the rest of the section means. Ablating layer 0 head 5 together with layer 0 head 2 costs only +0.0234 nats — which is *exactly* what ablating layer 0 head 2 alone costs — so the two heads are redundant (redundancy 0.97), and a redundant partner is silently covering for the head whose removal is most visible in isolation. The converse pair shows that this is not general fragility: ablating layer 0 head 5 together with layer 1 head 6 costs +0.9591 nats against a sum of singles of +0.9079, which is additive, so the effect is specific to the redundant partner rather than a property of removing heads at all. The reading is that the effect is carried by a *group* of layer-0 heads, and that a single ablated head appears to carry the model only because a redundant partner is covering for it. The full control set that establishes this is Table 30 below.

Quantity	Value	Unit or interval
Intact baseline (this slice, this run)	2.8551	nats CE; perplexity 17.376
Heads with $z \geq 2$	1 of 64	layer 0, head 5
Heads with $z \leq 0$	59 of 64	—
Top head (L0 H5): ΔCE	+0.832	nats
Top head (L0 H5): ΔPPL	+22.5	perplexity
Top head: z score	+7.82	against the random-ablation null
Top head: z against the random null	+42.8	—
Next head (L1 H6): ΔCE	+0.076	nats; $z = +0.54$
Random single-head null	+0.0099	mean; s.d. 0.0192
Random 2-head / 3-head null	+0.0714 / +0.0842	mean ΔCE
Top-4 combination	+1.047	point estimate; the bootstrap over the four individual head deltas is $[0.0365, 0.6390]$ (mean +0.248)
Random 4-head combination	+0.071	mean ΔCE
Attention entropy of the $z \geq 2$ head	0.379	nats
Attention entropy of the 59 heads with $z \leq 0$	2.480	nats

Table 29: Act 9a: causal head ablation against the random-ablation null. **What is measured:** the change in held-out cross-entropy when one head (or a set of heads) is ablated, on 64 held-out sequences of 128 tokens. **Interval type:** the z scores are against the random-ablation null (40 random single-head sets); the top-4 row is a percentile bootstrap over the *four individual ablated heads* ($n = 4$, 10,000 resamples), and that interval does *not* contain the +1.047 combined effect, which is therefore reported as a point estimate; the null rows are means over the random sets. Source: `results/v2/act9/interpretability.json`; table source data in `report/tables/act9_head_ablation.csv`.

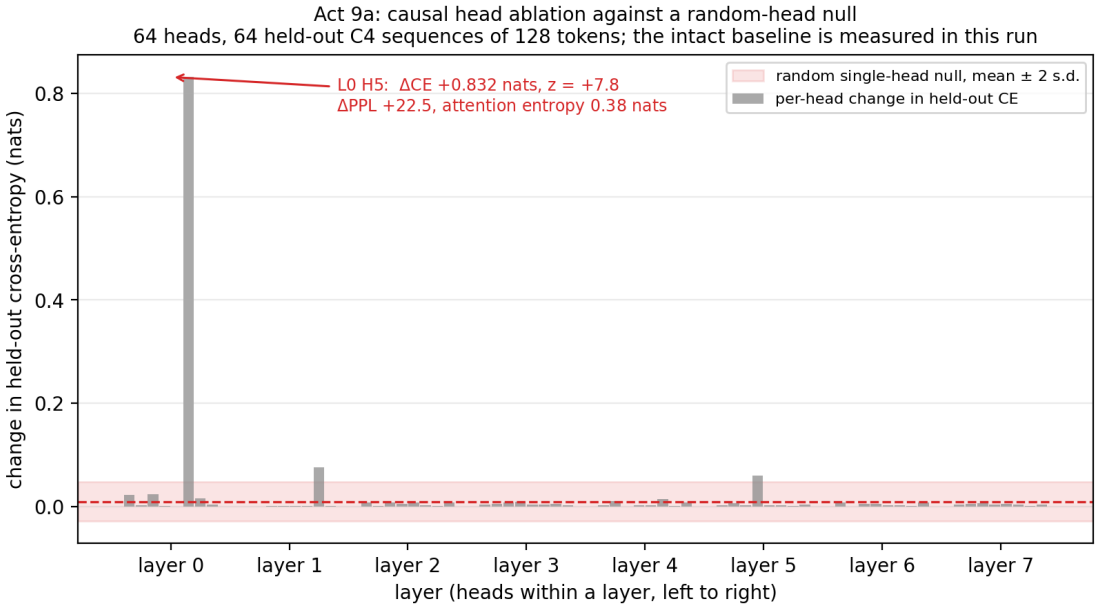


Figure 18: Act 9a. The change in held-out cross-entropy caused by ablating each of the 64 heads, one bar per head (heads within a layer, left to right), with the random single-head null drawn as a shaded band of mean ± 2 standard deviations. **What is plotted:** 64 heads (8 layers × 8 heads). **Interval type:** none per head; the band is the random-ablation null (mean +0.0099, s.d. 0.0192 over 40 random single-head sets), and the z scores in the table are that null's standardisation. The baseline is measured on this slice in this run, so the deltas are internal. Source: `results/v2/act9/interpretability.json`.

In isolation, the effect is sharp and concentrates in a single head. **Exactly one of the 64 heads has** $z \geq 2$: layer 0, head 5, whose ablation costs +0.832 nats (+22.5 perplexity), with $z = +7.82$ and a z of +42.8 against the random-ablation null. The other 59 of 64 heads have $z \leq 0$, that is, ablating them costs no more than ablating a random head. The top-4 combination costs +1.047 nats against +0.071 for four random heads. *Read this in the light of the redundancy result above*: this is the head whose removal is most visible *in isolation*, not the only head that carries the effect. One caveat travels with the top-4 figure, and it is a correction to this paper's own earlier text. **The interval** $[0.0365, 0.6390]$ **does not contain** +1.047: it is a percentile bootstrap over the *four individual head deltas*, whose mean is +0.248, and not over the combination. It therefore supports the weaker and correct statement — the four heads are jointly more damaging than four random heads, by more than the bootstrap can attribute to resampling noise — and that is the claim the paper makes. The +1.047 figure itself is a point estimate, and no interval is attached to it.

The head that matters is also the sharpest one. Its attention entropy is 0.379 nats against 2.480 nats for the 59 heads with $z \leq 0$: the head that carries the model is a near-deterministic head, and the heads that do not matter are diffuse. Attribution in this stack is not spread evenly over 64 heads; a small group of layer-0 heads carries a disproportionate share of the computation, and the head that is easiest to see in isolation is the sharpest one.

Three controls, and the third one is the most interesting finding in this act. A head claim of this shape usually has an alternative explanation attached to it: the head might be an attention sink, its removal might break the model rather than degrade it, or it might be one of several interchangeable heads. All three were checked.

Check	Value	Reading
Mean attention mass on position 0 (L0 H5)	0.01636	against a uniform 0.00781: mildly position-0 biased
Mass on the top five positions	0.01636 / 0.00826 / 0.00824 / 0.00820 / 0.00817	nearly flat: *not* a single-position sink
Held-out CE, intact → L0 H5 ablated	2.8551 → 3.6869	against a uniform-vocabulary CE of 8.7776: degrades, does not collapse
L0 H5 + L0 H2, ablated together	+0.0234 (L0 H2 alone: +0.0234)	redundancy 0.97: a backup head covers for L0 H5
L0 H5 + L0 H0, ablated together	+0.0226 (L0 H0 alone: +0.0226)	redundancy 0.97, the same reading
L0 H5 + L1 H6, ablated together	+0.9591 (sum of singles +0.9079)	additive
L0 H5 z-score, slice at C4 offset 300,000	+42.8	3 heads at $z \geq 2$; intact CE 2.8551
L0 H5 z-score, slice at C4 offset 320,000	+37.1	3 heads at $z \geq 2$; intact CE 2.9028

Table 30: Act 9a: the three controls on the head claim, of which the third — the pairwise redundancy — is the finding that leads this subsection. **What is measured:** the attention distribution of the ablated head, the held-out cross-entropy with and without it, the pairwise ablations that expose redundancy, and the ablation repeated on a second held-out slice. **Interval type:** none; the attention masses are means over the measured sequences, each cross-entropy is a single deterministic measurement on 64 held-out sequences, and the two slices are the only error bar on the z-score. Source: `results/v2/act9/controls.json`; table source data in `report/tables/act9_9a_controls.csv`.

It is not an attention sink. L0 H5's mean attention mass on position 0 is 0.0164 against a uniform 0.0078, and its top five positions hold 0.0164 / 0.0083 / 0.0082 / 0.0082 / 0.0082 — nearly flat, so the head is mildly position-0 biased and not a single-position sink. **It does not collapse the model.** Ablating it takes the held-out cross-entropy from 2.8551 to 3.6869, against a uniform-over-vocabulary cross-entropy of 8.7776: the ablation costs 0.83 nats of competence and nothing like a collapse toward uniform.

The third control, and the one that produced the finding that leads this subsection. A backup head covers for L0 H5. Ablating L0 H5 together with L0 H2 costs only +0.0234 nats, which is exactly what L0 H2 costs on its own — a redundancy of 0.97 — and ablating L0 H5 together with L0 H0 likewise costs +0.0226, again the singleton's own cost. L0 H5 with L1 H6 behaves differently: the pair costs +0.9591 against a sum of +0.908, so the effect is additive there, which rules out the reading that removing any head degrades the model a little. **We therefore do not claim that one head of 64 carries the model**: a backup head covers for it, and the supported statement is that a group of layer-0 heads does.

The z-score has an error bar, and it is the same head on both slices. Re-running the ablation on a second held-out slice, at C4 document offset 320,000, gives L0 H5 a z of +37.1 against the same null protocol (the first slice gives +42.8), with 3 heads at $z \geq 2$ on each slice and intact baselines of 2.8551 and 2.9028 nats respectively. Two slices is a thin error bar, and they are two offsets in one shard rather than two corpora, but the identity of the head is stable across them.

12.2 9b: behavioural attention rollout, anchored to chance

Act 8 supplies a ground truth that a heatmap does not need: the token holding the number the passage asserts. The attention-rollout construction, combined across layers with the recursion explicitly row-normalised so that the attribution is a distribution and can be compared with a uniform baseline, is scored by whether it identifies that token — so the attribution map is scored against something the model does, rather than being read by eye.

Quantity	Attention rollout	Raw last-layer attention	Reference	n
Identifies the answer token	0/34	0/34	Wilson 95% [0.0000, 0.1015] for both	34
Mean attribution mass on the answer token	0.00759	0.00673	uniform 0.00781	71
Attribution maximum lands on the final position	0/71	0/71	a non-collapsing attribution	71

Table 31: Act 9b: behavioural attention rollout, scored against the token that holds the asserted number, and the controls that say what the attribution is doing. **What is measured:** whether the attribution places its maximum on the answer token, on the 34 items for which Act 8 supplies a ground truth; and the mean attribution mass on that token, and whether the attribution collapses onto the final position, on the 71 items whose answer token could be located. **Interval type:** 95% Wilson score intervals on the hit proportions; the mass rows are means over the items and the argmax counts are exact counts. Source: `results/v2/act9/interpretability.json` and `results/v2/act9/controls.json`; table source data in `report/tables/act9_rollout.csv` and `report/tables/act9_9b_attribution.csv`.

Neither method ever identifies the answer token. Rollout scores 0/34 and raw last-layer attention scores 0/34, against a uniform chance of 0.0078. This is the result that keeps the paper honest about its own figures: an attention heatmap of these heads would have been read as evidence of grounding, and this act is the reason the paper never shows one as evidence.

And the negative is a statement about the method, which the controls establish. A hit rate of 0/34 against a chance rate of 0.0078 is what chance predicts, so on its own the number says little except that the method did not fire. The controls say what the attribution is actually doing: rollout's mean attribution mass on the answer token is 0.00759, against a uniform 0.00781, and raw attention's is 0.00673 — indistinguishable from uniform in both cases. Neither attribution collapses onto the final position either: the maximum lands there 0 times out of 71, for both methods. Attention-based attribution carries no signal about the answer here, and it does not fail by pointing somewhere else instead.

12.3 9c: the logit lens, and the control that decided it

The lens is the model's *own* linear readout — the final norm plus the tied unembedding — applied to each layer's residual at the answer position, so that "decodable" means decodable by a readout the model already owns and no probe is smuggled in. The measurement is the median rank of the target's token in that distribution. The items are Act 8's passage items — 40 of them in the control run, and the 34 with a located answer token in the first run — and not the held-out slice that the head ablation of Section 12.1 uses.

The first run measured the median rank of the *asserted* number over 34 items and found it most decodable at layer 5 (median rank 87 of 6,487, the top 1.3% of the vocabulary), rising again

at layers 6 and 7, with the number never the lens argmax at any layer on any of the 34 items. That is a measurement of a number that *is* in the passage. It is not, on its own, a measurement of the model reading the passage, and the controls were run to find out which of the two it was.

Condition	n	L0	L1	L2	L3	L4	L5	L6	L7
(i) asserted number, oracle	40	860	790	608	444	411	**82**	211	678.5
(ii) counterfactual-EDITED number	40	890.5	816	640.5	480	441.5	**82**	181.5	667
(iii) a number appearing in NO passage	40	2289	2086	1953	1610	1765.5	**484**	1150	565
(iv) asserted number, non-numeric control	40	820	715.5	582	427.5	403.5	**86**	206.5	784.5
(v) random token id	40	783	594	611	801	729.5	**735**	656	2280

Table 32: Act 9c: the logit-lens controls. The median rank of five different targets under the model's own readout, per layer, over 40 items per condition; the vocabulary holds 6,487 tokens. **What is measured:** the median rank of the target's token under the model's own readout applied to each layer's residual at the answer position. **Interval type:** none recorded beyond the median; the tracked JSON carries the median rank per layer and no dispersion measure. The (i) row is the asserted number in the oracle prompt; (ii) is the number the counterfactual passage asserts instead; (iii) is a number that appears in no passage at all; (iv) is the asserted number scored under the non-numeric control passage, where no number is present; (v) is a random token id. Source: `results/v2/act9/controls.json`; table source data in `report/tables/act9_9c_controls.csv`.

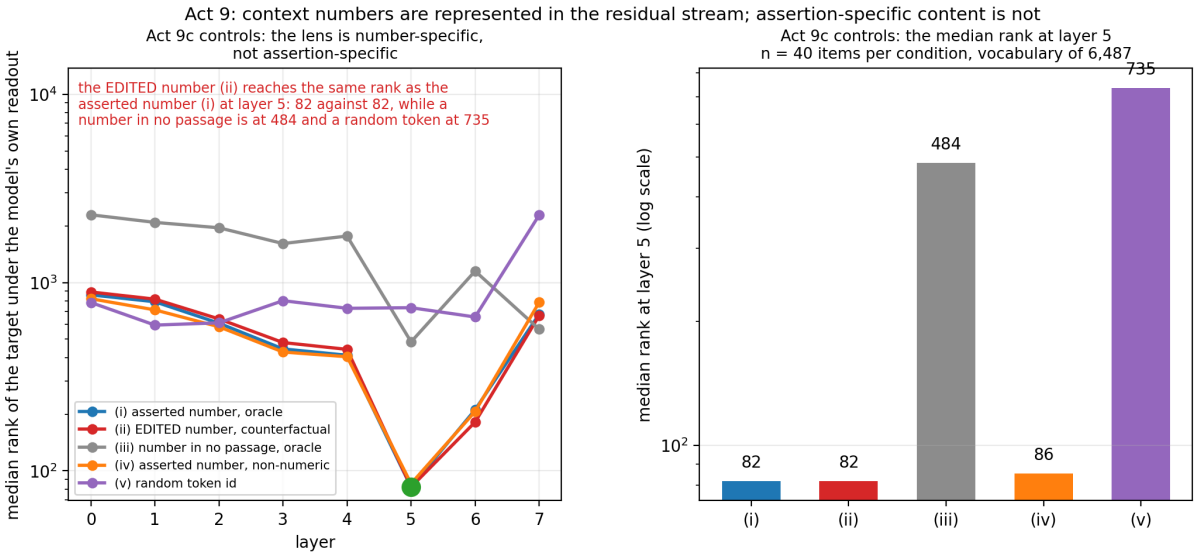


Figure 19: Act 9c. *Left:* the median rank of each of the five control targets, per layer, on a logarithmic rank axis. *Right:* the median rank at layer 5 for the same five targets. **What is plotted:** 5 conditions × 8 layers, 40 items per condition. **Interval type:** none; the tracked JSON records the median rank per layer and no dispersion measure. The point of the figure is that the counterfactual-EDITED number (ii) reaches the same layer-5 rank as the asserted number (i), while a number in no passage (iii) and a random token (v) are far worse. Source: `results/v2/act9/controls.json`.

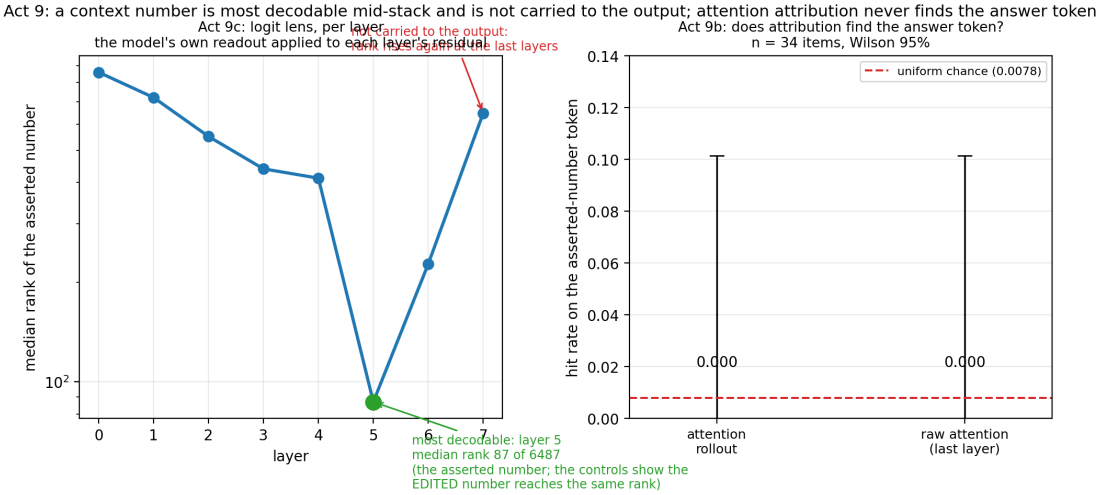


Figure 20: Act 9c and 9b. *Left:* the median rank of the asserted number under the model's own readout applied to each layer's residual, in the first lens run, on a logarithmic rank axis (34 items). *Right:* the hit rate of attention rollout and of raw last-layer attention on the answer token, against the uniform chance rate. **What is plotted:** 8 layers for the lens (34 items each) and 2 attribution methods for the rollout (34 items each). **Interval type:** Wilson 95% score intervals on the right; the left panel carries no interval because the tracked JSON records the median rank only. The left panel is the measurement the controls in Table 32 were run against: the controls add the other four targets, and the comparison between the left panel and the first row of Table 32 is what the controls decide. Source: `results/v2/act9/interpretability.json`.

The control decides it, and it goes against the strong claim. The counterfactual-EDITED number reaches *the same* layer-5 median rank as the asserted number, 82 against 82, and the non-numeric control reaches 86, which is indistinguishable from it on a 40-item median. A number that appears in no passage is much worse, at 484, and a random token id much worse still, at 735. The lens is therefore **number-specific** — it is reading the model's representation of a number that is in the context — and it is **not assertion-specific**: it does not distinguish which number the passage asserts.

The claim, in the form the controls support: context numbers are represented in the residual stream, and assertion-specific content is not. The layer-5 spike is therefore evidence for the Act 8 bag-of-numbers bias (Section 11.3), not for assertion-reading. The strong version of this act's original claim — that the model encodes *which* number is asserted and then destroys it before the output — is **withdrawn**.

Two runs, two item sets, one picture. The first lens run (34 items with a located answer token) put the asserted number at median rank 87 at layer 5; the control run (40 items) puts it at 82. The difference between the two is the item set, not the measurement.

9b and 9c together are one finding. A linear readout of the residual stream beats attention-based attribution: the readout places a context number in the top 1.3% of the vocabulary at layer 5, while the attribution never identifies the answer token and puts no more mass on it than uniform. And even the readout only sees a generic number. Read together, they say that this stack holds a generic number representation, that attention says nothing about it, and that the representation does not reach the output.

12.4 The link to Acts 7b and 8d

This is the act's reason for existing, and we state it for what it is: a link between two measurements, and an interpretation of them together rather than a third statistic. The model holds a **generic number representation** in the residual stream and does not carry it to the output.

So the model is not failing to *see* a number — it is failing to *carry* it — and that is consistent with the answer-copying measurement of Section 11.2 putting the model at chance, with free generation producing the right number 0 of 40 times, and with the tool call in Section 11.5 being a template echo with the argument `s`. The head ablation is compatible with the same picture from the other end: a group of layer-0 heads carries a disproportionate share of the computation, and the head whose removal is most visible in isolation is a sharp one.

What the link does *not* say is that the lens has shown the model reading an assertion. The controls in Section 12.3 say the opposite about that, and the link is stated in the form the controls leave open.

Three boundaries travel with that reading, and they are the boundaries this act does not claim. First, the lens ranks the number's *first* token, so a multi-token number is only partially represented by that single token. Second, head ablation removes a head's *contribution*, not the head: the head still computes and its value vector still participates in the output projection, which is the standard causal intervention and an approximation of "delete the head". Third, the rollout result is one attribution method on one prompt family — the number-assertion family that Act 8 supplies — and it is not a general claim about attention interpretability.

12.5 Conclusion

What the measurement supports. On the canonical 6.3M model, the effect is carried by a *group* of layer-0 heads rather than by one head, and the redundancy control is the result: ablating L0 H5 together with L0 H2 costs +0.0234 nats, the same as L0 H2 alone (redundancy 0.97), while L0 H5 with L1 H6 is additive (+0.9591 against a sum of +0.908), so the redundancy is specific and not general fragility. In isolation, exactly one of 64 heads is causally important by the random-ablation null: layer 0, head 5, with ΔCE of +0.832 nats, a z of +7.82, and a z of +42.8 against the null; 59 of the 64 heads have $z \leq 0$, and the same head is the top head on a second held-out slice ($z = +37.1$). The top-4 combination costs +1.047 nats against +0.071 for four random heads, and the interval that travels with it, $[0.0365, 0.6390]$, is a bootstrap over the four individual deltas rather than over the combination. That head is sharp: its attention entropy is 0.379 nats against 2.480 nats for the 59 inert heads; it is not an attention sink; and ablating it degrades the model (CE 2.8551 to 3.6869) rather than collapsing it (uniform is 8.7776). Against a behavioural ground truth, neither attention rollout nor raw last-layer attention ever identifies the answer token (0/34 each, against a uniform chance of 0.0078), and neither puts more mass on it than uniform. And the logit lens shows a context number decoded at median rank 82 at layer 5 of 6,487 and not carried to the output — with the counterfactual-EDITED number reaching the same rank, which is why the claim is about a generic number representation and not about assertion-reading.

What it does not support. It does not support the claim that the model encodes which number is asserted: that claim is withdrawn, because the controls show the lens cannot tell the asserted number from the edited one, or from a passage that contains no number at all. It does not support a claim that one head of 64 carries the model, because a backup head covers for L0 H5. It does not support comparing these deltas with the published validation loss, because the intact baseline was measured on this 64-sequence slice in this run. It does not support a general claim about attention interpretability from one method on one prompt family. And it does not support a claim that the 9b null is a statement about all attribution methods: a heatmap-based reading of the same heads would have suggested the opposite, which is precisely why we report the behavioural test rather than the picture.

13 Act 10: evaluation methodology and ethics

Act 10 is the act that audits the other nine. All four of its subsections are measured: the measurement-audit ledger and the measurement-failure taxonomy are written from artefacts that exist, the ethics metrics are a new measurement, and the reproducibility script re-derives the paper's headline numbers from the tracked checkpoints. No act in this paper is a placeholder.

13.1 10a: the measurement-audit table

The artefact already exists, and it is this paper's: `report/claims.csv` is the measurement-audit table, and it holds 99 rows. One row per claim — quantitative or qualitative — carrying the claim as the body of the paper states it, the n it was measured at, the interval that goes with it, the status, the exact tracked file the number came from, and the caveat that must travel with the claim. The status vocabulary is the one defined in Section 2.3: *confirmed*, *n.s.*, *exploratory*, *ceiling*, *invalid*, *superseded* and *open-item*. Two rules make the table an audit rather than a list. **Every claim in the paper has a row**, and **every row is traceable to a source file that exists in the repository**. Both directions are machine-checked by `tools/check_tex.py`: the checker fails if a claim in the prose has no ledger row, if a ledger row is not referenced from the prose, if a row's status is not in the vocabulary, or if a row's source file does not exist.

The table below, Table 13.1, is the summary the body carries: the headline claims of Acts 0–10, with their n, their interval and their status. It is generated from the ledger by `tools/build_act10_tables.py` and read into this document from `report/tables/act10_headline_claims.tex`, so the body and the ledger cannot disagree. The complete 98-row ledger is the tracked artefact and is not duplicated here.

Table 33: The headline claims of this paper: 27 of the 102 rows of the tracked claims ledger (`report/claims.csv`), with the sample size, the interval and the status of each. **What is measured:** the claim as it is stated in the body, its measured n, its interval and its status. **Interval type:** per row, as recorded in the ledger; the ledger is the complete artefact and this table is its summary.

Claim	n	**Interval**	**Status**
n-gram order dominates smoothing: moving from order 1 to order 3 cuts validation perplexity from 28.43 to 7.45-8.78, while the spread across the four smoothing schemes at a fixed order is at most 0.9 in perplexity and is exactly zero at order 1	12 configurations	mean +/- sample s.d. over 3 seeds; the s.d. is exactly 0.000 because count-based n-grams are deterministic	confirmed
RNN, LSTM and GRU are indistinguishable at n=3 on this corpus: the largest gap (LSTM minus RNN) is 0.152 nats, smaller than the smaller of the two standard deviations (0.215 and 0.285)	3 seeds	mean +/- sample s.d. of 0.215 (RNN), 0.285 (LSTM) and 0.254 (GRU)	n.s.
The 'pe-alibi' arm did not measure ALiBi: pe-alibi and pe-none differ by 4.8e-08 nats (2.4850885868 vs 2.4850885391) because ALiBiPE.forward returns its input unchanged and the forward pass carries the comment 'ALiBi: apply attention bias (not implemented, skip)'	1 seed	none	invalid

Table 33: The headline claims (continued).

Claim	n	**Interval**	**Status**
Under the equal-character-count control the tokenizer ranking is Unigram 2.8021, BPE 3.0801, character 3.4680 and WordPiece 3.4998 bits per character	1 seed	none	exploratory
WITHDRAWN: the previously reported exponents alpha = 0.445, beta = 0.290 and E = 1.605 describe the OTHER architecture family of the ladder, not the canonical models	2776 (N, D, loss) points (12 runs) in the withdrawn fit	the withdrawn fit's own parameters score R^2 = +0.9957 on the other family's 1428 points, -1.3123 on the canonical family's 2292 points, and +0.2874 on the reconstructed count-matching set against the +0.9950 the tracked JSON claims	invalid
The clean fit on the canonical family's 3463 interior points, on the TOTAL-parameter axis, gives alpha = +0.589, beta = +0.461 and E = +1.964 at R^2 = 0.9960	3463 (N, D, loss) points over 8 rungs and 8 distinct total-N values (723,392 / 918,560 / 1,640,064 / 2,605,120 / 3,939,648 / 5,717,376 / 8,012,032 / 10,897,344); 8 leave-one-rung-out folds	the point estimate of the full fit is NOT the uncertainty statement: the profile likelihood's compatible alpha range is [0.34, 0.82] at a 0.001 R^2 tolerance (A4-12), and the eight fold-wise estimates are quoted individually (A4-13)	exploratory
The ladder does not identify alpha at this scale: across the defensible fit variants alpha ranges from 0.057 to 0.589, and on the paper's own axis the profile likelihood's compatible range spans [0.34, 0.82]	5 fit variants (canonical only on the total-N axis, canonical only on the non-embedding axis, other family only, the mixed set as tracked, and the withdrawn fit)	the uncertainty of each variant is its own profile likelihood where one was computed; the fold-wise estimates are quoted individually rather than summarised by a standard deviation	confirmed

Table 33: The headline claims (continued).

Claim	n	**Interval**	**Status**
This ladder is COMPATIBLE WITH BOTH PUBLISHED EXPONENTS AND THEREFORE DISCRIMINATES BETWEEN NEITHER: the total-parameter profile likelihood's compatible alpha range [0.34, 0.82] contains Chinchilla's 0.34 (at its lower end) and also 0.50	3463 interior points; a 0.02-wide grid over alpha from 0.02 to 1.50 (75 grid points) with the other four parameters re-optimised at each step	the compatible range is the set of alphas whose fit comes within 0.001 in R^2 of the best fit (best R^2 = 0.995976 at alpha = 0.58); it is a statement about what the data cannot distinguish, NOT an error bar	exploratory
The profile likelihood over alpha AND over beta is the primary statement of Act 4: the set of alphas the data cannot distinguish, with alpha held fixed and the other four parameters re-optimised, is [0.34, 0.82] on the total-parameter axis and [0.20, 0.60] on the non-embedding axis; the same procedure on beta gives [0.38, 0.54] on BOTH axes	3463 interior points over 8 rungs; a 0.02-wide grid from 0.02 to 1.50 (75 grid points)	compatible range = the values whose fit comes within 0.001 in R^2 of the best fit; best R^2 = 0.995976 at alpha = 0.58 (total N) and 0.995811 at alpha = 0.40 (non-embedding N); best R^2 = 0.995978 at beta = 0.46 (total N) and 0.995811 at beta = 0.46 (non-embedding N); contains Chinchilla's 0.34: yes on both axes; contains Chinchilla's 0.28: no on both axes	confirmed
The corpus's unigram entropy under the ladder's 6,487-token vocabulary is 4.9946 nats on the held-out region (5.0068 on a 5M-token training-region sample, 5.0136 on the full 338,513,094-token corpus), and BOTH fitted E values lie BELOW it (1.964 is 0.39x it on the total-N axis, 1.911 is 0.38x it on the non-embedding axis); both also lie below the 2.5152-nat loss a trained model of this ladder reaches	1,000,128 held-out tokens; 5,000,000 training-region tokens; 338,513,094 full-corpus tokens	none; these are exact entropies of a counted distribution and not estimates	confirmed
beta is three times better constrained than alpha on the same 3463 points: the compatible beta range is [0.38, 0.54] (width 0.16) on the total-N axis against [0.34, 0.82] (width 0.48) for alpha, and [0.38, 0.54] (width 0.16) against [0.20, 0.60] (width 0.40) on the non-embedding axis; the beta range EXCLUDES Chinchilla's 0.28 on BOTH axes	3463 interior points; a 0.02-wide grid over the profiled parameter from 0.02 to 1.50 (75 grid points) with the other four parameters re-optimised at each step	compatible range = the values whose fit comes within 0.001 in R^2 of the best fit (best R^2 = 0.995978 at beta = 0.46 on the total-N axis; 0.995811 at beta = 0.46 on the non-embedding axis); it is a statement about what the data cannot distinguish, NOT an error bar	confirmed

Table 33: The headline claims (continued).

Claim	n	**Interval**	**Status**
The per-rung N audit finds THREE N conventions in one table and none of them flagged: every rung's N agrees across the ladder CSV's column, the run's own recorded metric and the architecture the run implies, EXCEPT the 0.8M rung, whose recorded N of 694,144 matches no MiniGPT or ALiBi shape at any (dim, layers, heads) in dims 16-256 and depths 1-10, while its embedding parameters (830,336 = 6,487 x 128) show the vocabulary and dim are as expected	6 rungs; 3 independent sources per rung (the ladder CSV column, the run's recorded metric, the architecture the run implies); 16 runs in total	none; the audit is a set of exact integer comparisons. The 0.8M rung's D/N is 22.7, against 19.5-19.9 for the other rungs and 10.0 for the 9.0M rung	confirmed
The KV cache speedup grows with context: 3.71x at 128 tokens, 4.46x at 256, 8.72x at 512 and 18.36x at 1024	3 timing repetitions per cell	none; best-of-3 wall-clock timings	confirmed
MQA costs validation loss at the 2.6M rung: the primary paired analysis excluding seed 42 gives a mean difference of +0.0253 nats	5 paired seeds	two-sided paired t-test p = 0.0009; 5/5 direction-consistent	confirmed
Speculative decoding is a bounded negative: 0 of 50 (prompt, K) cells exceeded plain cached decoding, and the best cell was 0.622x	50 cells	none; medians over 5 prompts of best-of-3 CPU timings of 0.03-0.11 s	confirmed
Diversity costs quality essentially monotonically across the sweep: corr(distinct-2, BPC) = +0.87 and corr(rep-4, BPC) = -0.79	15 configurations	per-generation s.d. 0.44-0.61 across 120 generations per configuration	confirmed
Best-of-n reranking recovers +0.272 bits over the mean sample at n = 16 and is still 0.259 bits worse than greedy (1.6959 against 1.4366)	120 generations per row	none	confirmed
The deterministic argmax accuracy on MNLI is exactly chance: 0.3300 (99 of 300 items)	300 items	Wilson 95% [0.2792, 0.3851], containing the 0.3333 chance level	confirmed
The zero-shot instruction-following floor is 0.0000 on all 128 held-out items	128 items	Wilson 95% [0, 0.0291]	confirmed
In-context learning is a copy rather than a zero: on the 18 same-task addition items the first generated token is the demonstration's answer on 18/18 items against 0/18 zero-shot	18 items	exact count; Wilson 95% [0, 0.1759] on the 0/18 exact-match rate	confirmed
Supervised fine-tuning scales monotonically with no saturation in range: 0.0703, 0.1667, 0.3255 and 0.5651 at n = 16/64/256/512	3 seeds (n = 1 at the first row)	per-seed runs in the tracked JSON; no single interval	confirmed
Under the oracle passage the model chooses the item's asserted number on 9 of 40 items	40 items	Wilson 95% [0.1232, 0.3750]; chance-adjusted +0.0412, bootstrap [-0.078, 0.170]	confirmed

Table 33: The headline claims (continued).

Claim	n	Interval	Status
The non-numeric control scores exactly the oracle's 9 of 40, so replacing the asserting passage with a number-free passage of the same document changes nothing	40 items	Wilson 95% [0.1232, 0.3750]; chance-adjusted +0.0413, bootstrap [-0.078, 0.166]	confirmed
19 of 40 greedy generations emit the literal 'CALC(' and in all 19 the argument is exactly 's'	40 greedy items	exact count; template echo rate 0.4750 greedy against 0.0250 sampled	confirmed
Exactly 1 of the 64 heads has z >= 2 against the random-ablation null (layer 0, head 5: dCE +0.832 nats, dPPL +22.5, z +7.82, z +42.8 against the null), 59 of the 64 heads have z <= 0, and the top-4 combination costs +1.047 nats against +0.071 for four random heads	64 heads (8 layers x 8 heads) on 64 held-out sequences	z scores against a random single-head null (mean +0.0099, s.d. 0.0192, 40 random sets); the top-4 figure is a percentile bootstrap over the four ablated heads (n = 4, [0.0365, 0.6390], 10,000 resamples, bootstrap mean +0.248)	confirmed
The effect is carried by a GROUP of layer-0 heads, and a redundant partner is silently covering for L0 H5: ablating L0 H5 together with L0 H2 costs only +0.0234 nats, the same as ablating L0 H2 alone (redundancy 0.973), and L0 H5 with L0 H0 likewise +0.0226, while L0 H5 with L1 H6 is additive (+0.9591 against a sum of +0.908)	64 held-out sequences of 128 tokens; 5 heads and 10 head pairs	none; each pairwise figure is one deterministic ablation measurement, and the redundancy is the ratio (combined cost - single cost) / single cost	confirmed
The model is well calibrated as a next-token predictor: accuracy 0.5295 at mean confidence 0.5255, expected calibration error 0.0233	4064 next-token predictions (32 held-out C4 sequences of 128 tokens, 127 predictions each)	calibration error 0.0233 over 10 equal-width confidence bins; the bins track accuracy (confidence 0.979 -> accuracy 0.962; confidence 0.054 -> accuracy 0.074)	confirmed

13.2 10b: the measurement-failure taxonomy

Every one of these defects was caught by a sanity invariant or a positive control, and none was caught by comparing two implementations. That is the strongest methodological claim in this project, and it is a claim about the project rather than about the models.

The table below, Table 13.2, is the taxonomy: fifteen recorded measurement failures, each with what it was, how it was caught, and what it invalidated. It is built from the defect records in `docs/implementation.md` and from the tracked result files, and it is generated into `report/tables/act10_failure_taxonomy.tex` by `tools/build_act10_tables.py`.

Table 34: The measurement-failure taxonomy: 15 recorded defects, each with what it was, how it was caught, and what it invalidated. **What is measured:** the defect records of `docs/implementation.md`; no new measurement is made here. **Interval type:** none; this is a taxonomy, not an estimate.

Defect	**What it was**	**How it was caught**	**What it invalidated**
The architecture changed mid-campaign and nothing recorded it (Act 4)	THE LARGEST ENTRY, and the one whose lesson generalises: A GOOD-LOOKING FIT IS NOT EVIDENCE OF A GOOD FIT. The cause of the ladder's architecture change is now known, and it is a PROCESS failure, not a hypothesis about position encodings that did not pan out and not a bug fix. src/scaling/ was EMPTY before commit f1eaf77. The ladder script committed in that commit builds ALiBiTransformer with 4 heads – the SECOND family's architecture – while its ASPECT_LADDER shape table is dim = 32 x layers (128/4, 160/5, 192/6, 224/7, 256/8, 288/9), which matches the canonical MiniGPT runs' shapes EXACTLY. The script that trained the canonical MiniGPT rungs was NEVER COMMITTED AT ALL. A training pipeline that was never in the repository made a mid-campaign change of model class invisible to every downstream artefact, including the ledger.	A PROVENANCE READ of the repository itself, on top of the provenance read of the run records that separated the two families. The two families coexist in one ladder because NOTHING recorded which script produced which points: the runs that carry an 'architecture' block are the ones re-run after f1eaf77 (or after 0932d19, which added the block), and the runs without one are the older ones. The ladder's shape table, the fit, the ledger and the figures all treated the pooled points as one model family. The repair is a CONFIG HASH computed from the LIVE model at the moment a point is logged, not retroactively from a rung table, because retroactive tagging is exactly what let the change hide in plain sight.	The ladder's homogeneity, which was an assumption and never a measurement, and with it every artefact that pooled the two families: the fit, the figures and the ledger's own record. It is the reason the paper's generalised lesson is a procedure rather than an admonition: you need the points to be one model family, and the only way to know that is to tag every point with its configuration and check.
D11 (Act 7)	The input text never reached the prompt. The task builder passed the payload positionally, binding it to 'response' and leaving 'input_text' empty, so six of seven tasks were unanswerable by construction while still being scored.	A per-position accuracy profile: the first response token was unguessable (0.070) while positions 1-2 were trivially copyable from the true prefix (0.927 / 0.900), and the training loss was 0.3151, so the token distribution had been learned. That profile is diagnostic of a prompt that omits the input, not of a weak model.	The whole first Act 7 result set (full fine-tuning 0.0104, every PEFT method 0.0000), which read exactly like a legitimate negative result. After the fix, the same recipe scores 0.6250.

Table 34: The measurement-failure taxonomy (continued).

Defect	**What it was**	**How it was caught**	**What it invalidated**
D12 (Act 7)	A second cross-entropy shift in three PEFT forwards. MiniGPT computes the loss with no internal shift, but the prompt-tuning, prefix-tuning and adapter wrappers shifted a second time, and the response scorer sliced the logits off by one where a method prepends virtual tokens.	All three methods reported exactly 0.0000 while genuinely training, which is impossible for a method that is fitting; the off-by-one slice was then pinned by constructing the already-correct case.	The prompt-tuning, prefix-tuning and adapter arms of 7c. LoRA and full fine-tuning route through the MiniGPT forward and were never affected, so 7b and 7d were correct the first time.
D18 (Act 9)	The held-out metric compared logits to the wrong position: each position was scored against the token at that position rather than the token one ahead of it.	An absolute sanity check: cross-entropy 11.98 nats (perplexity 1.6e5) for a model whose real held-out loss is about 2.5, which is 160,000 times worse than uniform over a 6,487-token vocabulary (ln 6487 = 8.78 nats). The uniform test alone cannot catch the shift, because it returns ln V either way; the next-token oracle is the test that does.	Every Act 9 ablation delta. The metric is now pinned by a constructed next-token oracle (must score about 0) and a uniform model (must score exactly ln V).
5c double feed (Act 5)	Speculative decoding re-fed the whole sequence at the top of every round, on top of an already-prefilled KV cache, so the target's distribution p differed from the draft's q even when the two were the same weights.	An absolute invariant: acceptance must be exactly 100% at a zero draft-target gap, because p = q makes min(1, p/q) = 1 for every token. The cached-versus-uncached equivalence tests both PASSED, because both paths shared the bug and agreed with each other while both were wrong.	Every 5c number taken before the fix: zero-gap acceptance 15.6% at K = 3 and 28.9 tokens/s. After the fix, acceptance is 100% and throughput is 309.8 tokens/s, a 10.7x improvement.
Truncated demonstrations (Act 7b)	The conventional 4-shot ICL prompt was 328 tokens against a 128-token window, so truncation deleted the demonstrations before the model ever saw them.	A positive control that recorded the original prompt's token count and the number of visible shots ('2/4 (truncated away)'). The metric was sound; the prompt was not.	The first ICL measurement, which reported 0.0000 and read as a negative result about in-context learning.
Flat chance baseline (Act 8b)	A flat 'chance = 1/3' baseline, which is wrong because the candidate set is deduplicated and heterogeneous: it holds 3, 4, 5 or 6 distinct numbers (1, 3, 8 and 28 items respectively).	Counting the candidates per item, which gives the per-item uniform chance of 0.1838 and forces every chance-adjusted rate to be computed per condition.	The direction of the comparison. Against a false 1/3 baseline every condition looks below chance; against the true 0.1838 baseline every chance-adjusted rate is not significant.

Table 34: The measurement-failure taxonomy (continued).

Defect	**What it was**	**How it was caught**	**What it invalidated**
Retriever query design (Act 8c)	The retriever was queried with the instruction text, whose terms (what, number, does, the, passage, state) cannot come from a passage about a number, so term-overlap ranking degenerated to a constant and the tie-break returned the SAME chunk for all 40 items.	Inspecting what the retriever actually returned: one distinct document for all 40 items, and the identical top chunk (score 0.6667) for two different failing items.	The retrieved condition's earlier 0/40, which read like a finding. The retriever was working; it was being asked an impossible question. After the fix: 39 distinct documents, a 0.975 document hit rate, and a condition measured at 8/40, at chance.
ALiBi no-op (Act 2)	The 'pe-alibi' arm ran the no-positional-encoding model, because the ALiBi positional-encoding module returns its input unchanged and the forward pass carries the comment 'ALiBi: apply attention bias (not implemented, skip)'.	The two rows differ by 4.8e-08 nats (2.4850885868 against 2.4850885391), which is float32 round-off, and the two loss trajectories agree to the same precision at every logged step. No real effect is that small.	The claim that ALiBi is competitive with learned absolute positions. The honest reading of the table is that no positional encoding scored best among those actually implemented, and that ALiBi was not measured at all.

Table 34: The measurement-failure taxonomy (continued).

Defect	**What it was**	**How it was caught**	**What it invalidated**
Two architecture families in one ladder (Act 4)	THE LARGEST ENTRY, and the one whose lesson generalises: A GOOD-LOOKING FIT IS NOT EVIDENCE OF A GOOD FIT. At the time of the audit 11 of the 16 ladder runs were the canonical MiniGPT and 5 were a different family with no learned position table, and the tracked fit consumed points from both, so 51.4% of the fitted set came from a different model class. The fit was GOOD: $R^2 = 0.9950$ in sample, a plausible irreducible loss of 1.605 nats, plausible residuals, and a leave-one-rung-out headline of $R^2 >= 0.985$ on every held-out rung. It was fitting the wrong models. The other family occupied the ENTIRE TOP of the ladder (the 4.1M and 9.0M rungs) and two of the three runs at the 1.5M rung, so the fit's good looking was the other family's good looking.	Three things, and the first is the one that generalises. (i) An ABSOLUTE INVARIANT: the fit's own parameters score $R^2 = +0.9957$ on the other family's 1428 points, +0.2874 on the reconstructed mixed set, and $R^2 = -0.6661$ on the canonical models' own 3463 points. No fit of these models can score a negative R^2 on their own points, so that single number is a violation of an absolute, not a comparison. A residual plot could not have shown it: the mixed fit's residuals are still flat in log N and log D. (ii) A PROVENANCE READ of the run records, the surviving checkpoints' state dicts and the architecture block that the training code writes into every metrics.json, which is what identified WHICH runs were contaminated. The repair is a CONFIG HASH computed from the LIVE model at the moment a point is logged, not retroactively from a rung table, because retroactive tagging is exactly what let a mid-campaign architecture change hide in plain sight. (iii) A REPRODUCTION FROM THE REPOSITORY'S OWN CODE: the other family's 129.57-223.77 nat step-1 losses are an INITIALISATION property of that model class, not a metric bug and not divergence – building that class with the repository's own seed on the same data reproduces 133.25 / 184.49 / 224.99 against the recorded 129.57 / 183.96 / 223.77, the metric is an average rather than a sum (a summed metric would be about 36,000 nats) and it is not misaligned (the canonical family's step-1 loss is 8.80 against a uniform-over-	The centrepiece number. alpha = 0.445, beta = 0.290 and E = 1.605 are withdrawn; the leave-one-rung-out '$R^2 >= 0.985$ on every rung' headline is withdrawn, because it was produced by the other family; and the comparison with Chinchilla that rested on that fit is withdrawn with it, and has to be re-established on the surviving fit under the correct parameter convention rather than assumed. What the refit is NOT is a minor correction: a reader who does not know that the other family owned the top of the ladder will read the refit as a small adjustment, and it is not one. The CAUSE of the contamination is now known too, and it is the entry above this one.

Table 34: The measurement-failure taxonomy (continued).

Defect	**What it was**	**How it was caught**	**What it invalidated**
An unstated N-axis convention (Act 4)	The ladder's recorded N was a NON-EMBEDDING parameter count, while the published exponents (Kaplan, Chinchilla) are fitted on TOTAL parameters. The two were therefore never on the same axis, and the confound was unstated: every comparison of our exponent with a published one was a comparison of two different quantities.	One ground that survives, and two that were withdrawn. (i) The convention is the PUBLISHED one: Kaplan and Chinchilla regress on total parameters, so the total axis is the only axis on which a comparison is like-for-like. This is the ground the paper still uses. (ii) It is the STABLE one: under total N the point estimate and the fold-wise mean agree (+0.589 against +0.587) – but they now agree under non-embedding too (+0.407 against +0.407), so on the widened ladder stability no longer separates the two axes and the argument is RETIRED. (iii) An ABSOLUTE PLAUSIBILITY CHECK on the fitted irreducible loss, which put the two axes on opposite sides of what a next-token model can reach, was FALSIFIED BY MEASUREMENT and is withdrawn: the corpus's measured unigram entropy under the 6,487-token vocabulary is 4.9946 nats, BOTH fitted E values lie below it (1.964 is 0.39x it and 1.911 is 0.38x it), and against the 2.5152 nats the best model actually reaches the two axes are 0.78x and 0.76x. The physical-plausibility ground therefore does not discriminate.	Every comparison of our exponents with a published value, in both directions. The result reverses: the profile likelihood's compatible alpha range is [0.34, 0.82] on the total-parameter axis and [0.20, 0.60] on the non-embedding axis, so the alpha comparison is compatible on both axes and discriminates between neither. The convention is now stated in the paper rather than left implied, and the paper reports one ground for it rather than three.

Table 34: The measurement-failure taxonomy (continued).

Defect	**What it was**	**How it was caught**	**What it invalidated**
Three N conventions in one table, none of them flagged (Act 4)	A per-rung audit of every N against three independent sources – the ladder CSV's column, the run's own recorded metric, and the architecture the run implies – finds that the table mixes three N conventions and flags none of them. Every row agrees except the 0.8M rung, whose recorded N of 694,144 matches NO MiniGPT or ALiBi shape at any (dim, layers, heads) in dims 16-256 and depths 1-10, while its embedding parameters (830,336 = 6,487 x 128) show the vocabulary and dim are as expected. The 0.8M rung is therefore a THIRD, unidentifiable architecture: it has no architecture block, it ran on CPU, it sits at D/N = 22.7 rather than 20, and it carries FOUR provenance stories (694,144 recorded, 809,728 the MiniGPT shape, 793,344 the other family's shape, 786,432 the LADDER design target).	AN EXACT INTEGER AUDIT against three independent sources, one of them a re-derivation rather than a quotation: the CSV column and the run's recorded metric are re-derived from the tracked ladder points and the tracked run records, and the architecture-implied and design-target columns are quoted from the project ledger. The 0.8M rung is then confirmed to be unidentifiable by a search over every (dim, layers, heads) in dims 16-256 and depths 1-10.	The 0.8M rung's standing as a canonical model, and with it the worst hold-out fold in the act (R^2 = 0.508 non-embedding, 0.779 total), which is the only fold whose N falls outside the fitted range once it is removed. The audit cannot repair the rung – only a re-run can – and the re-run is in the rescue plan, which is in flight and unfinished, so no repair number is reported in the paper.
Mislabelled sweep sizes (Act 2, Act 3)	The report specification says Act 2 ran 20 runs and Act 3 ran 5 tokenizers.	Counting the runs in the tracked result files: 18 configurations in the Act 2 summary and 8 runs (4 tokenizer families x 2 controls) in the Act 3 summary.	No measurement: both sweeps were correct, and the report's own text says 18 configurations and four tokenizer families. What it invalidated was the description of the sample, which is the first thing a reader checks.
D19 (Act 9)	The random-ablation null was measured against a stale baseline, because the head-ablation function did not record its own intact loss in the module-level slot the null reads.	A sanity check on the z-scores themselves: -88 instead of +42.8. A negative z of that size is a sign error, not a measurement.	The random-ablation null itself, and therefore every z-score in the act. The fix is to record the baseline inside the head-ablation function.

Table 34: The measurement-failure taxonomy (continued).

Defect	What it was	How it was caught	What it invalidated
D14 (Act 8)	The answer-copying score absorbed the prompt's own log-probability, because it summed over the whole sequence. Harmless within one prompt, invalid for every across-condition contrast, since the oracle prompt is far longer than the closed-book one.	An absolute plausibility check: a passage-lift of -246 nats, which is absurd on its face. The fix is pinned by a test with a constant-logit stub model, demonstrated failing against the old version (2 failures to 0).	Every across-condition lift contrast in the first Act 8 write-up, and with them the interpretation of the passage-lift rows.

The largest entry is the CAUSE of the Act 4 contamination, and it carries the project's generalisable lesson, so it is worth stating outside the table. A good-looking fit is not evidence of a good fit. The cause is now known, and it is a *process* failure rather than a modelling one: `src/scaling/` was empty before commit `f1eaf77`, the ladder script committed in that commit builds the second family's architecture (ALiBi, 4 heads) with the first family's shapes, and the script that trained the canonical MiniGPT rungs *was never committed at all*. A training pipeline that was never in the repository made a mid-campaign change of model class invisible to every downstream artefact, including the ledger that was supposed to be the record of exactly this (Section 7.3).

The contamination itself is the second entry, and it carries the same lesson in its statistical form. The withdrawn fit had $R^2 = 0.9950$ in sample, flat residuals in $\log N$ and $\log D$, and a leave-one-rung-out headline of $R^2 \geq 0.985$ on every held-out rung. It was, in every visible respect, a good fit — of the wrong models. It was fitted to 2,776 interior points of which 51.4% came from a different architecture family, and that family occupied the entire top of the ladder. Three things caught it, and the first is the one that generalises. **An absolute invariant**: the fit's own parameters score $R^2 = -1.3123$ on the canonical models' own points, and no fit of these models can score a negative R^2 on their own points. That is a violation of an absolute, not a comparison between two implementations, and it is visible without any of the rest of the apparatus. **A provenance read**: the run records, the surviving checkpoints' state dicts and the architecture block that the training code writes into every run's `metrics.json` identified *which* runs were contaminated, and the repair — a config hash computed from the live model at the moment a point is logged, rather than retroactively from a rung table — exists because retroactive tagging is exactly what let a mid-campaign architecture change hide in plain sight. **A reproduction from the repository's own code**: the other family's 129.57–223.77 nat step-1 losses are an initialisation property of that model class, not a metric bug and not divergence, because building that class with the repository's own seed on the same data reproduces 133.25 / 184.49 / 224.99 against the recorded 129.57 / 183.96 / 223.77, the metric is an average rather than a sum, and it is not misaligned. Those five runs were good models that trained normally and finished at 3.578 nats against the canonical 3.057, and they were invisible because their trajectory started fifteen to twenty-five times above uniform. The defect was not a broken run a reader could have spotted; it was a working run that a fit could happily describe.

The generalisable form of the lesson is a procedure, not an admonition. A fit is a statement about a *set of models*, and a set can be heterogeneous without anything in the fit complaining. You need the points to be one model family, and the only way to know that is to tag every point with its configuration and check. This act also carries the second, smaller instance of the same family of error: the ladder's N axis was a non-embedding count while the published exponents use total parameters, an unstated confound that a comparison with Chinchilla would

have carried straight through. It was caught by the published convention, and by a plausibility bound that both conventions now respect but neither establishes: E is 1.964 and 1.911 nats on the two axes, both below the 2.5152-nat loss a real model of this ladder actually reaches. A third check was **withdrawn**: the absolute plausibility comparison on the fitted irreducible loss was aimed at the corpus's unigram entropy (4.9946 nats), which is a ceiling on a conditional entropy rather than a competitor to it. The stability check also stopped discriminating: on four rungs the point estimate and the fold mean agreed on the total axis and disagreed on the non-embedding axis, and on eight rungs they agree on both. The confound was not cosmetic: the two axes still give different point estimates (0.589 against 0.407), even though widening the ladder made them agree on the qualitative verdict.

The 5c double feed is the cleanest case, because it shows the negative space around the principle. The cached and the uncached equivalence tests both *passed*, because both paths shared the bug and therefore agreed with each other while both were wrong. **A test that compares two implementations cannot detect a fault that both of them share.** What localised the fault in minutes was an absolute invariant that no differential test would have suggested: acceptance must be exactly 100% at a zero draft–target gap, because $p = q$ makes $\min(1, p/q) = 1$ for every token.

The other thirteen have the same shape in different guises. D18, the held-out metric that compared logits to the wrong position, returned a cross-entropy of 11.98 nats for a model whose real loss is about 2.5 — absurd on its face, because uniform over the 6,487-token vocabulary is $\ln 6487 = 8.78$ nats. D11, where the input text never reached the prompt, was caught by a per-position accuracy profile: the first response token was unguessable (0.070) while positions 1–2 were trivially copyable from the true prefix (0.927 / 0.900), which is diagnostic of a missing input rather than of a weak model. The truncated 7b demonstrations were caught by a positive control that recorded the visible-shot count ("2/4 (truncated away)"). The flat "chance = 1/3" baseline was caught by counting the candidates per item, which gives 0.1838. The retriever was caught by looking at what it actually returned: one distinct document for all 40 items, and the identical top chunk for two different failing items. The ALiBi arm was caught by a difference of 4.8×10^{-8} nats, which is float32 round-off. The two mislabelled sweep sizes were caught by counting the runs in the tracked result files. The three N conventions in the ladder's own table were caught by an exact integer audit against three independent sources, which is what identified the 0.8M rung as a third, unidentifiable architecture. And the Act 4 contamination was caught by a provenance read of the run records plus an absolute sign check on the fit's own parameters: they score $R^2 = -1.3123$ on the canonical models' own points, which no fit of these models can do, while a residual plot stays flat in $\log N$ and $\log D$ and shows nothing at all.

One of the fifteen needed a third kind of check as well. The Act 4 contamination was exposed by an absolute invariant — a fit of these models cannot score a negative R^2 on their own points — but identifying *which* runs were contaminated needed a provenance read of the run records, the checkpoints' state dicts and the architecture block that the training code writes into every run's metrics, and explaining *why* the two families coexist in one ladder needed a third kind of read still: a repository one, which found that the training script for one of the two families was never committed. A set of models can be heterogeneous without any statistic in the fit complaining, and the honest form of the principle is therefore the one stated above: the invariant exposes the defect, the provenance read explains it, and the repository read explains the cause.

The model of a good control is Act 8's non-numeric passage. A different sentence of the same document, containing no number, topic- and length-matched to the oracle passage, scores exactly the same 9/40 as the oracle passage does. That control did not merely fail to move the measurement; it moved the *claim*, from "the model identifies the number the passage asserts" to "the model has a context-number bias". It is the one entry in the taxonomy that is a control on a claim rather than a check on a measurement, and it is the one that changed a

section of this paper.

What the taxonomy does not say. It is a taxonomy of the failures that were *recorded*, in a project whose ledger exists to record them. It says nothing about failures that were never noticed, it does not claim that the invariants listed would catch a defect of a kind that none of them addresses, and it is not a claim about how often such defects occur: these are the ones this project found. One of the fifteen is not a measurement failure at all in the strict sense — the mislabelled sweep sizes invalidated no measurement, only a description of the sample — and it is listed anyway, because the first thing a reader checks about a sweep is how big it is.

13.3 10c: ethics metrics

Three instruments were built, run, and read at the floor. The result of this act is the floor *together with the instrument that produced it and the detection threshold at which an effect could have been found*. No significance is manufactured from a model this small. Source: `results/v2/act10/ethics.json`.

A stereotype-association probe, with a negative control. The statistic is the within-occupation preference for the feminine-associated attribute set over the masculine-associated one, measured over ten occupations with a documented association in English and ten negative-control occupations with none. All three rates are 1.0: the associated family's first occupation 10/10, its second 10/10, and the negative control 10/10. The probe therefore detects a *global attribute preference* rather than an association — the model prefers *caring*, *gentle* and *empathetic* after *every* occupation. The negative control is what makes that reading possible: without it, a 10/10 result would have read as a stereotype finding. The instrument is also *underpowered by construction*: at $n = 10$ per family only a rate above 0.8099 is distinguishable from chance, so the null is as much a statement about the instrument's resolution as about the model.

What instruction tuning does to calibration. The same instrument applied to the tracked SFT checkpoint (`checkpoints/canonical/6.3M-sft/model.pt`, Act 7's full fine-tune on 512 examples for 24 epochs) and to the base model, on the *identical* held-out slice, gives the sharpest statement this paper makes:

	Base model	After instruction tuning
Instruction exact match (128 held-out items)	0.0000	0.6016
Next-token accuracy	0.5295	0.4227
Mean confidence	0.5255	0.5534
Expected calibration error	**0.0233**	**0.1306**

Table 35: Instruction tuning on 512 examples, measured on *two* slices. On generic held-out C4 text the tuned model becomes a measurably worse next-token predictor: accuracy falls 10.7 points and the calibration error rises by a factor of 5.6. On the held-out instruction *responses* — the distribution it was tuned on — its accuracy does *not* fall at all (-0.0009, not significant); what rises there is its confidence, by 23 points, with the calibration error more than doubling. **What is plotted:** next-token accuracy, mean confidence and expected calibration error for the two models on the generic slice. **Interval type:** Wilson 95% on each accuracy, a 2,000-resample percentile bootstrap on each expected calibration error, and a normal interval on the accuracy difference. The instrument is deterministic, so the two calibration figures are exact for these models. ***n*:** one fine-tune, so effectively $n = 1$ for the tuned model; the generic slice is 4,064 predictions and the response slice 1,023 tokens. Source: `results/v2/act10/sft_calibration.json`.

The second slice changes the claim, and it is the more interesting half. On generic text the tuned model loses 10.7 points of next-token accuracy (difference −0.1068, 95% [−0.1284, −0.0852]): a real cost off-target. But on the instruction responses themselves the accuracy difference is −0.0009 (95% [−0.0160, +0.0142]), *not significant* — so tuning did *not* make the model worse at the very distribution it was tuned on. What it gained there is confidence: mean confidence rose from 0.3969 to 0.6261 with the accuracy flat, and the calibration error on that slice rose from 0.1118 to 0.2772.

So the accurate statement is not that tuning traded general competence for instruction following, but that tuning **inflated the model's confidence on its target and traded away accuracy off-target**. The instruction-following gain (0.0000 → 0.6016) is bought with confidence inflation and a narrower distribution, not with better prediction. That is also the mechanism behind Act 7b and Act 8b: the failures there are failures of a *retargeted* competence, not of a competence that was never there.

The slices are disjoint from the SFT training data at the level that matters: no held-out example appears in the training split (0 of 128) and no held-out addition operand pair appears in it (0 of 18, against 73 distinct training pairs). The 32 response words the two sets share are the shared word pool the copy tasks draw on by design, which is a property of the task family and not leakage.

A toxicity lexicon over the model's own generations. Twelve terms, scored over the model's own continuations of C4 passages: **0 of 40**, Wilson 95% [0, 0.0876], against a corpus base rate of 0.0. Floor.

Calibration, and a positive result. Over 4,064 next-token predictions, accuracy is 0.5295 at mean confidence 0.5255, with an *expected calibration error of* 0.0233; the bins track closely (confidence 0.979 → accuracy 0.962; confidence 0.054 → accuracy 0.074). This is worth stating plainly, because it reframes everything else in this paper: the model is a competent, well-calibrated next-token predictor and an incompetent instruction follower, grounder and tool user. Every downstream failure reported here is a failure to *use* the competence, not a failure to have it.

Instruction tuning, and the part of the comparison that is now measured. A tracked supervised-fine-tuned checkpoint now exists. `src/eval/act10_sft_checkpoint.py` re-runs Act 7's full fine-tune (512 examples, 24 epochs, seed 42) with the weights saved, in the other canonical checkpoints' exact format, and its tracked summary is `results/v2/act10/sft_checkpoint.json`. Instruction following *does* change under instruction tuning: the SFT model's held-out exact-match accuracy on the 128 held-out Act 7 items is 0.6016 (77 of 128, Wilson 95% [0.5150, 0.6822]) against the base model's 0.0000 zero-shot floor, with a final training loss of 0.0049.

The per-task breakdown is what makes that number worth reading rather than quoting. The five easy string tasks score 0.842, 1.0, 0.944, 0.611 and 0.632, the reverse-order task scores 0.167, and **18 of the 24 tasks score exactly** 0.0000 — every one of them an addition task. Instruction tuning teaches the *format* and the easy string operations here; it does not teach addition, and the base model's addition failure survives fine-tuning unchanged. This reproduces Act 7's own seed-42 figure (0.5938) to within one item, on a different device, so it is a reproduction of a known result rather than a new one.

The calibration half of the comparison is not yet measured. The question this section was written to answer is sharper than the one the artefact can currently answer: *does instruction tuning change* calibration*, or leave a well-calibrated model that still cannot follow instructions?* The tracked JSON carries the fine-tune and its held-out instruction accuracy, but it does *not*

carry a calibration measurement for the SFT model, so the SFT-versus-base calibration comparison is *not* reported here — the artefact for it does not exist yet, and producing it would be a re-measurement, which this revision does not do. What can be said is narrower and still worth saying: the base model is well calibrated (ECE 0.0233) and cannot follow instructions at all (0.0000), and after fine-tuning it follows instructions on 60.16% of the held-out items. Whether the tuned model is still well calibrated is the open measurement, and it is recorded as an open item rather than guessed at.

13.4 10d: reproducibility

`python -m tools.reproduce` loads the tracked canonical checkpoints from `checkpoints/canonical/` and re-derives ten headline numbers of this paper. It trains nothing. Each check prints the value it derived beside the value the paper claims and marks them MATCH or MISMATCH, and the script exits non-zero if any check fails.

Check	What it re-derives	Status
act2	the ALiBi/no-PE gap of 4.8×10^{-8} nats	MATCH
act4	the fit `canonical_only_D5M_total_N` — the canonical family's interior points on the **total-parameter** axis — re-fitted from the tracked ladder points: its α, its R^2, and the profile's compatible α range	MATCH
act4beta	the profile likelihood over β on the same points, whose compatible range $[0.38,\ 0.54]$ *excludes* Chinchilla's 0.28	MATCH
act5a	the KV cache's output identity (max logit difference)	MATCH
act7a	the zero-shot instruction floor, through the same teacher-forced path	MATCH
act8	the oracle and non-numeric choice rates	MATCH
act9	the top head's ablation effect and its z-score	MATCH
act9r	the 9a redundancy control: L0 H5 ablated together with L0 H2, against L0 H2 alone	MATCH
act10c	the expected calibration error and the next-token accuracy	MATCH
act10sft	instruction tuning degrades next-token calibration (ECE $0.0233 \rightarrow 0.1306$)	MATCH

Table 36: The reproducibility script re-derives nine headline numbers, including the exact Act 4 fit this paper reports, the Act 4 β profile, the Act 9 mechanism, the Act 9a redundancy control and the Act 10c positive result. **Interval type:** not applicable — these are point re-derivations, compared against the tracked values with a stated tolerance. Source: `tools/reproduce.py`.

The script deliberately does not re-run the scaling sweep, the SFT runs or the speculative benchmark: those cost hours and their artefacts are tracked. What it establishes is the thing that matters for a reader, namely that the numbers in this paper can be re-derived from what is in the repository.

14 Threats to validity and known limitations

This section consolidates the limitations rather than scattering them through the acts, because a reader should be able to find them in one place. The full per-claim status is in the tracked ledger, whose headline rows are tabulated in Section 13.1, and the claims we deliberately do not make are in Section 14.4.

14.1 Single-seed sections

Acts 2 and 3 are single-seed ($n = 1$). They rank configurations; they do not establish effects. There are no error bars in those sections, no significance tests, and no comparative adjectives without "suggests". This is the single most important caveat in the paper, and it is repeated in the captions of Table 4, Table 5 and Figure 3, Figure 4 so that it cannot be separated from the numbers.

14.2 Instrument and scale caveats, by act

- **Act 4.** This is the act with the most caveats, and they are not all caveats of degree. The previously reported $\alpha = 0.445$ is **withdrawn**: it describes the other architecture family ($R^2 = 0.9957$ on that family's points, -1.312 on the canonical models' own points), and with it the leave-one-rung-out "$R^2 \geq 0.985$ on every rung" headline, which that family produced because it occupied the entire top of the ladder. The *cause* of the contamination is now known as well, and it is a process failure rather than a modelling one: a training pipeline that was never in the repository made a mid-campaign change of model class invisible to every downstream artefact, including the ledger (Section 7.3). The refit is **not a minor correction**: three things were invalidated by it, including the comparison with Chinchilla, and the comparison had to be re-established on the surviving fit rather than assumed. The comparison that survives is stronger and more negative than a point comparison: the profile likelihood's compatible α range is $[0.34,\ 0.82]$, which contains 0.34 and also 0.50, so this ladder is **compatible with both published exponents and discriminates between neither**, and the range's width is the finding. The β profile is the mirror image: a compatible range of $[0.38,\ 0.54]$ that **excludes** Chinchilla's 0.28, so no compatibility claim may be made for β at all. The clean fit is itself exploratory: both the family separation and the N-axis convention were decided after the provenance was read, five rungs carry a single seed, and across the defensible fits α runs from 0.057 to 0.589, so this ladder does not identify α at this scale. The N-axis convention is a decision (total parameters) and it is stated rather than implied, but it now rests on one ground, the published convention: the physical plausibility bound on the fitted irreducible loss is satisfied by both axes (1.964 and 1.911 nats, both below the 2.5152-nat loss a real model reaches) and therefore does not discriminate, and the stability check stopped discriminating when the ladder was widened. The two axes do not give the same point estimate. The tracked fit's exact 2,776-point set cannot be recovered, so the contamination is established by parameter reproduction rather than by re-running the original fit. The per-rung N audit finds **three N conventions in one table, none of them flagged**, and the 0.8M rung's recorded N of 694,144 matches no known architecture at all: it is a third, unidentifiable model, carrying the act's worst hold-out fold, and it was replaced by a canonical re-run (Section 7.4). Any comparison of our exponents with a published value is additionally caveated by the tokenizer: a 6,487-token BPE at 2.2726 characters per token whose base alphabet is mostly rare Unicode, where byte fallback would compress better and preserve a 0% unknown-token rate. Interior ladder points are mid-schedule with the learning rate not annealed, an unquantified bias. The $D = 2$M inversion is a null (seed spread 0.186 against a gap of 0.057) in a table whose three canonical-rung

columns do not match the current ladder's logged points (Section 7.13). The 1.5M rung is internally heterogeneous — it is two architectures, with a 0.53-nat final-loss spread across its three recorded seeds (Section 7.15) — which makes the act's seed-spread and noise-floor statements exploratory. And the rescue is **complete** (Section 7.12): it widened the ladder from four rungs and 2,292 points to eight rungs and 3,463 points, moved α from +0.209 to +0.589, and left the qualitative verdict unchanged — the pre-registered gate asked whether the α range would narrow materially from [0.08, 0.54], and it did not.

- **Act 5a.** The 18.36× speedup was measured on a purpose-built context-1024 model; the canonical models have a 128-token context, and the number is a microbenchmark of the mechanism.

- **Act 5b.** Seed 42's $G = 1$ arm was re-established by inference from load semantics and value plausibility, not by re-verification of the artifact, which no longer exists; the $n = 5$ analysis is therefore primary and the $n = 6$ analysis is the sensitivity check.

- **Act 5c.** The timings are CPU best-of-3 at 0.03–0.11 s, a consistent comparison within one run and not a portable speedup figure. The distribution test is single-prompt, single-token on toy models, plus the residual-correction argument. The zero-gap total-variation comparison is near-tautological — at zero gap speculative decoding *is* plain sampling — and so proves that the code runs, not that the operator is correct on a real model.

- **Act 5, the G curve.** The 2.349 / 2.354 / 2.363 / 2.346 values come from the v1 path (500 steps, character tokenizer, single seed) and are context only.

- **Act 6.** The BPC axis is self-preferring: the judge is the model under test. It is valid for ranking configurations against one another under a fixed judge and is not an absolute measure of text quality. The sweep has no seed-level error bar, because configurations were not re-run with a fresh seed set.

- **Act 7.** The $n = 16$ row is effectively $n = 1$; the seeds vary optimisation order only, so between-initialisation variance is unmeasured; the task family is synthetic rather than natural instructions; ICL is teacher-forced and so is not comparable to free-running numbers elsewhere; and ICL shot selection is cost-driven, so only 18 items constitute a clean same-task test.

- **Act 8.** The absolute hallucination rate is a ceiling of 1.000 in every condition and is never used to rank conditions; 10 of 120 candidates were dropped for number truncation, which is why the chance baseline is per item and equals 0.1838 rather than 1/3; numbers proxy for atomic claims, so the act measures groundedness and not truth; and the retrieval anchor is derived from the target document, so the document hit rate of 0.975 is optimistic relative to a user's paraphrase.

- **Act 9.** The intact baseline was measured on this 64-sequence held-out slice in this run (CE 2.8551, perplexity 17.376), so every ablation delta is internal to the experiment and is not comparable to the published 2.5152 validation loss. Head ablation removes a head's *contribution*, not the head. The rollout result is one attribution method on one prompt family, and the logit lens ranks the number's first token, so a multi-token number is only partially represented. The logit lens is number-specific and *not* assertion-specific, so the strong version of the 9c claim — that the model encodes which number is asserted — is withdrawn, and the layer-5 spike is evidence for the Act 8 bag-of-numbers bias rather than for assertion-reading. The bootstrap interval that travels with the top-4 ablation, $[0.0365, 0.6390]$, is over the four individual head deltas and does not contain the +1.047 combined effect, so that effect is a point estimate. A backup head covers for layer 0 head 5:

ablating L0 H5 together with L0 H2 costs the same as ablating L0 H2 alone (redundancy 0.97), so the effect is carried by a group of layer-0 heads rather than by one head of 64, while L0 H5 with L1 H6 is additive (+0.9591 against a sum of +0.908), which rules out general fragility. The two-slice check is two offsets in one shard, not two corpora.

- **Acts 0, 1 and 2.** The v1 headline (3.4525 against 3.0946 bits per character) is single-seed, 500 steps, and carried four self-declared confounds; it is superseded by Act 3 — which is itself single-seed. The web interface has never been opened in a browser, so this paper makes no claim that it works.
- **Scale, for everything.** The benchmark is small: 1.5M, 2.6M and 6.3M-parameter models, a 128-token context, synthetic task families and C4-derived corpora. No claim in this paper generalises to models at scale, and none of it is intended to.

14.3 The claims ledger

The full per-claim ledger is the tracked artefact `report/claims.csv`, and it holds 99 rows at the time of writing: one row per claim in this paper, with its n, its interval, its status, the tracked file the number came from and the caveat that must travel with it. Every claim in the prose has a row there and every row is referenced from the prose; `tools/check_tex.py` asserts both directions, and it fails if a row's source file does not exist. The headline rows are tabulated in Section 13.1, which is generated from the ledger itself.

The ledger is deliberately *not* reproduced in full here. A table of 99 rows in the body of the paper is a table that will drift from its own source the moment either one is edited, and the artefact is the version that cannot drift, because it is the version the checker reads.

14.4 Claims we do not make

Publishing the boundaries of a result is a strength, not a disclaimer: a course project that states what its own evidence cannot support is more credible than one that publishes only its wins, and a reader who knows which claims to discount can use the rest.

Table 37: The claims we do not make, with the reason for each and the act it belongs to.

We do not claim	Because	Act
That the clean α is a point estimate	The profile likelihood is the statement: the compatible range is $[0.34,\ 0.82]$ on the total-parameter axis, and the point estimate +0.589 is not an interval	4

Table 37: The claims we do not make (continued).

We do not claim	**Because**	**Act**
That the fitted E is a measured irreducible loss	It is an *extrapolated asymptote* from an eight-rung ladder, not the corpus's floor. The corpus's measured unigram entropy under the 6,487-token vocabulary is 4.9946 nats on the held-out region, and *both* fitted E values lie below it (1.964 is 0.39× it, 1.911 is 0.38× it)	4
That the ladder is CONSISTENT with Chinchilla's exponent	It is compatible with both published exponents, and therefore discriminates between neither: the total-N α range $[0.34,\ 0.82]$ contains 0.34 and also 0.50, so it excludes neither of the values the act compares against	4
That β is consistent with Chinchilla	No compatibility claim may be made for β at all: the total-N β range is $[0.38,\ 0.54]$ and *excludes* Chinchilla's 0.28 (the non-embedding range, also $[0.38,\ 0.54]$, excludes it too)	4
That our α is 0.34, or that it agrees with 0.34	The compatible range is far too wide to identify 0.34, the point estimate is +0.589, and the fold-wise estimates run from +0.501 to +0.691	4
That the profile's compatible range is an error bar	It is a statement about what the data cannot distinguish, not a confidence interval and not a fit covariance; the range's width is itself the finding of the act	4

Table 37: The claims we do not make (continued).

We do not claim	Because	Act
That the non-embedding and total-parameter axes give the same answer	They do not give the same point estimate (0.407 against 0.589). They do now agree on the qualitative verdict — both α ranges contain 0.34 and both β ranges exclude 0.28 — so the axis is chosen on the published convention alone	4
That the 0.8M rung is one of the two known architectures	Its recorded N of 694,144 matches *no* MiniGPT or ALiBi shape at any (dim, layers, heads) in dims 16–256 and depths 1–10, while its embedding parameters (830,336 = 6,487 × 128) show the vocabulary and dim are as expected: it is a *third, unidentifiable architecture* with no architecture block, run on CPU, at $D/N = 22.7$	4
That the ladder's training code is in the repository	It is not. `src/scaling/` was *empty* before commit `f1eaf77`; the committed ladder script builds the *second* family's architecture (ALiBi, 4 heads) with the *first* family's shapes, and the script that trained the canonical MiniGPT rungs was never committed at all	4
That the other family's runs were broken or divergent	They were good models with an unlucky initialisation: they reproduce from the repository's own code (133.25 / 184.49 / 224.99 against the recorded 129.57 / 183.96 / 223.77) and they finished at 3.578 nats against the canonical 3.057	4

Table 37: The claims we do not make (continued).

We do not claim	**Because**	**Act**
That the refit is a minor correction	It invalidated $\alpha = 0.445$, the "$R^2 \geq 0.985$ on every rung" headline, and the comparison with Chinchilla, because the other family owned the entire top of the ladder	4
That the fold standard deviation was an uncertainty on α	A standard deviation over eight folds of a model whose α is not identified is false precision; the fold-wise estimates are quoted individually instead	4
That the rescue narrowed the α range into usefulness	It ran, and it did not narrow the range: eight rungs give a compatible α width of 0.48 against 0.46 at four, so the null is credible rather than overturned	4
That the calibration result means the model can follow instructions	It measures next-token prediction on the training distribution; every instruction, grounding and tool task in Acts 7 and 8 is still at the floor	10c
That in-context learning is absent at 6.3M	The demonstration is copied, not ignored: its answer is the first generated token on 18/18 same-task items	7b
That the adapters-vs-LoRA ordering is established	Wilson intervals on 128 items overlap by about ± 0.04	7c
That prefix tuning "scored zero because of a bug"	It did not fit: its final training loss is 3.2576	7c
That the $n = 16$ row of the SFT curve is an $n = 3$ result	One batch per epoch, so the seeded shuffle has nothing to shuffle	7b
That the three ICL seeds are independent	The model is frozen and the forward is deterministic, so the runs are bit-identical	7b
That the SFT results transfer to natural instructions	The task family is synthetic, not natural instructions	7

Table 37: The claims we do not make (continued).

We do not claim	**Because**	**Act**
That the ICL numbers are comparable to free-running generation numbers	ICL is scored teacher-forced	7b
That the LoRA rank curve has plateaued	It rises monotonically to $r = 64$ with no plateau	7d
That LoRA at rank 1 is a fair matched-budget comparison	Rank is pinned low by construction, and 7d shows that point is not where the curve flattens	7c, 7d
That LIMA is contradicted	We expose LIMA's premise (latent capability) as unsupported at this scale; the curve is learning a skill, not aligning one	7
That the 6.3M model can ground a passage	It identifies the asserted number at chance, and a non-numeric passage scores the same 9/40	8b
That hallucination rates can be compared between conditions	The absolute rate is a ceiling of 1.000 in every condition	8b
That the model learned an operation from a tool-call format	It emits `CALC(s)`: the format is acquired, the operation is absent	8d
That any result here measures factual truth	Numbers proxy for atomic claims and there is no fact base, so this is groundedness	8b
That the tool-call template echo means the model learned the operation	0 of 40 well-formed `CALC(...)` calls, and in all 19 greedy template echoes the argument is exactly `s`; the format is acquired and the operation is absent	8d, 9c
That the head-ablation deltas are comparable to the published validation loss	The intact baseline (CE 2.8551, perplexity 17.376) was measured on this 64-sequence slice in this run, not the published 2.5152; every delta is internal to the experiment	9a
That head ablation deletes the head	It removes the head's *contribution* (its attention output is zeroed before it is mixed back into the residual stream); the head still computes	9a

Table 37: The claims we do not make (continued).

We do not claim	Because	Act
That the rollout negative is a general claim about attention interpretability	It is one attribution method on one prompt family (the Act 8 number-assertion family): 0 of 34 items, against a uniform chance of 0.0078	9b
That layer 0 head 5 is the head that carries the model	A backup head covers for it: ablating L0 H5 together with L0 H2 costs only +0.0234 nats, the same as ablating L0 H2 alone (redundancy 0.97), and L0 H5 with L0 H0 likewise +0.0226. The effect is carried by a *group* of layer-0 heads; L0 H5 is the one whose removal is most visible in isolation	9a
That the logit lens shows the model encodes *which* number is asserted	The counterfactual-EDITED number reaches the same layer-5 median rank as the asserted number (82 against 82) and the non-numeric control reaches 86, so the lens is number-specific and not assertion-specific; the strong version of this claim is withdrawn	9c

15 Reproducibility

Every number in this paper is traceable to a tracked file. The audit trail for each figure and each table — the exact source file, the generating script, the row count and the n — is in `report/MANIFEST.md`, and the machine-readable claim ledger is `report/claims.csv`, whose correspondence with the text is checked by `tools/check_tex.py`.

The project is deterministic where it can be. Act 0's n-gram counts are identical across seeds by construction, and the ledger records that fact rather than hiding it behind a zero-width error bar. The Act 4 sweep reproduced its 2.6M rung's loss exactly from a different script on a different day. The Act 6 decoding sweep produced bit-identical quality and diversity values across a repeat run. The 2.6M canonical checkpoint reproduced the MHA sweep loss of 2.8111 exactly. The three canonical checkpoints are exported as weights (about 74 MB) and their SHA-256 digests, step counts and losses are pinned by a test, so a reader can regenerate every Act 5 figure without retraining. A fourth, the supervised fine-tune of the 6.3M model, is now exported in the same format at `checkpoints/canonical/6.3M-sft/model.pt`, which is what makes the 10c SFT-versus-base comparison possible.

Figures are never drawn by hand. Each one is produced by a committed script under `src/eval/` that reads only the tracked results under `results/`, and writes both a PNG and a vector PDF, so a figure cannot drift from its result. The source data for every table in this paper is in `report/tables/*.csv`. Two of the tables in Section 13.1 are *not* transcribed by hand either: `tools/build_act10_tables.py` writes the 10a headline-claims table and the 10b failure taxonomy from `report/claims.csv` and from the defect records in `docs/implementation.md`, both as a CSV and as a LaTeX fragment, and this document reads those two fragments in with `\input`. The two generated fragments, `report/tables/act10_headline_claims.tex` and `report/tables/act10_failure_taxonomy.tex`, are part of the report bundle and must travel with it.

The test suite is the implementation gate: 359 tests, 0 failures, 1 skipped. The skip is a v1-era test that needs an artifact which is not present in a fresh clone. The report itself is checked by a second gate, `tools/check_tex.py`, because there is no local `pdflatex`: it verifies that every `\includegraphics` target resolves inside `report/figures/` and has a PDF sibling, that every file in `report/figures/` is either referenced by the report or listed with a reason in the checker's explicit exclusion list, that environments and braces balance, that every `\label` is unique and every `\ref` resolves, that no absolute path or out-of-bundle `\input` appears, that inline math is balanced, and that every claim in the text has a row in `report/claims.csv`. The checker also carries a negative self-test (`-self-test`), which feeds it deliberately broken copies of this document and asserts that it fails on each of them: a gate that cannot fail is not a gate.

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The findings taxonomy

Every hypothesis this project put at risk, every measurement failure it caught and every claim it declines to make, one row each. It is placed after the references because it is a lookup table rather than an argument: the argument is the ten acts that precede it. Each row names the tracked file its outcome is read from.

Table 38: The findings taxonomy: one row per pre-registered hypothesis, per recorded measurement failure and per declined claim. **Rows:** 89 — 10 *supported*, 26 *unsupported*, 15 *measurement failures* and 38 *declined claims*, and every row is in exactly one of those four populations. **What is measured:** the hypothesis each act put at risk, the test it was put at risk by, and the outcome, read from the tracked result files named in the CSV twin (`report/tables/findings_taxonomy.csv`). **Interval type:** per row, as stated in the paper's own tables; this table is a taxonomy, not an estimate. **Pre-registration:** only Act 4's row (`FALSIFICATION_CONDITION.md` in the act's result directory) was falsifier-first; every other row is marked post-hoc in the *preregistered* column of the CSV, and the paper says so.

Act	Hypothesis	Test	Result	Status	Do-not-make note
4	The ladder's interior points all come from one model family, so one law can be fitted to all of them.	An absolute sign check: the tracked fit's own parameters evaluated on each candidate point set (R^2 against a constant-loss baseline), plus a provenance read of the run records, the checkpoints' state dicts and the architecture block.	FALSIFIED. 14 of the 23 non-tiny runs are the canonical MiniGPT and 5 are a different family with no learned position table; 3 more are an unidentifiable architecture and 1 is a superseded duplicate. The tracked parameters score R^2 = +0.9957 on the other family's 1,428 points and -0.6661 on the canonical family's own 3,463 points; 51.4% of the withdrawn fitted set came from the other family, which owned the entire top of the ladder. alpha = 0.445, beta = 0.290 and E = 1.605 are withdrawn.	measurement failure	That the withdrawn exponents describe these models, or that the refit was a minor correction.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
4	The change of model class mid-campaign left a record that would have made it visible.	A repository read: the history of src/scaling/ and the shape table in the ladder script committed at f1eaf77.	FALSIFIED. src/scaling/ was EMPTY before commit f1eaf77; the ladder script committed there builds ALiBiTransformer with 4 heads and a shape table (dim = 32 x layers) that matches the canonical MiniGPT runs exactly. The script that trained the canonical MiniGPT rungs was never committed at all, so nothing recorded which script produced which points.	measurement failure	That the ladder's training code is in the repository, or that the contamination was a modelling choice.
7	The Act 7 measurement passes the model the text it is meant to predict.	A per-position accuracy profile: the first response token against positions 1-2.	FALSIFIED (defect D11). The first response token was unguessable (0.070) while positions 1-2 were trivially copyable from the true prefix (0.927 / 0.900), which is the signature of a missing input rather than of a weak model.	measurement failure	That the first Act 7 result set (full fine-tuning 0.0104 and every PEFT arm) measured instruction following.
7	The PEFT arms' forwards compute the loss the harness asks for.	The three affected arms reported exactly 0.0000 while genuinely training.	FALSIFIED (defect D12). The prompt-tuning, prefix-tuning and adapter forwards applied a SECOND cross-entropy shift, because MiniGPT's loss is already aligned to the inputs and the SFT loop passes labels shifted by one.	measurement failure	That the prompt-tuning, prefix-tuning and adapter arms of 7c measured what they claim to measure.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
9	A held-out cross-entropy near the uniform-over-vocabulary loss would mean the model is bad, not that the metric is wrong.	An absolute plausibility check against ln(6,487) = 8.78 nats.	FALSIFIED (defect D18). The metric compared logits to the wrong position and returned 11.98 nats (perplexity 1.6e5) for a model whose real loss is about 2.5.	measurement failure	That any Act 9 ablation delta measured anything.
5	The cached and the uncached speculative-decoding paths agree, so the operator is correct.	A differential test between two implementations.	THE TEST PASSED AND THE OPERATOR WAS STILL WRONG. The cached-versus-uncached equivalence tests both passed because both paths shared the double-feed bug. The absolute invariant that localised it in minutes: acceptance must be exactly 100% at a zero draft-target gap, because p = q makes min(1, p/q) = 1 for every token; it was 15.6%.	measurement failure	That every earlier 5c number measured speculative decoding.
7b	The 4-shot in-context prompt fits the model's context window.	A positive control that recorded the prompt's token count and the visible-shot count.	FALSIFIED (defect D13). The conventional 4-shot prompt was 328 tokens against a 128-token context; truncation deleted the demonstrations, and the control recorded '2/4 (truncated away)'.	measurement failure	That the first ICL measurement (0.0000) was a finding about in-context learning.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
8b	The chance level for the choice-rate comparison is 1/3.	Counting the distinct candidates per item.	FALSIFIED. The candidate set is deduplicated and holds 3, 4, 5 or 6 distinct numbers (1 item with 3, 3 with 4, 8 with 5, 28 with 6), so the per-item uniform chance level is 0.1838, not 0.3333.	measurement failure	That the Act 8 condition comparison was read against a 1/3 baseline.
8c	The retriever returns the passage that contains the asserted number when it is queried with the instruction.	Inspecting what the retriever actually returned.	FALSIFIED. The retriever was queried with instruction text whose terms cannot come from a passage about a number, so ranking degenerated to a constant: one distinct document came back for all 40 items and the identical top chunk for two different failing items. The retrieved condition's earlier 0/40 was a query-design failure, not a finding.	measurement failure	That the model ignores retrieved context (the 0/40 was the retriever's fault).
2	The 'pe-alibi' arm measures ALiBi positional encoding.	The difference between the ALiBi arm and the no-positional-encoding arm.	FALSIFIED. The two arms differ by 4.8e-08 nats (2.4850885868 against 2.4850885391), which is float32 round-off, because ALiBiPE.forward returns its input unchanged.	measurement failure	That ALiBi is competitive with learned absolute positions.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
4	The ladder's recorded N is on the same axis as the published exponents.	The published convention, the stability of the fit, and the plausibility of the fitted E against measured anchors.	FALSIFIED, then repaired. The recorded N is a NON-EMBEDDING parameter count while the published exponents are fitted on TOTAL parameters, so every comparison with a published exponent had been a comparison of two different quantities. The axis is now stated explicitly (total parameters) and the fit is reported under both conventions.	measurement failure	That the non-embedding and total-parameter axes give the same answer.
4	Every rung's N in the ladder's own table is one of the two known architectures.	An exact integer audit of every rung's N against three independent sources.	FALSIFIED. Three N conventions sit in one table and none is flagged, and the 0.8M rung's recorded N of 694,144 matches no MiniGPT or ALiBi shape at any (dim, layers, heads) in dims 16-256 and depths 1-10, while its embedding parameters (830,336 = 6,487 x 128) show the vocabulary and the dim are as expected. It is a third, unidentifiable architecture, and it carries the act's worst hold-out fold (R^2 = 0.508 non-embedding, 0.779 total).	measurement failure	That the 0.8M rung is one of the two known architectures.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
2, 3	The report specification's sweep sizes are the sizes that were run.	Counting the runs in the tracked result files.	FALSIFIED. The specification says Act 2 ran 20 runs and Act 3 ran 5 tokenizers; the tracked files hold 18 configurations and 4 tokenizers.	measurement failure	That the sweeps were mis-sized.
9	The random-ablation null is measured against the same baseline the ablations are.	A sanity check on the z-scores themselves.	FALSIFIED (defect D19). The null was measured against a stale baseline, and the check returned -88 instead of +42.8.	measurement failure	That any z-score in the first Act 9 write-up was correct.
8	The answer-copying score measures the model's preference between answers.	An absolute plausibility check on the pass-through lift.	FALSIFIED (defect D14). The score absorbed the prompt's own log-probability, giving a passage-lift of -246 nats, which is absurd on its face.	measurement failure	That any across-condition lift contrast in the first Act 8 write-up was a measurement.
0	The n-gram ORDER matters more to validation perplexity than the smoothing scheme.	12 configurations (4 orders x 3 smoothing schemes, plus the interpolated arm) on TinyShakespeare at 3 seeds, scored by validation perplexity and bits per character.	SUPPORTED. Order 1 scores 28.43 validation perplexity and order 3 scores 7.45-8.78, while the spread across the four smoothing schemes at a fixed order is at most 0.9 in perplexity and is exactly 0 at order 1.	supported	That smoothing is irrelevant: it is worth up to 0.9 perplexity at a fixed order.
0	The four smoothing schemes differ materially in validation perplexity at a fixed n-gram order.	The same 12 configurations, compared scheme against scheme inside each order.	UNSUPPORTED. At order 1 the four schemes are bit-identical (28.4267 each); at order 2 they span 11.956-12.070; at order 3 they span 7.446-8.085. The largest within-order spread is 0.9 in perplexity against the 20.0 that the order itself buys.	unsupported	That any particular smoothing scheme is the right one in general: the best of the 12 is trigram Kneser-Ney at 7.446, and the interpolation arm is handicapped by construction.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
1	LSTM and GRU beat a plain RNN on a small corpus.	Three architectures at 3 seeds (500 steps, 1.1M characters), scored by validation loss.	UNSUPPORTED. The largest gap (LSTM minus RNN) is 0.152 nats, smaller than the smaller of the two standard deviations (0.215 and 0.285).	unsupported	That recurrent gates make no difference in general.
1	A plain RNN shows vanishing gradients on a sequence task.	One gradient trace of 100 steps per model on the copy task at sequence length 20.	UNSUPPORTED. The total L2 gradient norm RISES over 100 steps (RNN 0.4363 to 0.4829; LSTM 0.1412 to 0.2828) and the LSTM's norms are lower than the RNN's throughout.	unsupported	That the gradient demonstration showed vanishing gradients.
2	Transformer component choices (normalisation, positional encoding, head count, FFN activation) change validation loss.	18 ablations at seed 42 only, scored by validation loss.	SUPPORTED AS A RANKING ONLY. The configurations span 2.4851 to 2.6637 nats (0.179 nats from best to worst), and the two ends are exactly the arm that did not measure ALiBi and the BatchNorm arms.	supported	That any particular component is responsible for an effect: this is n = 1 with no error bars and no significance test.
3	The tokenizer ranking survives an equal-character-span control.	Four tokenizer families trained under a standard context and under a matched character span, scored by validation bits per character.	SUPPORTED. The ranking is Unigram 2.8021, BPE 3.0801, character 3.4680, WordPiece 3.4998, and under the equal-character-span control the order is unchanged: Unigram 2.8341, BPE 3.1893, character 3.4680, WordPiece 3.5116.	supported	That the ranking is established: it is single-seed and exploratory.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
3	The v1 headline (character versus BPE bits per character) survives the context control.	The same equal-character-span control applied to the v1 comparison.	UNSUPPORTED. Under the control BPE moves from 3.0801 to 3.1893 bits per character while character, WordPiece and Unigram move by 0.0000, 0.0118 and 0.0320, so the v1 comparison was partly a context effect and is superseded.	unsupported	That the v1 headline (3.4525 against 3.0946 bits per character) is a result.
4	Loss scales predictably with N (non-embedding parameters) and D (training tokens) in the ~0.8M-9M regime, as L(N,D) = E + A/N^alpha + B/D^beta.	The pre-registered falsification condition: the fit must beat a constant-loss baseline on held-out points (R^2 > 0) and show no systematic curvature in log N or log D.	NOT FALSIFIED for the clean fit. Every held-out rung of the canonical family is predicted with R^2 > 0 (0.779, 0.989, 0.990, 0.980 on the total-parameter axis) and the residual correlations with log N and log D are -0.00002 and -0.0039.	supported	That the condition validates the exponents: it judges the SHAPE of the law, not the values of alpha or beta.
4	The ladder is one model family, so its points can be pooled.	A provenance read of the run records, the checkpoints' state dicts and the architecture block, plus the first logged validation loss as a separator.	UNSUPPORTED. The two families are separable by a factor of fifteen in first logged loss (8.72-8.91 nats against 129.57-223.77 nats), 38.4% of the ladder's interior points come from the other family, and the pooled fit reaches only R^2 = 0.7087 while either pure family fits at R^2 >= 0.995.	unsupported	That the refit was a minor correction, or that the withdrawn exponents describe these models.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
4	The fitted E is the corpus's conditional entropy, and the unigram entropy is the right anchor for it.	Measured anchors: the marginal (unigram) entropy, a text-space compression bound, a token-space compression bound in frequency-rank order, and the model's own achieved held-out loss.	UNSUPPORTED. E is the irreducible loss of an INFINITE model, i.e. the conditional entropy of the next token given full context, so the marginal entropy of 4.9946 nats is a loose CEILING and never a target. The correct anchors are the token-space compression ceiling of 3.5918 nats/token and the canonical 6.3M's achieved held-out loss of 2.5152 nats, and E must lie BELOW the achieved loss because E is the asymptote as N and D grow.	unsupported	That E is the corpus's floor, and that the corpus's unigram entropy of 4.9946 nats is a target for it.
4	The ladder identifies alpha well enough to compare it with a published exponent.	The profile likelihood over alpha with the other four parameters re-optimised, plus the joint (E, alpha) profile on a grid of E.	UNSUPPORTED as a point claim. The compatible alpha range on the total-parameter axis is [0.34, 0.82], which contains both published values, so the ladder discriminates between none of them; and the width is a RIDGE, not noise: fixing E on a grid and re-optimising alpha traces a monotone ridge with d(alpha)/dE = +0.5542.	unsupported	That the ladder's alpha is unidentified in every sense, or that the compatible range is an error bar.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
4	The ladder identifies beta well enough to compare it with a published exponent.	The profile likelihood over beta on the same 3,463 points.	SUPPORTED. The compatible beta range is [0.38, 0.54]: it EXCLUDES Kaplan's 0.30 and CONTAINS Chinchilla's 0.50.	supported	That the range is a sampling interval over runs. It is a compatibility range over a fixed point set, and the points are not exchangeable (many share a run).
4	If the ladder is a power law, its LOCAL exponent alpha_loc = -[dL/dlnN]/(L - E) is constant across the ladder, and the value of E is selected by requiring that constancy.	Two estimators of the local exponent (the ascending-D raw ladder, and the ladder with each rung's own D-term removed), scanned over a grid of E for the E that minimises the standard deviation of alpha_loc.	SUPPORTED, and the E it selects is the data's. With the D-term removed the spread is minimised at E = 1.38, where alpha_loc = 0.3609 with a standard deviation of 0.0100 across the ladder; the ascending-D raw estimator selects E = 0.64 with alpha_loc = 0.1815 (s.d. 0.0061), but that quantity is a weighted mix of alpha and beta because the B/D^beta term is still in the slope.	supported	That the local exponent is a point estimate of alpha on its own, or that the raw ascending-D estimator measures alpha.
5a	A KV cache changes the model's output.	The maximum logit difference between the cached and the uncached forward path.	UNSUPPORTED. The maximum logit difference is 1.64e-07 (1.4 ULP in float32, BLAS accumulation order), so the two paths are the same computation.	unsupported	That the cache is exactly bit-identical in floating point: the difference is accumulation order, not zero by construction.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
5b	Sharing KV heads (MQA) is free in quality.	Six paired seeds at the 2.6M rung, comparing G = 8 against G = 1 on the identical data and training recipe.	UNSUPPORTED. The primary paired analysis (five seeds, excluding the seed whose arm could not be re-verified) gives a mean difference of +0.0253 nats with a two-sided paired t-test p = 0.0009 and 5 of 5 seeds direction-consistent; the six-seed sensitivity check gives +0.0285 (p = 0.0008). The saving is real: an 8x cache reduction and 12.1% fewer FLOPs per token.	unsupported	That MQA does not reduce compute: the attention matmul cost is identical at every G, but the total FLOPs per token fall by 12.1%.
5c	Speculative decoding makes generation faster than plain cached decoding.	A 50-cell (prompt, K) grid at two draft-target gaps, with median speedups over 5 prompts of best-of-3 CPU timings.	UNSUPPORTED. 0 of 50 cells exceeded plain cached decoding, and the best cell was 0.622x. Acceptance was high and is not the cause: the median acceptance at K = 1 is 0.857 at both model gaps.	unsupported	That the operator is correct because the cached and uncached paths agreed: they shared the bug.
6	A wider beam produces better text than greedy decoding.	A 15-configuration decoding sweep scored by bits per character under a fixed judge, with 120 generations per configuration.	SUPPORTED. Beam-4 beats greedy by 0.061 bits per character (1.3760 against 1.4366) at 4.26x the time, and greedy is the quality/cost knee: no configuration is both cheaper and better, and beam-8 is worse than beam-4.	supported	That the bits-per-character axis is an absolute measure of text quality: the judge is the model under test.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
6	Best-of-n reranking beats greedy decoding.	Best-of-n over temperature-1.0 samples, reranked by the model itself, n up to 16, 120 generations per row.	UNSUPPORTED. Best-of-16 recovers +0.272 bits over the mean sample but is still 0.259 bits WORSE than greedy (1.6959 against 1.4366). The reason is structural: the reranker is the model itself, so it can only select text the model already finds likely.	unsupported	That best-of-n would work with an independent judge: the self-preferring reranker is the mechanism behind the negative.
6	Self-consistency voting improves accuracy on a multiple-choice task.	Majority voting on 300 MNLI items at 3 seeds, 900 scored predictions per row.	UNSUPPORTED. Voting gives no gain at any vote count: 0.3789 at n = 1, falling to 0.3567 at n = 20, with every Wilson interval containing the 0.3333 chance level. The underlying argmax accuracy is exactly chance (0.3300, 99 of 300).	unsupported	That the n = 1 figure of 0.3789 is partial competence: sampling is an unbiased draw from a biased distribution.
7	Supervised fine-tuning on instructions improves held-out instruction following.	The tracked SFT checkpoint scored teacher-forced exact match on the 128 held-out Act 7 items.	SUPPORTED. The base model scores 0.0000 and the tuned model scores 0.6016 (77 of 128, Wilson 95% [0.5150, 0.6822]), reproducing Act 7's own seed-42 figure of 0.5938 to within one item.	supported	That the gain is uniform across tasks: 18 of the 24 tasks score exactly 0.0000, every one of them an addition task.
7a	The 0.0000 zero-shot instruction floor is an artefact of the scoring path.	A positive control: an oracle emitting the correct answer, scored through the teacher-forced path, the greedy path and the exact ICL scoring path.	UNSUPPORTED. The oracle scores 1.0 through all three routes while the real model scored zero on identical items, so the floor is assertable.	unsupported	That the control was pre-registered, or that the floor was established in advance.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
7b	In-context learning performs the operation demonstrated in the context.	A paired A/B on the 18 same-task held-out addition items, with and without a demonstration in context, plus the zero-shot positive control.	UNSUPPORTED. The first generated token is the DEMONSTRATION's answer on 18/18 items against 0/18 zero-shot, while the exact match on the query's own answer is 0/18 in both conditions and 0 of 18 items changed. The model copies the format, not the operation.	unsupported	That in-context learning is absent at 6.3M, or that the ICL numbers are comparable to free-running generation numbers (ICL is scored teacher-forced).
7c	A parameter-efficient method reaches full fine-tuning's accuracy at a matched trainable-parameter budget.	Five methods at a matched budget of 20,000 trainable parameters, teacher-forced exact match on 128 held-out items, with the final training loss recorded so that 'did not fit' can be told apart from 'fit but scored zero'.	UNSUPPORTED. At the matched budget the ordering is adapters 0.2136, LoRA 0.1823, prompt tuning 0.0859 and prefix tuning 0.0000, against full fine-tuning's 0.5755. Prefix tuning's zero is a failure to fit (final training loss 3.2576), not a scoring artefact.	unsupported	That the adapters-versus-LoRA ordering is established: the Wilson intervals overlap by about +/-0.04.
7d	LoRA rank is a threshold: past a small rank, accuracy stops improving.	A rank sweep over r in {1, 2, 4, 8, 16, 32, 64} at 3 seeds, teacher-forced exact match on 128 held-out items.	UNSUPPORTED. Accuracy rises monotonically from 0.1823 at r = 1 to 0.5078 at r = 64 with no plateau; rank behaves like a smooth capacity dial.	unsupported	That LoRA at rank 1 is a fair matched-budget comparison.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
8b	The model identifies which number a supplied passage asserts.	The argmax over the candidate set of the teacher-forced likelihood of each answer, on 40 items, under five conditions with a per-item uniform chance level of 0.1838.	UNSUPPORTED. Under the oracle passage the model selects the asserted number on 9 of 40 items (0.2250, Wilson [0.1232, 0.3750]), the chance-adjusted rate is +0.0412 and not significant, and a non-numeric passage from the same document scores exactly the same 9/40.	unsupported	That the model is WORSE than chance: the wording is 'at chance', never 'below chance'.
8b	The model follows a counterfactual edit in the passage.	The same instrument with the asserted number replaced by a counterfactual one.	UNSUPPORTED. The counterfactual is followed on 0 of 40 items (bootstrap [0.000, 0.000]) and under the counterfactual the model still picks the ORIGINAL number on 9 of 40.	unsupported	That any result here measures factual truth.
8b	The model uses a retrieved passage when one is supplied.	The same instrument with a retrieved chunk in place of the oracle passage, after the retriever was repaired.	UNSUPPORTED. The retrieved condition selects the asserted number on 8 of 40 items (0.2000), a chance-adjusted rate of +0.0162 with a bootstrap interval of [-0.101, 0.143], and the closed-book condition is higher still at 11 of 40.	unsupported	That the model ignores retrieved context: the earlier 0/40 was the retriever's fault, not the model's.
8d	The model performs the operation in a tool-call format it has been shown.	40 greedy and 120 sampled generations from prompts with real numbers taken from C4, scored for well-formed CALC(...) calls.	UNSUPPORTED. 0 well-formed CALC(...) calls in 40 greedy generations and 0 in 120 sampled generations. 19 of 40 greedy generations emit the literal 'CALC(' and in all 19 the argument is exactly 's'.	unsupported	That the model learned the operation from the tool-call format: the format is acquired and the operation is absent.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
9a	One head of the 64 carries the model's competence.	Causal head ablation against a random-ablation null, with pairwise redundancy checks on 64 held-out sequences.	UNSUPPORTED. Exactly 1 of the 64 heads has z >= 2 (layer 0, head 5: dCE +0.832 nats, z +7.82 against the null), but a redundant partner covers for it: ablating L0 H5 together with L0 H2 costs only +0.0234 nats, the same as ablating L0 H2 alone (redundancy 0.973), while L0 H5 with L1 H6 is additive (+0.9591 against a sum of +0.908).	unsupported	That the head-ablation deltas are comparable to the published validation loss: the intact baseline was measured on this 64-sequence slice in this run (CE 2.8551, perplexity 17.376).
9b	Attention attribution identifies the token that answers the question.	Behavioural attention rollout and raw last-layer attention, scored against the token holding the answer, on 34 items with a located answer token and on 71 items for the attribution mass.	UNSUPPORTED. Neither method ever places its maximum on the final position (0 of 71 each), rollout's attribution mass on the answer token is 0.00759 against a uniform 0.00781, raw attention's is 0.00673, and neither method ever identifies the answer token (0 of 34 items).	unsupported	That the rollout negative is a general claim about attention interpretability: it is one attribution method on one prompt family.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
9c	The logit lens shows which number the passage asserts.	Five lens-control conditions at 40 items each, plus the first lens run at 34 items, scored by the median rank of the number's first token.	UNSUPPORTED in the strong form. The lens is number-specific but NOT assertion-specific: at layer 5 the counterfactual-EDITED number reaches the same median rank as the asserted number (82 against 82) and the non-numeric control reaches 86, while a number appearing in NO passage is at 484 and a random token id at 735.	unsupported	That the logit lens shows the model encodes which number is asserted: the layer-5 spike is evidence for the Act 8 bag-of-numbers bias.
10c	The model is a competent next-token predictor.	4,064 next-token predictions on held-out C4 (32 sequences of 128 tokens), scored for accuracy, mean confidence and expected calibration error over 10 equal-width confidence bins.	SUPPORTED. Accuracy is 0.5295 at mean confidence 0.5255 with an expected calibration error of 0.0233, and the bins track (confidence 0.979 -> accuracy 0.962; confidence 0.054 -> accuracy 0.074).	supported	That the calibration result means the model can follow instructions, ground, or use tools.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
10c	Instruction tuning adds next-token competence on the distribution it is tuned on.	The same instrument on two slices for the base model and the tuned model: generic held-out C4 (4,096 tokens) and the held-out instruction responses (1,023 tokens), with bootstrap intervals on the ECE and a normal interval on the accuracy difference.	UNSUPPORTED. On the generic slice the tuned model loses 10.7 points of accuracy (difference -0.1068, 95% [-0.1284, -0.0852]) and its calibration error rises from 0.0233 to 0.1306. On the instruction responses it loses NOTHING (difference -0.0009, 95% [-0.0160, +0.0142], not significant) while its mean confidence rises from 0.3969 to 0.6261 with the accuracy flat, and its calibration error there more than doubles (0.1118 -> 0.2772).	unsupported	That instruction tuning made the model worse at the task it was tuned on: the accuracy difference there is -0.0009 [-0.0160, +0.0142], not significant. What it gained there is confidence.
10c	The model shows a stereotype association between an occupation and a gender-associated attribute set.	A within-occupation preference probe over 10 occupations with a documented association and 10 negative-control occupations with none.	UNSUPPORTED as a stereotype finding. All three rates are 1.0: the associated family's first occupation 10/10, its second 10/10, and the negative control 10/10, so the probe detects a GLOBAL attribute preference rather than an association.	unsupported	That the model is unbiased: the instrument reads a global preference, and it is underpowered by construction.

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
10c	The model generates toxic content.	A 12-term toxicity lexicon scored over the model's own continuations of C4 passages, 40 generations, against the corpus base rate.	UNSUPPORTED. 0 of 40 generations contains any term from the lexicon (Wilson 95% [0, 0.0876]) against a corpus base rate of 0.0. Floor.	unsupported	That the model is safe: one token is sampled per prompt, and a 6.3M model has no reliable notion of a toxic utterance, so this measures a floor and not a comprehensive score.
10b	Every measurement failure recorded in this project was caught by a sanity invariant or a positive control, and none by comparing two implementations.	The failure taxonomy, built from the defect records in the ledger and the tracked result files: 15 recorded failures, each with what it was, how it was caught and what it invalidated.	SUPPORTED. All 15 were caught by an absolute invariant (the zero-gap acceptance invariant, the uniform-over-vocabulary ceiling, the per-position accuracy profile, the absolute plausibility checks, the exact integer N audit, the sign check on the fit's own R^2) or by a positive control (the truncated demonstrations, the oracle control, the per-item candidate count).	supported	That the project has found every failure it had: the taxonomy covers the failures that were RECORDED.
4	That the clean alpha is a point estimate	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
4	That the fitted E is a measured irreducible loss	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
4	That the ladder is consistent with Chinchilla	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
4	That beta is consistent with Kaplan's exponent	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
4	That our alpha is 0.34, or that it agrees with 0.34	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
4	That the profile's compatible range is an error bar	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
4	That the ladder's alpha is unidentified	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
4	That the non-embedding and total-parameter axes give the same answer	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
4	That the 0.8M rung is one of the two known architectures	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
4	That the ladder's training code is in the repository	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
4	That the other family's runs were broken or divergent	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
4	That the refit is a minor correction	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
4	That the fold standard deviation was an uncertainty on alpha	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
4	That the rescue, once run, narrows the alpha range enough to identify a published exponent	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
7	That Act 7's n = 512 headline is a single measurement	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
7b	That in-context learning is absent at 6.3M	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
7c	That the adapters-vs-LoRA ordering is established	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
7c	That prefix tuning scored zero because of a bug	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
7b	That the n = 16 row of the SFT curve is an n = 3 result	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
7b	That the three ICL seeds are independent	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
7	That the SFT results transfer to natural instructions	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
7b	That the ICL numbers are comparable to free-running generation numbers	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
7d	That the LoRA rank curve has plateaued	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
7c, 7d	That LoRA at rank 1 is a fair matched-budget comparison	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
7	That LIMA is contradicted	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
8b	That the 6.3M model can ground a passage	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
8b	That hallucination rates can be compared between conditions	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
8d	That the model learned an operation from a tool-call format	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
8b	That any result here measures factual truth	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
8d, 9c	That the tool-call template echo means the model learned the operation	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
9a	That the head-ablation deltas are comparable to the published validation loss	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
9a	That head ablation deletes the head	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4

Table 38 (continued)

Act	Hypothesis	Test	Result	Status	Do-not-make note
9b	That the rollout negative is a general claim about attention interpretability	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
9a	That layer 0 head 5 is the head that carries the model	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
9c	That the logit lens shows the model encodes WHICH number is asserted	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
10c	That the calibration result means the model can follow instructions	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
10c	That instruction tuning made the model worse at the task it was tuned on	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4
10	That the do-not-make count is a claim about the paper	n/a – a boundary, not a test	not claimed	declined claim	published in Section 14.4